

Squidstat API User Manual

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Chapter 1

Squidstat API User Manual

The Admiral Instruments API provides you with more control of [our potentiostats](#). Through [asynchronous signals](#) you can capture events such as experiments starting, stopping, and pausing to seamlessly integrate these controls into your pipeline and workflow automation.

Some common use cases include:

- Chaining multiple experiments together for long term testing.
- Manipulating experiment parameters based on the result of the previous experiment.
- Starting and stopping an experiment when an external device detects a specific temperature.

You can find it on our [GitHub repository](#) including examples for [C++](#) and [Python](#).

This documentation will guide you through how to install and set up our API as well as provide you with examples with explanations to help support your workflow.

If you have any difficulty using our API, please reach out to us through the [discussions page](#) on our GitHub.

1.0.0.1 Where to go from here

- [Install and Setup C++](#)
- [Install and Setup Python](#)
- [Guided Examples](#)
- [API Objects](#)

Chapter 2

Advanced Experiment Logic

This example demonstrates how to create a custom experiment using loops and subexperiments for added complexity.

2.0.1 Experiment Description

The API allows us to build experiments with multiple elements, which can loop individually or as a group. Let's consider building an experimental sequence with the following structure:

1. Open Circuit Potential (OCP) element
2. Constant Potential Element (loops 4 times, increasing the voltage by 100 mV each loop)
3. Galvanostatic EIS element and OCP element (loops 3 times)
4. Constant Current element that loops 2 times

• C++

```
auto customExperiment = std::make_shared<AisExperiment>();

AisOpenCircuitElement ocpElement(1, 10);
success &= customExperiment->appendElement(ocpElement);

int voltage = 0;
for (int i = 0; i < 4; i++) {
    AisConstantPotElement cvElement(voltage, 0.1, 5);
    success &= customExperiment->appendElement(cvElement, 1);
    voltage += 0.1; // Adding 100 mV
}

AisExperiment eisSubExperiment;

AisEISGalvanostaticElement galvEISElement(10, 10000, 10, 0.01, 0.1);
AisOpenCircuitElement ocpElement2(1, 5);

success &= eisSubExperiment.appendElement(galvEISElement, 1);
success &= eisSubExperiment.appendElement(ocpElement2, 1);

success &= customExperiment->appendSubExperiment(eisSubExperiment, 3);

AisConstantCurrentElement ccElement(0.1, 1, 10);
success &= customExperiment->appendElement(ccElement, 2);

if (!success) {
    qDebug() << "Error building experiment";
    return 0;
}
```

- **Python**

```

customExperiment = AisExperiment()
# Step 1

ocpElement = AisOpenCircuitElement(1, 10)
success &= customExperiment.appendElement(ocpElement)

voltage = 0
for i in range(0, 4):
    cvElement = AisConstantPotElement(voltage, 0.1, 5)
    success &= customExperiment.appendElement(cvElement, 1)
    voltage = voltage + 0.1

eisSubExperiment = AisExperiment()

galvEISElement = AisEISGalvanostaticElement(10, 10000, 10, 0.01, 0.1)
ocpElement2 = AisOpenCircuitElement(1, 5)

success &= eisSubExperiment.appendElement(galvEISElement, 1)
success &= eisSubExperiment.appendElement(ocpElement2, 1)

success &= customExperiment.appendSubExperiment(eisSubExperiment, 3)

ccElement = AisConstantCurrentElement(0.1, 1, 10)
success &= customExperiment.appendElement(ccElement, 2)

if not success:
    print("Error building experiment")
    sys.exit()

```

Let's go over the code in more detail:

2.0.2 Building the Experiment

Here, we create the Open Circuit Element and add it to the experiment without specifying the number of loops, so it runs once.

- **C++**

```

AisOpenCircuitElement ocpElement(1, 10);
success &= customExperiment->appendElement(ocpElement);

```

- **Python**

```

ocpElement = AisOpenCircuitElement(1, 10)
success &= customExperiment.appendElement(ocpElement)

```

Next, we create the Constant Potential Elements, using a for loop to increment the voltage for each element, and then adding it to the created subExperiment.

- **C++**

```

int voltage = 0;
for (int i = 0; i < 4; i++) {
    AisConstantPotElement cvElement(voltage, 0.1, 5);
    success &= customExperiment->appendElement(cvElement, 1);
    voltage += 0.1; // Adding 100 mV
}

```

- **Python**

```

voltage = 0
for i in range(0, 4):
    cvElement = AisConstantPotElement(voltage, 0.1, 5)
    success &= customExperiment.appendElement(cvElement, 1)
    voltage = voltage + 0.1

```

For the looped EIS and OCP elements, we create another [AisExperiment](#) (eisSubExperiment), add the elements, and use [AisExperiment::appendSubExperiment](#) with 3 passed as the loop argument. This makes the two elements loop consecutively 3 times.

- C++

```
AisExperiment eisSubExperiment;

AisEISGalvanostaticElement galvEISElement(10, 10000, 10, 0.01, 0.1);
AisOpenCircuitElement ocpElement2(1, 5);

success &= eisSubExperiment.appendElement(galvEISElement, 1);
success &= eisSubExperiment.appendElement(ocpElement2, 1);

success &= customExperiment->appendSubExperiment(eisSubExperiment, 3);
```

- Python

```
eisSubExperiment = AisExperiment()

galvEISElement = AisEISGalvanostaticElement(10, 10000, 10, 0.01, 0.1)
ocpElement2 = AisOpenCircuitElement(1, 5)

success &= eisSubExperiment.appendElement(galvEISElement, 1)
success &= eisSubExperiment.appendElement(ocpElement2, 1)

success &= customExperiment.appendSubExperiment(eisSubExperiment, 3)
```

Lastly, we create the Constant Current Element, adding it to the main experiment using `AisExperiment::appendElement` with the loop argument set to 2. This loops the element twice at the end.

- C++

```
AisExperiment eisSubExperiment;

AisEISGalvanostaticElement galvEISElement(10, 10000, 10, 0.01, 0.1);
AisOpenCircuitElement ocpElement2(1, 5);

success &= eisSubExperiment.appendElement(galvEISElement, 1);
success &= eisSubExperiment.appendElement(ocpElement2, 1);

success &= customExperiment->appendSubExperiment(eisSubExperiment, 3);
```

- Python

```
eisSubExperiment = AisExperiment()

galvEISElement = AisEISGalvanostaticElement(10, 10000, 10, 0.01, 0.1)
ocpElement2 = AisOpenCircuitElement(1, 5)

success &= eisSubExperiment.appendElement(galvEISElement, 1)
success &= eisSubExperiment.appendElement(ocpElement2, 1)

success &= customExperiment.appendSubExperiment(eisSubExperiment, 3)
```

2.0.3 Starting the Experiment

The experiment is now ready and can be started like any other experiment.

- C++

```
QObject::connect(tracker, &AisDeviceTracker::newDeviceConnected, [=](const QString& deviceName) {
    qDebug() << "New Device Connected: " << deviceName;
    auto& handler = tracker->getInstrumentHandler(INSTRUMENT_NAME);

    connectSignals(handler);

    AisErrorCode error = handler.uploadExperimentToChannel(CHANNEL, customExperiment);
    if (error) {
        qDebug() << error.message();
        return 0;
    }

    error = handler.startUploadedExperiment(CHANNEL);
    if (error) {
        qDebug() << error.message();
        return 0;
    }
});

AisErrorCode error = tracker->connectToDeviceOnComPort(COMPORT);
if (error != error.Success) {
    qDebug() << error.message();
    return 0;
}
```

- **Python**

```
def startExperiment():
    handler = tracker.getInstrumentHandler(INSTRUMENT_NAME)

    connectSignals(handler)

    error = handler.uploadExperimentToChannel(CHANNEL, customExperiment)
    if error.value() != AIS_ERROR_CODE_SUCCESS:
        print(error.message())
        app.quit()

    error = handler.startUploadedExperiment(CHANNEL)
    if error.value() != AIS_ERROR_CODE_SUCCESS:
        print(error.message())
        app.quit()

tracker.newDeviceConnected.connect(startExperiment)

error = tracker.connectToDeviceOnComPort(COMPORT)
if error.value() != AIS_ERROR_CODE_SUCCESS:
    print(error.message())
    sys.exit()
```

See the full example [here](#)

Now that we have seen how create complex custom experiments, lets move on to the next example, where we will utilize manual mode operations to provide real-time control of the Squidstat.

Previous	Next
Handling Signals	Manual Experiments

Chapter 3

Finding COM Ports

COM ports are used by the API to communicate with Squidstats. It is crucial to know which serial port the instrument is connected to in order to establish the correct communication pathway. Locate the serial port in the Squidstat User Interface ([SUI](#)) software in the "More Options" tab under "Device Information." If the SUI is not installed, you can determine the serial port using different methods based on your operating system

3.1 Windows

1. Connect the device via USB and power it on.
2. Open Device Manager using the search function (Windows key > then type "Device Manager") or through Control Panel via: Control Panel > Hardware and Sound > Devices and Printers > Device Manager
3. Expand 'Ports' to view the connected devices (e.g. USB Serial Port (COM3)). If multiple COM ports are listed, power cycle or disconnect the device to identify the correct one.
4. When referring to the serial port from the example above in the API, the format for the entry would be COM3.

3.2 Mac

1. Connect the device via USB and power it on.
2. Open Terminal via Applications > Utilities > Terminal or use Spotlight Search (Cmd + Space, then type "Terminal").
3. List serial devices by entering `ls /dev/cu.*` in Terminal.
4. Identify the USB serial port from the list.

Example output:

```
/dev/cu.Bluetooth-Incoming-Port
/dev/cu.usbmodem14201
/dev/cu.usbserial-Admiral_1409
```

In this example, `/dev/cu.usbmodem14201` and `/dev/cu.usbserial-Admiral_1409` are the USB serial ports associated with the connected devices. If there are multiple entries, power cycle or disconnect the device and rerun the command.

5. When referring to a serial port from the example above in the API, the format for the entry would be `cu.usbmodem14201`.

3.3 Linux

1. Connect the device via USB and power it on.
2. Open Terminal (Ctrl + Alt + T).
3. Run `ls /dev | grep tty` in Terminal
4. Identify the USB serial port. Look for the lines that refer to USB devices (e.g., "`ttyACM0`" or "`ttyUSB1`"). For multiple entries, power cycle or disconnect the device and rerun the command.
5. When referring to the serial port from the example above in the API, the format for the entry would be `tty↔ACM0`.

Chapter 4

Extra Examples

The previous examples cover all the API features required to run most experiments.

Below are additional examples showcasing specific features and use cases of the API.

	C++	Python
Writing Data to File	Link	Link
Nonblocking Experiment	Link	Link
Cycler Linked Channels	Link	Link
Handle Pulse Data	Link	Link
Advanced Control Flow	Link	
Async Example		Link
Remote Squidstat Operation		Client & Server

4.1 Discussions

If you have any other questions or requests regarding the API, please refer to the Admiral Instruments [discussions page](#). Here you can ask questions, share your ongoing projects, and engage in discussions with other users and the Admiral Instruments team.

Chapter 5

Updating Firmware

5.0.1 Introduction

Each API version includes the necessary firmware for all supported Squidstats to connect properly. When a new API is released, it may include updated firmware with new features or fixes. However, firmware updates are not automatic. They must be initiated manually. A Squidstat won't connect to the API until its firmware is compatible. Once updated, the firmware stays on the device until another update is performed. No separate files or tools are needed, and updates are only required when the API version changes.

This page explains how to detect and handle outdated firmware.

Note

The latest Squidstat API and Squidstat User Interface should always have matching firmware versions for all devices. If switching between them requires a firmware update, please verify that you have both the latest SUI and API.

5.0.2 Checking for and Updating Outdated Firmware

If [AisDeviceTracker::connectToDeviceOnComPort](#) returns the [AisErrorCode::FirmwareNotSupported](#) error code, a firmware update is required to use the device with the API. Here is an example of how to update the firmware in this scenario.

```
• C++
#include "AisInstrumentHandler.h"
#include "AisDeviceTracker.h"

#include <QCoreApplication>
#include <QThread>
#include <QDebug>

#define COMPORT "COM1"

int main()
{
    int args;
    QCoreApplication a(args, nullptr);

    auto tracker = AisDeviceTracker::Instance();

    QObject::connect(tracker, &AisDeviceTracker::firmwareUpdateNotification, [=](const QString&
message) {
        qDebug() << message;
        if (message.contains("firmware is updated.)) {
            // Now instrument is ready to go
        }
    });
}
```

```

});

// Attempt to connect to the device
auto error = tracker->connectToDeviceOnComPort (COMPORT);
if (error == AISErrorCode::FirmwareNotSupported) {
    error = tracker->updateFirmwareOnComPort (COMPORT);

    // Some other error occurred
    if (error != AISErrorCode::Success) {
        qDebug() << "Error: " << error.message();
    }

} else if (error != AISErrorCode::Success) {
    qDebug() << "Error: " << error.message();
} else {
    qDebug() << "Device firmware is up to date";
}

return a.exec();
}

```

• Python

```

import sys
from PySide6.QtCore import QApplication

from SquidstatPyLibrary import AISDeviceTracker
from SquidstatPyLibrary import AISInstrumentHandler
from SquidstatPyLibrary import AISErrorCode

# Define relevant device information, for easy access
COMPORT = "COM16"

app = QApplication()

tracker = AISDeviceTracker.Instance()

def onProgressMessage(message):
    print(message)
    if message.__contains__("firmware is updated"):
        app.quit()

tracker.firmwareUpdateNotification.connect(onProgressMessage)

# Attempt to connect to the device
error = tracker.connectToDeviceOnComPort (COMPORT)
if error.value() == AISErrorCode.FirmwareNotSupported:
    error = tracker.updateFirmwareOnComPort (COMPORT)

    # Some other error occurred
    if error.value() != AISErrorCode.Success:
        print(f"Error: {error.message()}")
        sys.exit()
elif error.value() != AISErrorCode.Success:
    print(f"Error: {error.message()}")
    sys.exit()
else:
    print("Device is already up to date.")
    sys.exit()

sys.exit(app.exec())

```

The `AISDeviceTracker::firmwareUpdateNotification` signal will display the progress of the firmware update as a percentage while the instrument is updating. Once the update is complete, this notification will send the message "Firmware is updated", signaling that the instrument can now be connected to the API.

• C++

```

QObject::connect(tracker, &AISDeviceTracker::firmwareUpdateNotification, [=](const QString&
message) {
    qDebug() << message;
    if (message.contains("firmware is updated.)) {
        // Now instrument is ready to go
    }
});

```

• Python

```

tracker.firmwareUpdateNotification.connect(onProgressMessage)

```

See the full example [here](#)

Note that we check the [AisErrorCode](#) first before performing the firmware update. Alternatively, you can use [AisDeviceTracker::updateFirmwareOnAllAvailableDevices](#) all connected devices that require an update.

Previous	Next
Guided Examples	Running an Experiment

Chapter 6

Guided Examples

In this section, we will walk through examples to demonstrate how to connect to instruments, build experiments, and handle data effectively. Each walkthrough references examples for Python and C++. Full versions of these can be found in the Raw Example Code tab of the Reference section. You can use these examples to test out code and explore the functionality of the API.

- [Updating Firmware](#)
- [Running an Experiment](#)
- [Handling Signals](#)
- [Advanced Experiment Logic](#)
- [Manual Experiments](#)

See the Reference section for more examples and documentation

6.1 About Qt

The SquidstatLibrary relies heavily on Qt's signals and slots mechanism for asynchronous communication between the API and Squidstats. This feature allows the API to operate in a non-blocking manner, enabling Squidstat control without blocking the main thread of your application. This ensures that the API can manage Squidstat operations seamlessly, without hindering the execution of other parts of your program or any additional applications running alongside it.

The SquidstatLibrary relies heavily on Qt's signal and slot mechanism for asynchronous communication between the API and Squidstats. This feature allows the API to operate in a non-blocking manner, enabling Squidstat control without interrupting the main thread of your application. This is essential for applications that need to control Squidstats via the API while keeping the main application responsive.

Here are some useful links to Qt documentation:

- [Signals and Slots](#)
- [Qt's Event Loop](#)

Previous	Next
Main	Updating Firmware

Chapter 7

Manual Experiments

Here, we will cover how to operate a Squidstat in Manual Mode. The Squidstat API provides a set of manual mode commands as functions, which allow the user to control the squidstat while an experiment is running. Which can be useful for scenarios where the user needs to change parameters on the fly (i.e. in a GUI application). See the manual experiments tab in the Squidstat User Interface software for an example use case of this feature.

7.1 Manual Mode Commands

The following commands are provided by the [AisInstrumentHandler](#) to control the Squidstat.

```
AisErrorCode AisInstrumentHandler::setManualModeSamplingInterval(uint8_t channel, double value) const;

AisErrorCode AisInstrumentHandler::setManualModeOCP(uint8_t channel) const;

AisErrorCode AisInstrumentHandler::setManualModeConstantVoltage(uint8_t channel, double value) const;

AisErrorCode AisInstrumentHandler::setManualModeCurrentRange(uint8_t channel, int currentRangeIndex) const;

AisErrorCode AisInstrumentHandler::setManualModeCurrentAutorange(uint8_t channel) const;

AisErrorCode AisInstrumentHandler::setManualModeVoltageRange(uint8_t channel, int voltageRangeIndex) const;

AisErrorCode AisInstrumentHandler::setManualModeVoltageAutorange(uint8_t channel) const;

AisErrorCode AisInstrumentHandler::setManualModeConstantCurrent(uint8_t channel, double value) const;
```

7.1.1 Start the Experiment

Before calling any of the above commands, the manual experiment must be started.

- C++

```
//The default starting mode is always Open Circuit Potential
QDebug() << "Starting manual mode at open circuit potential";
AisErrorCode error = handler.startManualExperiment(CHANNEL);
if (error != AisErrorCode::Success) {
    qDebug() << error.message();
    QCoreApplication::quit();
}
```

- Python

```
# The default starting mode is always Open Circuit Potential.

print("Starting manual mode at open circuit potential")
error = handler.startManualExperiment(CHANNEL)
if error.value() != AisErrorCode.Success:
    print(error.message())
    app.quit()
```

Now our experiment will run indefinitely, until [AisInstrumentHandler::stopExperiment](#) is called.

7.1.2 Manual Modes

The three main operational modes are OCP (Open Circuit Potential), Potentiostatic, and Galvanostatic, defined below:

- OCP: Measures the open circuit potential of the system.
- Potentiostatic: Maintains a constant potential while measuring current.
- Galvanostatic: Maintains a constant current while measuring potential.

Here is an example of how to set each at timed intervals after an experiment has started.

• C++

```
// In this section we connect singleshot QTimer to lambda functions that call our mode
// changing functions.
// These lambdas are called asynchronously when the QTimer expires.

// 5 seconds after starting experiment, change to Constant Current at .1A
QTimer::singleShot(5000, [=, &handler]() {
    qDebug() << "Switching to constant current at .1A";
    AisErrorCode error = handler.setManualModeConstantCurrent(CHANNEL, .1);
    if (error != AisErrorCode::Success) {
        qDebug() << error.message();
    }
});

// 15 seconds after starting experiment, change to Constant Voltage at 1V
QTimer::singleShot(15000, [=, &handler]() {
    qDebug() << "Switching to constant voltage at 1V";
    AisErrorCode error = handler.setManualModeConstantVoltage(CHANNEL, 1);
    if (error != AisErrorCode::Success) {
        qDebug() << error.message();
    }
});

// 25 seconds after starting experiment, change back into Open Circuit Potential mode.
QTimer::singleShot(25000, [=, &handler]() {
    qDebug() << "Switching to open circuit potential";
    AisErrorCode error = handler.setManualModeOCP(CHANNEL);
    if (error != AisErrorCode::Success) {
        qDebug() << error.message();
    }
});
```

• Python

```
# In this section we create wrapper functions for the manual mode changing functions.
# These wrappers are called asynchronously when singleshot QTimer expires.

# This function changes the instrument to Constant Current at .1A
def setConstantCurrent(channel):
    print("Switching to constant current at .1A")
    error = handler.setManualModeConstantCurrent(channel, .1)
    if error.value() != AisErrorCode.Success:
        print(error.message())
# It is called 5 seconds after the experiment starts
QTimer.singleShot(5000, lambda:setConstantCurrent(CHANNEL))

# This function changes the instrument to Constant Voltage at 1V
def setConstantVoltage(channel):
    print("Switching to constant voltage at 1V")
    error = handler.setManualModeConstantVoltage(channel, 1)
    if error.value() != AisErrorCode.Success:
        print(error.message())
# It is called 15 seconds after the experiment starts
QTimer.singleShot(15000, lambda:setConstantVoltage(CHANNEL))

# This function changes the instrument to Open Circuit Potential
def setOpenCircuit(channel):
    print("Switching to open circuit potential")
    error = handler.setManualModeOCP(channel)
    if error.value() != AisErrorCode.Success:
        print(error.message())
# It is called 25 seconds after the experiment starts
QTimer.singleShot(25000, lambda:setOpenCircuit(CHANNEL))
```

Note: As a reminder, it is always a good idea to check the error code returned by each command to ensure successful execution. See [AisErrorCode](#) for more information.

7.1.3 Stop the Experiment

Stopping the manual experiment uses the same command as stopping a normal experiment ([AisInstrumentHandler::stopExperiment](#)). In this case, we use a SingleShot timer ([C++](#) | [Python](#)) to stop the experiment after 30 seconds.

- **C++**

```
// Stop the experiment after 30 seconds
QTimer::singleShot(30000, [=, &handler]() {
    qDebug() << "Stopping experiment";
    if (handler.stopExperiment(CHANNEL) != AisErrorCode::Success) {
        qDebug() << error.message();
    }
});
```

- **Python**

```
# Stop experiment after 30 seconds
def stopExperiment(channel):
    print("Stopping experiment.")
    error = handler.stopExperiment(channel)
    if error.value() != AisErrorCode.Success:
        print(error.message())
QTimer.singleShot(30000, lambda:stopExperiment(CHANNEL))
```

See the full example [here](#)

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Chapter 8

Running an Experiment

The following example demonstrates how to set up and run a basic experiment using the API. This example simply outlines the steps needed to connect to a Squidstat, build an experiment, and start it. Notably, it does not cover handling and reporting data from the Squidstat, which will be addressed in the [Example](#) on handling signals.

8.0.1 Setup

To begin, we need to include all necessary headers or libraries, and setup the Qt event loop. At this stage, you can also retrieve the [AisDeviceTracker](#) Instance.

- **C++**

```
#include "AisDeviceTracker.h"
#include "AisExperiment.h"
#include "AisInstrumentHandler.h"

#include "experiments/builder_elements/AisConstantPotElement.h"

#include <QCoreApplication>
#include <QDebug>

// Define relevant device information, for easy access
#define COMPORT "COM1"
#define CHANNEL 0

int main()
{
    char** test = nullptr;
    int args;

    QCoreApplication a(args, test);

    auto tracker = AisDeviceTracker::Instance();
```

- **Python**

```
import sys
from PySide6.QtCore import QCoreApplication
from SquidstatPyLibrary import AisDeviceTracker
from SquidstatPyLibrary import AisExperiment, AisErrorCode
from SquidstatPyLibrary import AisInstrumentHandler
from SquidstatPyLibrary import AisConstantPotElement

# Define relevant device information, for easy access
COMPORT = "COM16"
CHANNEL = 0

app = QCoreApplication()

tracker = AisDeviceTracker.Instance()
```

Reference: <https://doc.qt.io/qt-5/qcoreapplication.html>

8.0.2 Building the experiment

Next, create an element (e.g. [AisConstantPotElement](#)) and pass in the required parameters. Then, we will create the experiment object [AisExperiment](#) and add the element to it using [AisExperiment::appendElement](#).

- **C++**

```
//          Voltage = 1V, Sampling Interval = 1s, Duration = 30s
AisConstantPotElement cvElement(1, 1, 30);

auto customExperiment = std::make_shared<AisExperiment>();
// Append the constant potential element, and tell it to execute that element 1 time
customExperiment->appendElement(cvElement, 1);
```

- **Python**

```
cvElement = AisConstantPotElement(1, 1, 30)

# After this point, the experiment is empty, so we need to add some elements to it
experiment = AisExperiment()
# Append the constant potential element, and tell the experiment to execute that element 1 time
success = experiment.appendElement(cvElement, 1)

# Check if the element was added successfully
if not success:
    print("Error adding element to experiment")
    app.quit()
```

Reference: https://en.cppreference.com/w/cpp/memory/shared_ptr

8.0.3 Connecting to the Device

Connect to the instrument by calling [AisDeviceTracker::connectToDeviceOnComPort](#) with the desired COM port to link the [AisDeviceTracker](#) to the device. Alternatively, use [AisDeviceTracker::connectAllPluggedInDevices](#) to connect to all available devices.

- **C++**

```
AisErrorCode error = tracker->connectToDeviceOnComPort(COMPORT);
if (error != error.Success) {
    qDebug() << error.message();
    return 0;
}
```

- **Python**

```
error = tracker.connectToDeviceOnComPort(COMPORT)
# Check if connection was successful
if error.value() != AisErrorCode.Success:
    print(error.message())
    sys.exit()
```

Note: Any command to the instrument should always be checked for errors via the returned [AisErrorCode](#)

8.0.4 Starting Experiment

Finally, to start the experiment:

1. Grab the instrument handler from the device tracker after establishing a connection to the instrument.
 - See [Next Example](#) for more details on establishing connections
2. Upload the experiment to the desired channel and check for errors
3. Start the experiment, again checking for any errors

4. Run the QApplication event loop to allow all relevant signals to be processed (see the next example "[Handling Signals](#)")

- **C++**

```
auto& handler = tracker->getInstrumentHandler(deviceName);

connectSignals(handler);

AisErrorCode error = handler.uploadExperimentToChannel(CHANNEL, customExperiment);
if (error) {
    qDebug() << error.message();
    return;
}

// Start the previously uploaded experiment on the same channel
error = handler.startUploadedExperiment(CHANNEL);

// Exit the application if there is any error starting the experiment
if (error) {
    qDebug() << error.message();
    return;
}
```

- **Python**

```
handler = tracker.getInstrumentHandler(deviceName)

connectSignals(handler)

error = handler.uploadExperimentToChannel(CHANNEL, experiment)
# Exit the application if there is any error uploading experiment
if error.value() != AisErrorCode.Success:
    print(error.message())
    app.quit()

error = handler.startUploadedExperiment(CHANNEL)
# Exit the application if there is any error starting experiment
if error.value() != AisErrorCode.Success:
    print(error.message())
    app.quit()
```

See the full example [here](#)

Now that we have covered the basics of connecting to a Squidstat, building an experiment, and starting it, we can proceed to the next example. It demonstrates how to combine these steps and handle the signals emitted by the Squidstat.

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Chapter 9

Installing with C++

You can download the C++ version of the Squidstat API from the Admiral Instruments GitHub page or clone the repository using Git on your machine.

9.1 Cloning the Repository

1. To clone the repository, you will need to install Git. Depending on your platform, you can download the Git tool from [this link](#).
 - To verify if Git is properly installed, you can follow these steps:
 - Open the command prompt.
 - Type `git -v` and press Enter.
 - If Git is installed correctly, you should see the version information displayed in the terminal. ex:
`git version 2.41.0.windows.1`
2. Open a command prompt and navigate to the directory where you want the API to be. **Note:** A new folder will be created automatically after the next step.
3. Clone the API from the [Git repository](#) by typing the following command in the command prompt and pressing Enter.

```
git clone https://github.com/Admiral-Instruments/AdmiralSquidstatAPI
```

The result in the command prompt should look like this:

```
Cloning into 'AdmiralSquidstatAPI'...
remote: Enumerating objects: 1846, done.
remote: Counting objects: 100% (508/508), done.
remote: Compressing objects: 100% (159/159), done.
remote: Total 1846 (delta 440), reused 360 (delta 349), pack-reused 1338
Receiving objects: 100% (1846/1846), 79.18 MiB | 10.10 MiB/s, done.

Resolving deltas: 100% (1044/1044), done.
```

The `AdmiralSquidstatAPI` folder will now be created, and it contains all the files required to get started with the API using a compatible compiler. We will refer to this folder as the "Root Directory" of the API.

9.1.1 Mac ARM64 (Apple Silicon)

If you are using a newer Mac with an ARM64 processor (M1, M2, etc.), you must perform additional steps to set up the Squidstat API.

1. Open the cloned repository in a Finder window and navigate down a level to `./SquidstatLibrary`.
 - In this folder you should see `Linux`, `mac`, `windows`, `Linux.tar`, and `mac.tar.gz`.
2. Delete the `mac` directory. It contains the library for Intel-based Mac processors.
3. Extract the tar file by double clicking `mac.tar.gz` in the Finder window.
 - You should see a new `mac/` directory get automatically created.
4. At this point, your build environment is set up, and you can proceed normally.

9.1.2 Linux

If you are on a device running Linux, before using the API, run `sudo ./SquidstatLibrary/↵Linux/install_dependencies.sh` from the root directory of the SquidstatAPI. This will install all dependencies and configuration files needed to run the API.

Note: This will run `apt-get update` && `sudo apt-get upgrade -y` which will update all packages currently installed on your device.

9.1.2.1 Linux ARM64

If you are using a newer Linux with an ARM64 processor, you must perform additional steps to set up the Squidstat API.

1. Open the cloned repository in a Finder window and navigate down a level to `./SquidstatLibrary`.
 - In this folder you should see `Linux`, `mac`, `mac.tar.gz`, `windows`, and `Linux.tar`.
2. Delete the `Linux` directory. It contains the library for Intel-based x86_64 architecture.
3. Extract the tar file by double clicking `Linux.tar` in the Finder window.
 - You should see a new `Linux/` directory get automatically created.
4. At this point, your build environment is set up, and you can proceed normally.

9.1.3 CMake Installation

The API uses CMake natively to build the project. Provided is a `CMakeLists.txt` file which you can include in your project to build the API.

1. To utilize the `CMakeLists.txt` file, you need to install CMake. You can download CMake from [here](#).
 - Note:** When installing, be sure to add the CMake executable to your environment variables.
2. To verify if CMake is installed correctly, type `cmake` in the command prompt and press Enter.

9.1.3.1 Building the project

1. To use the API on Windows you must install MSVC64 through Visual Studio. For Mac and Linux install Clang.
2. Open a command prompt or terminal in the root directory of the API.
3. Run the following command to generate the build files for the compiler installed in your environment.

```
cmake -B ./build -S ./SquidstatLibrary/
```

Note: If you want to use a different build generator, use the `-G` option.

4. Build the project using the following command. This will compile all examples present in the Squidstat↵
Library directory.

```
cmake --build ./build
```
5. You can now run any of the examples through their compiled executable. ex: `./build/examples/↵
FirmwareUpdateDemo/Debug/FirmwareUpdateDemo.exe`

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Chapter 10

Getting Started: LabVIEW (Beta)

10.0.1 Introduction

This is a proof of concept for Squidstat integration into LabVIEW. It is not feature complete, but should include enough to get started using our Squidstats in LabVIEW.

Note: **Control via LabVIEW** is still in beta. We cannot guarantee the accuracy or stability of the system while using it. If you see any issues or if you would like to request that we implement a missing feature, please create a new discussion on our [GitHub Discussions page](#). We currently only support Windows 64bit versions of LabVIEW for this beta release.

10.0.2 How to Use

1. Follow the "Cloning the Repository" section of our documentation [here](#).
2. Checkout the beta branch using the git command `git checkout LabVIEW-beta`
3. Create or open an existing LabVIEW project.
4. In the project explorer, right-click `My Computer` and navigate to `Add->Folder (Auto-populating)...`
5. In the file browser pop-up, navigate to the cloned API and then into the `SquidstatLibrary/windows/↔ LabVIEW-beta` directory in the API and select it.
6. Open the `.vi` located at `LabVIEW-beta/Demo/Demo.vi`. Here you will find an example of handling AC and DC data. Our API utilizes LabVIEW User Events to facilitate the transfer of data into the rest of the system.
 - The event system in LabVIEW adds some overhead to the library overall. If you need faster response time, you can look at `Squidstat/SquidstatEventLoop.vi` for an example of the polling loop that creates these events. You can copy this loop where desired and remove the event structures. For future releases, this may become simplified.
7. At the point you may choose to import the palette file located at `SquidstatLibrary/windows/↔ LabVIEW-beta/SquidstatPalette.mnu`. This can be done via `Tools->Advanced->Edit Palette Set...`

Chapter 11

Installing the Python Wrapper

The Squidstat API provides a Python wrapper, which can be downloaded from the Admiral Instruments GitHub page. The Python wrapper gives access to all the functions of the Squidstat API in Python.

11.0.1 Mac ARM64 (Apple Silicon)

If you are using a newer Mac with an ARM64 processor (M1, M2, etc.), you must perform additional steps to set up the Squidstat API.

1. Open the cloned repository in a Finder window and navigate down a level to `./SquidstatLibrary`.
 - In this folder you should see `Linux`, `mac`, `windows`, `Linux.tar`, and `mac.tar.gz`.
2. Delete the `mac` directory. It contains the library for Intel-based Mac processors.
3. Extract the tar file by double clicking `mac.tar.gz` in the Finder window.
 - You should see a new `mac/` directory get automatically created.
4. At this point, your build environment is set up, and you can proceed normally.

11.1 Linux

If you are on a device running Linux, before using the API, run `sudo ./SquidstatLibrary/Linux/install_dependencies.sh` from the root directory of the SquidstatAPI. This will install all dependencies and configuration files needed to run the API.

Note: This will run `apt-get update && sudo apt-get upgrade -y` which will update all packages currently installed on your device.

11.1.0.1 Linux ARM64

If you are using a newer Linux with an ARM64 processor, you must perform additional steps to set up the Squidstat API.

1. Open the cloned repository in a Finder window and navigate down a level to `./SquidstatLibrary`.
 - In this folder you should see `Linux`, `mac`, `mac.tar.gz`, `windows`, and `Linux.tar`.
2. Delete the `Linux` directory. It contains the library for Intel-based x86_64 architecture.
3. Extract the tar file by double clicking `Linux.tar` in the Finder window.
 - You should see a new `Linux/` directory get automatically created.
4. At this point, your build environment is set up, and you can proceed normally.

11.2 How To Use The SquidstatLibrary with Python

1. To use the SquidstatPyLibrary, you need to install a Python version ≥ 3.9 and ≤ 3.12 .
 - Visit the official Python website at <https://www.python.org/downloads/>.
 - Download the installer for the desired Python version (≥ 3.9 and ≤ 3.12).
 - Run the installer and follow the installation wizard's instructions.
 - Make sure to select the option to add Python to the system environment (PATH) variables during the installation process. This will enable you to run Python from any location on your computer.
2. Make sure you have installed Python correctly by checking the Python version using the `python -V` command.
3. Now you can choose to install the library in either the global environment or a virtual environment. If you want to install the library in the global environment, you can skip this step.
 - If you prefer using a [Python virtual environment](#), you can create one by running the command: `python -m venv VIRTUAL_ENVIRONMENT_NAME`
 - Open the command prompt and activate the virtual environment by typing: `./VIRTUAL_ENVIRONMENT_NAME/Scripts/activate.bat` in Windows or `source ./VIRTUAL_ENVIRONMENT_NAME/bin/activate` in Linux and Mac.
4. Now you can proceed to install the SquidstatPyLibrary. Download the .whl file from [here](#). After downloading, you can install it using the command `pip install YOUR_DOWNLOADED_FILE.whl`
5. Now, let's run an example script. If you have an Experiment.py file that you created to run an experiment, you can execute it by using the command `python Experiment.py`.

Check out the [Guided Examples](#) for help getting started.

Chapter 12

Handling Signals

This example reviews the basic signals emitted by the [AisDeviceTracker](#) and [AisInstrumentHandler](#), which notify users of instrument events, errors, experiment data, and progress. For simplicity, all signals in this example are connected to lambda functions. However, they can be connected to any slot function compatible with the signal's signature.

12.0.1 Tracker Signals

The [AisDeviceTracker](#) emits signals regarding device connection, disconnection, and firmware update status. The following example demonstrates how these signals are connected. For details on firmware update signals, refer to the [Firmware Update Example](#).

- **C++**

```
QObject::connect(tracker, &AisDeviceTracker::deviceDisconnected, [=](const QString& deviceName) {
    qDebug() << "New Device Connected: " << deviceName;
});
QObject::connect(tracker, &AisDeviceTracker::newDeviceConnected, [=](const QString& deviceName) {
    qDebug() << "New Device Connected: " << deviceName;
```
- **Python**

```
tracker.newDeviceConnected.connect(startExperiment)
tracker.deviceDisconnected.connect(lambda deviceName: print(f"Device Disconnected: {deviceName}"))
```

12.0.2 Experiment Data Signals

The [AisInstrumentHandler](#) emits signals containing experiment data and progress information during active experiments. The signals [AisInstrumentHandler::activeDCDataReady](#) and [AisInstrumentHandler::activeACDataReady](#) provide relevant **DC** and **AC**, respectively, and should be handled for each experiment.

- **C++**

```
QObject::connect(&handler, &AisInstrumentHandler::activeDCDataReady, [=](uint8_t channel,
const AisDCData& data) {
    qDebug() << "Timestamp: " << data.timestamp << " Current: " << data.current << " Voltage: " <<
data.workingElectrodeVoltage << " CE Voltage : " << data.counterElectrodeVoltage;
});
QObject::connect(&handler, &AisInstrumentHandler::activeACDataReady, [=](uint8_t channel,
const AisACData& data) {
    qDebug() << "Timestamp: " << data.timestamp << " Frequency: " << data.frequency << " " <<
data.absoluteImpedance;
});
```
- **Python**

```
handler.activeDCDataReady.connect(lambda channel, data: print(f"Timestamp: {data.timestamp}
Current: {data.current} Voltage: {data.workingElectrodeVoltage} CE Voltage :
{data.counterElectrodeVoltage}"))
handler.activeACDataReady.connect(lambda channel, data: print(f"Timestamp: {data.timestamp}
Frequency: {data.frequency} Absolute Impedance: {data.absoluteImpedance}"))
```

12.0.3 Other Signals

The [AisInstrumentHandler](#) also emits other signals, such as notifications for the start of new elements, the end of an experiment, instrument errors, and more. Below are some examples:

- **C++**

```
// Whenever a new node in the element starts, note: some Ais Elements contain multiple logical
nodes
// i.e AisCyclicVoltammatoryElement contains 4 nodes for each linear segment of each cycle plus
a quiet time node if enabled
// So this lambda would be executed atleast 4 times for each cycle
QObject::connect(&handler, &AisInstrumentHandler::experimentNewElementStarting, [=](uint8_t
channel, const AisExperimentNode& info) {
    qDebug() << "New element starting: " << info.stepName;
});
// Whenever an experiment completes or is manually stopped, this will execute
QObject::connect(&handler, &AisInstrumentHandler::experimentStopped, [=](uint8_t channel,
const QString& reason) {
    qDebug() << "Experiment Stopped Signal " << channel << "Reason : " << reason;
});
QObject::connect(&handler, &AisInstrumentHandler::deviceError, [=](uint8_t channel, const
QString& error) {
    qDebug() << "Device Error: " << error;
});
```

- **Python**

```
# Whenever a new node in the element starts, note: some Ais Elements contain multiple logical
nodes
# i.e AisCyclicVoltammatoryElement contains 4 nodes for each linear segment of each cycle plus a
quiet time node if enabled
# So this lambda would be executed atleast 4 times for each cycle
handler.experimentNewElementStarting.connect(lambda channel, info: print(f"New element starting:
{info.stepName}"))
# Whenever an experiment completes or is manually stopped, this will execute
handler.experimentStopped.connect(lambda channel, reason: (print(f"Experiment Stopped Signal
{channel}, {reason}"), app.quit()))
handler.deviceError.connect(lambda channel, error: print(f"Device Error: {error}"))
```

For other signals emitted: [AisInstrumentHandler signals](#)

12.0.4 Connecting Signals

To connect data and handler signals to a valid [AisInstrumentHandler](#) object, ensure the connection to the instrument is fully established. This can be done by connecting signals when the [AisDeviceTracker::newDeviceConnected](#) signal is emitted.

Here, a lambda function is created to connect relevant signals to the handler when the device is connected:

- **C++**

```
auto connectSignals = [=](const AisInstrumentHandler& handler) {
    QObject::connect(&handler, &AisInstrumentHandler::activeDCDataReady, [=](uint8_t channel,
const AisDCData& data) {
        qDebug() << "Timestamp: " << data.timestamp << " Current: " << data.current << " Voltage: " <<
data.workingElectrodeVoltage << " CE Voltage : " << data.counterElectrodeVoltage;
    });
    QObject::connect(&handler, &AisInstrumentHandler::activeACDataReady, [=](uint8_t channel,
const AisACData& data) {
        qDebug() << "Timestamp: " << data.timestamp << " Frequency: " << data.frequency << " " <<
data.absoluteImpedance;
    });
    // Whenever a new node in the element starts, note: some Ais Elements contain multiple logical
nodes
    // i.e AisCyclicVoltammatoryElement contains 4 nodes for each linear segment of each cycle plus
a quiet time node if enabled
    // So this lambda would be executed atleast 4 times for each cycle
    QObject::connect(&handler, &AisInstrumentHandler::experimentNewElementStarting, [=](uint8_t
channel, const AisExperimentNode& info) {
        qDebug() << "New element starting: " << info.stepName;
    });
    // Whenever an experiment completes or is manually stopped, this will execute
```

```

        QObject::connect(&handler, &AisInstrumentHandler::experimentStopped, [=](uint8_t channel,
const QString& reason) {
            qDebug() << "Experiment Stopped Signal " << channel << "Reason : " << reason;
        });
        QObject::connect(&handler, &AisInstrumentHandler::deviceError, [=](uint8_t channel, const
QString& error) {
            qDebug() << "Device Error: " << error;
        });
    };
};

```

• Python

```

def connectSignals(handler):

    handler.activeDCDataReady.connect(lambda channel, data: print(f"Timestamp: {data.timestamp}
Current: {data.current} Voltage: {data.workingElectrodeVoltage} CE Voltage :
{data.counterElectrodeVoltage}"))
    handler.activeACDataReady.connect(lambda channel, data: print(f"Timestamp: {data.timestamp}
Frequency: {data.frequency} Absolute Impedance: {data.absoluteImpedance}"))

    # Whenever a new node in the element starts, note: some Ais Elements contain multiple logical
nodes
    # i.e AisCyclicVoltammetryElement contains 4 nodes for each linear segment of each cycle plus a
quiet time node if enabled
    # So this lambda would be executed atleast 4 times for each cycle
    handler.experimentNewElementStarting.connect(lambda channel, info: print(f"New element starting:
{info.stepName}"))
    # Whenever an experiment completes or is manually stopped, this will execute
    handler.experimentStopped.connect(lambda channel, reason: (print(f"Experiment Stopped Signal
{channel}, {reason}"), app.quit()))
    handler.deviceError.connect(lambda channel, error: print(f"Device Error: {error}"))

```

Then, call this function after connecting to the device but before starting the experiment:

• C++

```

auto& handler = tracker->getInstrumentHandler(deviceName);

connectSignals(handler);

AisErrorCode error = handler.uploadExperimentToChannel(CHANNEL, customExperiment);
if (error) {
    qDebug() << error.message();
    return;
}

// Start the previously uploaded experiment on the same channel
error = handler.startUploadedExperiment(CHANNEL);

// Exit the application if there is any error starting the experiment
if (error) {
    qDebug() << error.message();
    return;
}

```

• Python

```

handler = tracker.getInstrumentHandler(deviceName)

connectSignals(handler)

error = handler.uploadExperimentToChannel(CHANNEL, experiment)
# Exit the application if there is any error uploading experiment
if error.value() != AisErrorCode.Success:
    print(error.message())
    app.quit()

error = handler.startUploadedExperiment(CHANNEL)
# Exit the application if there is any error starting experiment
if error.value() != AisErrorCode.Success:
    print(error.message())
    app.quit()

```

See the full example [here](#)

12.0.5 References

For information on syntax and related concepts used in this example, refer to the following documentation links:

- Signals and Slots: [C++](#) | [Python](#)
- Lambda Functions [C++](#) | [Python](#)

With all the relevant code to run an experiment in place, we can now explore how the API supports building more complex experiment logic.

Previous	Next
Running an Experiment	Advanced Experiment Logic

Chapter 13

Deprecated List

Member [AisDiffPulseVoltammetryElement::isAutoRange \(\)](#) const

This function is deprecated and no longer supports auto range for this element. Specify the current using [setApproxMaxCurrent\(\)](#). The device will determine the appropriate range based on the current value.

Member [AisDiffPulseVoltammetryElement::setAutoRange \(\)](#)

This function is deprecated. Use [setApproxMaxCurrent\(\)](#) to specify the current range instead.

Member [AisNormalPulseVoltammetryElement::isAutoRange \(\)](#) const

This function is deprecated and no longer supports auto range for this element. Specify the current using [setApproxMaxCurrent\(\)](#). The device will determine the appropriate range based on the current value.

Member [AisNormalPulseVoltammetryElement::setAutoRange \(\)](#)

This function is deprecated. Use [setApproxMaxCurrent\(\)](#) to specify the current range instead.

Member [AisSquareWaveVoltammetryElement::AisSquareWaveVoltammetryElement \(double startVoltage, double endVoltage, double voltageStep, double pulseAmp, double pulseFrequency\)](#)

Use the constructor with the approxMaxCurrent parameter instead.

Member [AisSquareWaveVoltammetryElement::isAutoRange \(\)](#) const

This function is deprecated and no longer supports auto range for this element. Specify the current using [setApproxMaxCurrent\(\)](#). The device will determine the appropriate range based on the current value.

Member [AisSquareWaveVoltammetryElement::setAutoRange \(\)](#)

This function is deprecated. Use [setApproxMaxCurrent\(\)](#) to specify the current range instead.

Chapter 14

Topic Index

14.1 API Objects by Category

Objects in the SquidstatAPI are grouped into categories to make it easier to find related items. Click any of the categories below to see the list of all included objects.

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Helpers	44

Chapter 15

Class Index

15.1 All Classes

All classes and structures in the SquidstatAPI in alphabetical order.

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AisCompRange	This class has advanced options controlling the device stability including the bandwidth index and the stability factor	48
AisConstantCurrentElement	Experiment that simulates a constant current flow with more advance options for stopping the experiment. 50	
AisConstantPotElement	Experiment that simulates a constant applied voltage. 57	
AisConstantPowerElement	This experiment simulates a constant power, charge or discharge". 66	
AisConstantResistanceElement	This element/experiment simulates a constant resistance load. 74	
AisCumulativeCapacityData	A structure containing DC cumulative capacity data collected from the instrument	81
AisCyclicVoltammetryElement	This experiment sweeps the potential of the working electrode back and forth between the first voltage-limit and the second voltage-limit at a constant scan rate (dE/dt) for a specified number of cycles	81
AisDataManipulator	This class offers advanced control over pulse data collection and manipulation. It provides methods to manipulate AIS primary data for all three pulse voltammetry experiments types, namely Differential Pulse Voltammetry (DPV), Square Wave Voltammetry (SWV), and Normal Pulse Voltammetry (NPV)	91
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A class representing an experiment to apply the stepped current.	206
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A class representing an experiment to apply the stepped voltage	211

Chapter 16

Topic Documentation

16.1 Experiment Elements

Represents all elements used to create and configure experiments in the API.

Classes

- class [AisConstantCurrentElement](#)
an experiment that simulates a constant current flow with more advance options for stopping the experiment.
- class [AisConstantPotElement](#)
an experiment that simulates a constant applied voltage.
- class [AisConstantPowerElement](#)
This experiment simulates a constant power, charge or discharge".
- class [AisConstantResistanceElement](#)
This element/experiment simulates a constant resistance load.
- class [AisCyclicVoltammetryElement](#)
*This experiment sweeps the potential of the working electrode back and forth between the **first voltage-limit** and the **second voltage-limit** at a constant **scan rate (dE/dt)** for a specified number of **cycles**.*
- class [AisDCCurrentSweepElement](#)
*this experiment performs a DC current sweep from the **starting current** to the **ending current** which progresses linearly according to the **scan rate**.*
- class [AisDCPotentialSweepElement](#)
*this experiment performs a DC potential sweep from the **starting current** to the **ending current** which progresses linearly according to the **scan rate**.*
- class [AisDiffPulseVoltammetryElement](#)
*In this experiment, the working electrode holds at a **starting potential** during the **quiet time**. Then it applies a train of pulses superimposed on a staircase waveform, with a uniform **potential step** size. The potential continues to step until the **final potential** is reached.*
- class [AisEISGalvanostaticElement](#)
*This experiment records the complex impedance of the experimental cell in galvanostatic mode, starting from the **start frequency** and sweeping through towards the **end frequency**, with a fixed number of frequency **steps per decade**.*
- class [AisEISPotentiostaticElement](#)

This experiment records the complex impedance of the experimental cell in potentiostatic mode, starting from the **start frequency** and sweeping through towards the **end frequency**, with a fixed number of frequency **steps per decade**.

- class [AisNormalPulseVoltammetryElement](#)

This experiment holds the working electrode at a **baseline potential** during the **quiet time**, then applies a train of pulses, which increase in amplitude until the **final potential** is reached.

- class [AisOpenCircuitElement](#)

This experiment observes the **open circuit potential** of the working electrode for a specific period of time.

- class [AisSquareWaveVoltammetryElement](#)

This experiment holds the working electrode at the **starting potential** during the **quiet time**. Then it applies a train of square pulses superimposed on a staircase waveform with a uniform **potential step** magnitude.

- class [AisStaircasePotentialVoltammetryElement](#)

[AisStaircasePotentialVoltammetryElement](#) class represents an element for staircase potential voltammetry experiments. It inherits from [AisAbstractElement](#).

- class [AisSteppedCurrentElement](#)

A class representing an experiment to apply the stepped current.

- class [AisMottSchottkyElement](#)

This class performs Mott-Schottky analysis on the working electrode for a specified range of potentials.

- class [AisSteppedVoltageElement](#)

A class representing an experiment to apply the stepped voltage.

16.2 Instrument Control

Classes used to communicate with and control connected devices.

Classes

- class [AisDeviceTracker](#)

This class is used track device connections to the computer. It also provides instrument handlers specific to each connected device which provide control of the relevant device.

- class [AisErrorCode](#)

This class contains the possible error codes returned to the user when working with the API. Error codes can help diagnose issues such as invalid parameters, communication failures, or device malfunctions. By handling errors properly, you can ensure reliable operation of your experiments.

- class [AisExperiment](#)

this class is used to create custom experiments. A custom experiment contains one or more elements. Once you create elements and set their parameters, you can add them to the container.

- class [AisInstrumentHandler](#)

this class provides control of the device including starting, pausing, resuming and stopping an experiment on a channel as well as reading the data and other controls of the device.

16.3 Helpers

Miscellaneous objects used to represent device configurations as well as manipulate and hold data.

Classes

- class [AisCompRange](#)
This class has advanced options controlling the device stability including the bandwidth index and the stability factor.
- struct [AisCumulativeCapacityData](#)
A structure containing DC cumulative capacity data collected from the instrument.
- struct [AisDCData](#)
A structure containing DC data collected from the instrument.
- struct [AisACData](#)
A structure containing AC data collected from the instrument.
- struct [AisExperimentNode](#)
A structure containing some information regarding the running element.
- class [AisDataManipulator](#)
This class offers advanced control over pulse data collection and manipulation. It provides methods to manipulate AIS primary data for all three pulse voltammetry experiments types, namely Differential Pulse Voltammetry (DPV), Square Wave Voltammetry (SWV), and Normal Pulse Voltammetry (NPV).

Enumerations

- enum [AisPulseType](#) { **DifferentialPulse** = 0 , **NormalPulse** = 1 , **SquarewavePulse** = 2 }
This enum represents different pulse types, it is intended to tell the [AisDataManipulator](#) class how to calculate various pulse experiment parameters.

16.3.1 Enumeration Type Documentation

16.3.1.1 AisPulseType

enum [AisPulseType](#)

Enumerator

NormalPulse	Corresponds to AisDiffPulseVoltammetryElement .
SquarewavePulse	Corresponds to AisNormPulseVoltammetryElement .

Chapter 17

Class Documentation

17.1 AisACData Struct Reference

A structure containing AC data collected from the instrument.

```
#include <AisDataPoints.h>
```

```
#include <AisDataPoints.h>
```

Public Attributes

- double **timestamp**
the time at which the AC data arrived.
- double **frequency**
the applied frequency in Hz.
- double **absoluteImpedance**
the magnitude of the complex impedance.
- double **realImpedance**
the real part of the complex impedance.
- double **imagImpedance**
the imaginary part of the complex impedance.
- double **phaseAngle**
the phase angle between the real and the imaginary parts of the impedance.
- double **totalHarmonicDistortion**
the percentage of the total harmonic distortion in the AC signal.
- double **numberOfCycles**
the number of cycles specific to the reported frequency.
- double **workingElectrodeDCVoltage**
the DC working electrode voltage in volts.
- double **DCCurrent**
the DC electric current value in Amps
- double **currentAmplitude**
the amplitude of the AC current.
- double **voltageAmplitude**
the amplitude of the AC voltage.

17.1.1 Member Data Documentation

17.1.1.1 numberOfCycles

```
double AisACData::numberOfCycles
```

Note

In EIS, we run a range of frequencies. For each frequency, a specific number of cycles are run. The higher the frequency, the more number of cycles.

The documentation for this struct was generated from the following file:

- AisDataPoints.h

17.2 AisCompRange Class Reference

This class has advanced options controlling the device stability including the bandwidth index and the stability factor.

```
#include <AisCompRange.h>
```

```
#include <AisCompRange.h>
```

Public Member Functions

- [AisCompRange](#) (const QString &compRangeName, uint8_t bandwidthIndex, uint8_t stabilityFactor)
constructor for the compensation-range object.
- **AisCompRange** (const [AisCompRange](#) &)
copy constructor for the compensation-range object.
- uint8_t [getBandwidthIndex](#) () const
get the value set for the bandwidth index.
- void [setBandwidthIndex](#) (uint8_t index)
set the index value for the bandwidth.
- uint8_t [getStabilityFactor](#) () const
get the value set for the stability factor.
- void [setStabilityFactor](#) (uint8_t factor)
set a value for the stability factor.
- void [setCompRangeName](#) (const QString &compRangeName)
set a name for the compensation range for reference purposes.
- const QString & [getCompRangeName](#) () const
get the name set for the compensation range.

17.2.1 Constructor & Destructor Documentation

17.2.1.1 AisCompRange()

```
AisCompRange::AisCompRange (
    const QString & compRangeName,
    uint8_t bandwidthIndex,
    uint8_t stabilityFactor) [explicit]
```

Parameters

<i>compRangeName</i>	a name to set for the compensation range for reference purposes.
<i>bandwidthIndex</i>	the index value for the bandwidth.
<i>stabilityFactor</i>	the factor value for the stability.

See also

[setBandwidthIndex](#)[setStabilityFactor](#)

17.2.2 Member Function Documentation

17.2.2.1 getBandwidthIndex()

```
uint8_t AisCompRange::getBandwidthIndex () const
```

Returns

the set value for the bandwidth index.

See also

[setBandwidthIndex](#)

17.2.2.2 getCompRangeName()

```
const QString & AisCompRange::getCompRangeName () const
```

Returns

the name set for the compensation range.

17.2.2.3 getStabilityFactor()

```
uint8_t AisCompRange::getStabilityFactor () const
```

Returns

the value set for the stability factor.

17.2.2.4 setBandwidthIndex()

```
void AisCompRange::setBandwidthIndex (  
    uint8_t index)
```

Usually, the device's default index value is optimal for running experiments. You may still increase the index within the range 0-10 as you run higher frequency experiments to see what best fits.

Parameters

<i>index</i>	the index value for the bandwidth (0-10).
--------------	---

17.2.2.5 setCompRangeName()

```
void AisCompRange::setCompRangeName (  
    const QString & compRangeName)
```

Parameters

<i>compRangeName</i>	the name to set for the compensation range.
----------------------	---

17.2.2.6 setStabilityFactor()

```
void AisCompRange::setStabilityFactor (  
    uint8_t factor)
```

Usually, the device's default factor value is optimal for running experiments. You may still increase the factor within the range 0-10 as you run experiments with more oscillations to see what best fits.

Parameters

<i>factor</i>	the stability-factor value (0-10)
---------------	-----------------------------------

The documentation for this class was generated from the following file:

- AisCompRange.h

17.3 AisConstantCurrentElement Class Reference

an experiment that simulates a constant current flow with more advance options for stopping the experiment.

```
#include <AisConstantCurrentElement.h>
```

```
#include <AisConstantCurrentElement.h>
```

Public Member Functions

- [AisConstantCurrentElement](#) (double current, double samplingInterval, double duration)
the constant current element constructor.
- [AisConstantCurrentElement](#) (const [AisConstantCurrentElement](#) &)
copy constructor for the [AisConstantCurrentElement](#) object.
- [AisConstantCurrentElement](#) & **operator=** (const [AisConstantCurrentElement](#) &)
overload equal to operator for the [AisConstantCurrentElement](#) object.
- QString [getName](#) () const override
get the name of the element.
- QStringList [getCategory](#) () const override
get a list of applicable categories of the element.
- double [getCurrent](#) () const
get the value set for the current.
- void [setCurrent](#) (double current)
set the value for the current.
- double [getSamplingInterval](#) () const
get how frequently we are sampling the data.
- void [setSamplingInterval](#) (double samplingInterval)
set how frequently we are sampling the data.
- double [getMinSamplingVoltageDifference](#) () const
get the minimum sampling voltage difference for reporting the data.
- void [setMinSamplingVoltageDifference](#) (double minVoltageDifference)
set a minimum sampling voltage difference for reporting the voltage.
- double [getMaxVoltage](#) () const
get the value set for the maximum voltage. The experiment will end when it reaches this value.
- void [setMaxVoltage](#) (double maxVoltage)
set a maximum voltage to stop the experiment.
- double [getMinVoltage](#) () const
get the value set minimum for the voltage in volts.
- void [setMinVoltage](#) (double minVoltage)
set a minimum voltage to stop the experiment.
- double [getMaxDuration](#) () const
get the maximum duration set for the experiment. The experiment will end when the duration of the experiment reaches this value.
- void [setMaxDuration](#) (double maxDuration)
set the maximum duration for the experiment.
- double [getMaxCapacity](#) () const
get the value set for the maximum capacity / cumulative charge.
- void [setMaxCapacity](#) (double maxCapacity)
set the value for the maximum capacity / cumulative charge in Coulomb.
- bool [isAutoRange](#) () const
tells whether the current range is set to auto-select or not.
- void [setAutoRange](#) ()
set to auto-select the current range.
- double [getApproxMaxCurrent](#) () const
get the value set for the expected maximum current.
- void [setApproxMaxCurrent](#) (double approxMaxCurrent)
set maximum current expected, for manual current range selection.
- bool [isAutoVoltageRange](#) () const
tells whether the voltage range is set to auto-select or not.

- void `setAutoVoltageRange ()`
set to auto-select the voltage range.
- double `getApproxMaxVoltage ()` const
get the value set for the expected maximum voltage.
- void `setApproxMaxVoltage (double approxMaxVoltage)`
set maximum voltage expected, for manual voltage range selection.

17.3.1 Constructor & Destructor Documentation

17.3.1.1 AisConstantCurrentElement()

```
AisConstantCurrentElement::AisConstantCurrentElement (
    double current,
    double samplingInterval,
    double duration) [explicit]
```

Parameters

<i>current</i>	the value for the current in Amps.
<i>samplingInterval</i>	the data sampling interval value in seconds.
<i>duration</i>	the maximum duration for the experiment in seconds.

17.3.2 Member Function Documentation

17.3.2.1 getApproxMaxCurrent()

```
double AisConstantCurrentElement::getApproxMaxCurrent () const
```

Returns

the value set for the expected maximum current in Amps.

Note

if nothing was manually set, the device will auto-select the current range and the return value will be positive infinity.

17.3.2.2 getApproxMaxVoltage()

```
double AisConstantCurrentElement::getApproxMaxVoltage () const
```

Returns

the value set for the expected maximum Voltage in volt.

Note

if nothing was manually set, the device will auto-select the voltage range and the return value will be positive infinity.

17.3.2.3 getCategory()

```
QStringList AisConstantCurrentElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Galvanostatic Control", "Basic Experiments").

17.3.2.4 getCurrent()

```
double AisConstantCurrentElement::getCurrent () const
```

Returns

the value for the current in Amps.

17.3.2.5 getMaxCapacity()

```
double AisConstantCurrentElement::getMaxCapacity () const
```

Returns

the value set for the maximum capacity in Coulomb.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity.

17.3.2.6 getMaxDuration()

```
double AisConstantCurrentElement::getMaxDuration () const
```

Returns

the maximum duration for the experiment in seconds.

17.3.2.7 getMaxVoltage()

```
double AisConstantCurrentElement::getMaxVoltage () const
```

Returns

the value set for the maximum voltage.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity

17.3.2.8 getMinSamplingVoltageDifference()

```
double AisConstantCurrentElement::getMinSamplingVoltageDifference () const
```

get the value set for the minimum sampling voltage difference.

Returns

the value set for the minimum sampling voltage difference.

See also

[setMinSamplingVoltageDifference](#)

Note

this is an optional parameter. If no value has been set, the default value is negative infinity.

17.3.2.9 getMinVoltage()

```
double AisConstantCurrentElement::getMinVoltage () const
```

Returns

the value set for the minimum voltage in volts.

Note

this is an optional parameter. If no value has been set, the default value is negative infinity

17.3.2.10 getName()

```
QString AisConstantCurrentElement::getName () const [override]
```

Returns

The name of the element: "Constant Current, Advanced".

17.3.2.11 getSamplingInterval()

```
double AisConstantCurrentElement::getSamplingInterval () const
```

Returns

the data sampling interval value in seconds.

17.3.2.12 isAutoRange()

```
bool AisConstantCurrentElement::isAutoRange () const
```

Returns

true if the current range is set to auto-select and false if a range has been selected.

17.3.2.13 isAutoVoltageRange()

```
bool AisConstantCurrentElement::isAutoVoltageRange () const
```

Returns

true if the voltage range is set to auto-select and false if a range has been selected.

17.3.2.14 setApproxMaxCurrent()

```
void AisConstantCurrentElement::setApproxMaxCurrent (
    double approxMaxCurrent)
```

This is an **optional** parameter. If nothing is set, the device will auto-select the current range.

Parameters

<i>approxMaxCurrent</i>	the value for the maximum current expected in Amps.
-------------------------	---

17.3.2.15 setApproxMaxVoltage()

```
void AisConstantCurrentElement::setApproxMaxVoltage (
    double approxMaxVoltage)
```

This is an **optional** parameter. If nothing is set, the device will auto-select the voltage range.

Parameters

<i>approxMaxVoltage</i>	the value for the maximum current expected in V.
-------------------------	--

17.3.2.16 setAutoRange()

```
void AisConstantCurrentElement::setAutoRange ()
```

This option is set by default. There is no need to call this function to auto-select if the range was not manually set.

17.3.2.17 setAutoVoltageRange()

```
void AisConstantCurrentElement::setAutoVoltageRange ()
```

This option is set by default. There is no need to call this function to auto-select if the range was not manually set.

17.3.2.18 setCurrent()

```
void AisConstantCurrentElement::setCurrent (
    double current)
```

Parameters

<i>current</i>	the value for the current in Amps.
----------------	------------------------------------

17.3.2.19 setMaxCapacity()

```
void AisConstantCurrentElement::setMaxCapacity (
    double maxCapacity)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit cumulative charge value. If a maximum capacity is set, the experiment will continue to run as long as the cumulative charge is below that value.

Parameters

<i>maxCapacity</i>	the value to set for the cell maximum capacity.
--------------------	---

17.3.2.20 setMaxDuration()

```
void AisConstantCurrentElement::setMaxDuration (
    double maxDuration)
```

The experiment will continue to run as long as the time passed is less than the value to set.

Parameters

<i>maxDuration</i>	the maximum duration for the experiment in seconds.
--------------------	---

17.3.2.21 setMaxVoltage()

```
void AisConstantCurrentElement::setMaxVoltage (
    double maxVoltage)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit voltage value. If a maximum voltage is set, the experiment will continue to run as long as the measured voltage is below that value.

Parameters

<i>maxVoltage</i>	the maximum voltage value in volts at which the experiment will stop.
-------------------	---

17.3.2.22 setMinSamplingVoltageDifference()

```
void AisConstantCurrentElement::setMinSamplingVoltageDifference (
    double minVoltageDifference)
```

This is an **optional** condition. If nothing is set, then the experiment will report the data at time sampling interval. When this is set, then the voltage is reported when there is a voltage difference of at least the given minimum sampling voltage difference. So, when one voltage data point is reported (at the minimum possible time sampling interval), the next data point is not reported unless the difference between the two voltage data points exceeds this given minimum sampling voltage difference value.

Note

when this is set, this overrides the set value for the sampling interval.

Parameters

<i>minVoltageDifference</i>	the minimum sampling voltage difference value in volts.
-----------------------------	---

17.3.2.23 setMinVoltage()

```
void AisConstantCurrentElement::setMinVoltage (
    double minVoltage)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an lower-limit voltage value. If a maximum voltage is set, the experiment will continue to run as long as the measured voltage is above that value.

Parameters

<i>minVoltage</i>	the minimum voltage value in volts at which the experiment will stop.
-------------------	---

17.3.2.24 setSamplingInterval()

```
void AisConstantCurrentElement::setSamplingInterval (
    double samplingInterval)
```

Parameters

<i>samplingInterval</i>	the data sampling interval value in seconds.
-------------------------	--

The documentation for this class was generated from the following file:

- AisConstantCurrentElement.h

17.4 AisConstantPotElement Class Reference

an experiment that simulates a constant applied voltage.

```
#include <AisConstantPotElement.h>
```

```
#include <AisConstantPotElement.h>
```

Public Member Functions

- [AisConstantPotElement](#) (double voltage, double samplingInterval, double duration)
the constant potential element constructor.
- [AisConstantPotElement](#) (const [AisConstantPotElement](#) &)
copy constructor for the [AisConstantPotElement](#) object.
- [AisConstantPotElement](#) & **operator=** (const [AisConstantPotElement](#) &)
overload equal to operator for the [AisConstantPotElement](#) object.
- QString [getName](#) () const override
get the name of the element.
- QStringList [getCategory](#) () const override
get a list of applicable categories of the element.
- double [getPotential](#) () const
get the value set for the potential in volts.
- void [setPotential](#) (double potential)
set the value for the potential in volts.
- bool [isVoltageVsOCP](#) () const
tells whether the specified voltage is set against the open-circuit voltage or the reference terminal.
- void [setVoltageVsOCP](#) (bool vsOCP)
set whether to reference the specified voltage against the open-circuit voltage or the reference terminal.
- double [getSamplingInterval](#) () const
get how frequently we are sampling the data.
- void [setSamplingInterval](#) (double samplingInterval)
set how frequently we are sampling the data.
- double [getMaxDuration](#) () const
get the maximum duration set for the experiment. The experiment will end when the duration of the experiment reaches this value.
- void [setMaxDuration](#) (double maxDuration)
set the maximum duration for the experiment.
- double [getMaxAbsoluteCurrent](#) () const
get the maximum value set for the absolute current in Amps. The experiment will end when the absolute current reaches this value.
- void [setMaxAbsoluteCurrent](#) (double maxCurrent)
set the maximum value for the absolute current in Amps.
- double [getMaxCurrent](#) () const
get the maximum value set for the absolute current in Amps. The experiment will end when the absolute current reaches this value.
- void [setMaxCurrent](#) (double maxCurrent)
set the maximum value for the absolute current in Amps.
- double [getMinAbsoluteCurrent](#) () const
get the minimum value set for the absolute current in Amps. The experiment will end when the absolute current falls down to this value.
- void [setMinAbsoluteCurrent](#) (double minCurrent)
set the minimum value for the absolute current in Amps.
- double [getMinCurrent](#) () const
get the minimum value set for the absolute current in Amps. The experiment will end when the absolute current falls down to this value.
- void [setMinCurrent](#) (double minCurrent)
set the minimum value for the absolute current in Amps.
- double [getMaxCapacity](#) () const
get the value set for the maximum capacity / cumulative charge.
- void [setMaxCapacity](#) (double maxCapacity)

- set the value for the maximum capacity / cumulative charge in Coulomb.*

 - double `getMindIdt` () const

get the value set for the minimum current rate of change with respect to time (minimum di/dt).

- void `setMindIdt` (double mindIdt)

set the minimum value for the current rate of change with respect to time (minimum di/dt).

- bool `isAutoRange` () const

tells whether the current range is set to auto-select or not.

- void `setAutoRange` ()

set to auto-select the current range.

- double `getApproxMaxCurrent` () const

get the value set for the expected maximum current.

- void `setApproxMaxCurrent` (double approxMaxCurrent)

set maximum current expected, for manual current range selection.

- int `getVoltageRange` () const

get the value set for the voltage range.

- void `setVoltageRange` (int idx)

manually set the voltage control range.

17.4.1 Constructor & Destructor Documentation

17.4.1.1 AisConstantPotElement()

```
AisConstantPotElement::AisConstantPotElement (
    double voltage,
    double samplingInterval,
    double duration) [explicit]
```

Parameters

<i>voltage</i>	the value set for the voltage/potential in volts.
<i>samplingInterval</i>	the data sampling interval value in seconds.
<i>duration</i>	the maximum duration for the experiment in seconds.

17.4.2 Member Function Documentation

17.4.2.1 getApproxMaxCurrent()

```
double AisConstantPotElement::getApproxMaxCurrent () const
```

Returns

the value set for the expected maximum current in Amps.

Note

if nothing was manually set, the device will auto-select the current range and the return value will be positive infinity.

17.4.2.2 getCategory()

```
QStringList AisConstantPotElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Potentiostatic Control", "Basic Experiments")

17.4.2.3 getMaxAbsoluteCurrent()

```
double AisConstantPotElement::getMaxAbsoluteCurrent () const
```

Returns

the maximum absolute current value in Amps.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity.

17.4.2.4 getMaxCapacity()

```
double AisConstantPotElement::getMaxCapacity () const
```

Returns

the value set for the maximum capacity in Coulomb.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity.

17.4.2.5 getMaxCurrent()

```
double AisConstantPotElement::getMaxCurrent () const
```

Returns

the maximum current value in Amps.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity.

Attention

Deprecation Warning: This function may be modified or changed in a future version. Use [getMaxAbsoluteCurrent](#) instead.

17.4.2.6 getMaxDuration()

```
double AisConstantPotElement::getMaxDuration () const
```

Returns

the maximum duration for the experiment in seconds.

17.4.2.7 getMinAbsoluteCurrent()

```
double AisConstantPotElement::getMinAbsoluteCurrent () const
```

Returns

the minimum absolute current value in Amps.

Note

this is an optional parameter. If no value has been set, the default value is zero.

17.4.2.8 getMinCurrent()

```
double AisConstantPotElement::getMinCurrent () const
```

Returns

the minimum absolute current value in Amps.

Note

this is an optional parameter. If no value has been set, the default value is zero.

Attention

Deprecation Warning: This function may be modified or changed in a future version. Use `getMinAbsoluteCurrent` instead.

17.4.2.9 getMindIdt()

```
double AisConstantPotElement::getMindIdt () const
```

Returns

the value set for the minimum current rate of change with respect to time (minimum di/dt).

Note

this is an optional parameter. If no value has been set, the default value is zero.

17.4.2.10 getName()

```
QString AisConstantPotElement::getName () const [override]
```

Returns

The name of the element: "Constant Potential, Advanced".

17.4.2.11 getPotential()

```
double AisConstantPotElement::getPotential () const
```

Returns

the value set for the potential in volts.

17.4.2.12 getSamplingInterval()

```
double AisConstantPotElement::getSamplingInterval () const
```

Returns

the data sampling interval value in seconds.

17.4.2.13 getVoltageRange()

```
int AisConstantPotElement::getVoltageRange () const
```

Returns

the index set for the voltage range. (see [AisInstrumentHandler::getManualModeVoltageRangeList\(\)](#))

17.4.2.14 isAutoRange()

```
bool AisConstantPotElement::isAutoRange () const
```

Returns

true if the current range is set to auto-select and false if a range has been selected.

17.4.2.15 isVoltageVsOCP()

```
bool AisConstantPotElement::isVoltageVsOCP () const
```

Returns

true if the specified voltage is set against the open-circuit voltage and false if it is set against the reference terminal.

See also

setVsOcp

17.4.2.16 setApproxMaxCurrent()

```
void AisConstantPotElement::setApproxMaxCurrent (
    double approxMaxCurrent)
```

This is an **optional** parameter. If nothing is set, the device will auto-select the current range.

Parameters

<i>approxMaxCurrent</i>	the value for the maximum current expected in Amps.
-------------------------	---

17.4.2.17 setAutoRange()

```
void AisConstantPotElement::setAutoRange ()
```

This option is set by default. There is no need to call this function to auto-select if the range was not manually set.

17.4.2.18 setMaxAbsoluteCurrent()

```
void AisConstantPotElement::setMaxAbsoluteCurrent (
    double maxCurrent)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit current value. If a maximum absolute current is set, the experiment will continue to run as long as the absolute measured current is below that value.

Parameters

<i>maxCurrent</i>	the maximum absolute current value in Amps.
-------------------	---

17.4.2.19 setMaxCapacity()

```
void AisConstantPotElement::setMaxCapacity (
    double maxCapacity)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit cumulative charge value. If a maximum capacity is set, the experiment will continue to run as long as the cumulative charge is below that value.

Parameters

<i>maxCapacity</i>	the value to set for the cell maximum capacity.
--------------------	---

17.4.2.20 setMaxCurrent()

```
void AisConstantPotElement::setMaxCurrent (
    double maxCurrent)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit current value. If a maximum current is set, the experiment will continue to run as long as the measured current is below that value.

Parameters

<i>maxCurrent</i>	the maximum current value in Amps.
-------------------	------------------------------------

Attention

Deprecation Warning: This function may be modified or changed in a future version. Use [setMaxAbsoluteCurrent](#) instead.

17.4.2.21 setMaxDuration()

```
void AisConstantPotElement::setMaxDuration (
    double maxDuration)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an duration. If a maximum duration is set, the experiment will continue to run as long as the passed time is less than that value.

Parameters

<i>maxDuration</i>	the maximum duration for the experiment in seconds.
--------------------	---

17.4.2.22 setMinAbsoluteCurrent()

```
void AisConstantPotElement::setMinAbsoluteCurrent (
    double minCurrent)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an lower-limit current value. If a minimum absolute current is set, the experiment will continue to run as long as the measured absolute current is above that value.

Parameters

<i>minCurrent</i>	the value to set for the minimum absolute current.
-------------------	--

17.4.2.23 setMinCurrent()

```
void AisConstantPotElement::setMinCurrent (
    double minCurrent)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an lower-limit current value. If a minimum absolute current is set, the experiment will continue to run as long as the measured absolute current is above that value.

Parameters

<i>minCurrent</i>	the value to set for the minimum absolute current.
-------------------	--

Attention

Deprecation Warning: This function may be modified or changed in a future version. Use [setMinAbsoluteCurrent](#) instead.

17.4.2.24 setMindIdt()

```
void AisConstantPotElement::setMindIdt (
    double mindIdt)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an lower-limit rate of change value. If a minimum value is set, the experiment will continue to run as long as the rage of change is above that value.

Parameters

<i>mindIdt</i>	the minimum value for the current rate of change with respect to time (minimum di/dt).
----------------	--

17.4.2.25 setPotential()

```
void AisConstantPotElement::setPotential (
    double potential)
```

Parameters

<i>potential</i>	the value to set for the potential in volts.
------------------	--

17.4.2.26 setSamplingInterval()

```
void AisConstantPotElement::setSamplingInterval (
    double samplingInterval)
```

Parameters

<i>samplingInterval</i>	the data sampling interval value in seconds.
-------------------------	--

17.4.2.27 setVoltageRange()

```
void AisConstantPotElement::setVoltageRange (
    int idx)
```

This is an **optional** parameter. If this function is not called, the device will auto-select the voltage range by default.

Parameters

<i>idx</i>	the corresponding voltage range index (see AisInstrumentHandler::getManualModeVoltageRangeList())
------------	--

17.4.2.28 setVoltageVsOCP()

```
void AisConstantPotElement::setVoltageVsOCP (
    bool vsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<code>vsOCP</code>	true to set the specified voltage to reference the open-circuit voltage and false to set against the reference terminal.
--------------------	--

The documentation for this class was generated from the following file:

- `AisConstantPotElement.h`

17.5 AisConstantPowerElement Class Reference

This experiment simulates a constant power, charge or discharge".

```
#include <AisConstantPowerElement.h>
```

```
#include <AisConstantPowerElement.h>
```

Public Member Functions

- [AisConstantPowerElement](#) (double power, double duration, double samplingInterval)
the constant power element constructor
- [AisConstantPowerElement](#) (bool [isCharge](#), double power, double duration, double samplingInterval)
the constant power element constructor that supports the isCharge parameter
- **AisConstantPowerElement** (const [AisConstantPowerElement](#) &)
copy constructor for the [AisConstantPowerElement](#) object.
- [AisConstantPowerElement](#) & **operator=** (const [AisConstantPowerElement](#) &)
overload equal to operator for the [AisConstantPowerElement](#) object.
- QString [getName](#) () const override
get the name of the element.
- QStringList [getCategory](#) () const override
get a list of applicable categories of the element.
- bool [isCharge](#) () const
tells whether the experiment is set to simulate charge or discharge.
- void [setCharge](#) (bool [isCharge](#))
set whether the experiment is to simulate charge or discharge.
- double [getPower](#) () const
get the value set for the power.
- void [setPower](#) (double power)
set the value for the power.
- double [getSamplingInterval](#) () const
get how frequently we are sampling the data.
- void [setSamplingInterval](#) (double samplingInterval)
set how frequently we are sampling the data.
- double [getMaxVoltage](#) () const
get the value set for the maximum voltage. The experiment will end when it reaches this value.
- void [setMaxVoltage](#) (double maxVoltage)
set a maximum voltage to stop the experiment.
- bool [isMaximumVoltageVsOCP](#) () const

- tells whether the specified maximum voltage is set against the open-circuit voltage or the reference terminal.*
- void [setMaximumVoltageVsOCP](#) (bool vsOCP)
set whether to reference the specified maximum voltage against the open-circuit voltage or the reference terminal.
- double [getMinVoltage](#) () const
get the minimum value set for the voltage in volts. The experiment will end when it reaches down this value.
- void [setMinVoltage](#) (double minVoltage)
set a minimum value for the voltage. The experiment will end when it reaches down this value.
- bool [isMinimumVoltageVsOCP](#) () const
tells whether the specified minimum voltage is set against the open-circuit voltage or the reference terminal.
- void [setMinimumVoltageVsOCP](#) (bool vsOCP)
set whether to reference the specified minimum voltage against the open-circuit voltage or the reference terminal.
- double [getMaxCurrent](#) () const
get the value set maximum for the current in amps.
- void [setMaxCurrent](#) (double maxCurrent)
set a maximum current to stop the experiment.
- double [getMinCurrent](#) () const
get the value set minimum for the current in amps.
- void [setMinCurrent](#) (double maxCurrent)
set a minimum current to stop the experiment.
- double [getMaxDuration](#) () const
get the maximum duration set for the experiment. The experiment will end when the duration of the experiment reaches this value.
- void [setMaxDuration](#) (double maxDuration)
set the maximum duration for the experiment.
- double [getMaxCapacity](#) () const
get the value set for the maximum capacity / cumulative charge.
- void [setMaxCapacity](#) (double maxCapacity)
set the value for the maximum capacity / cumulative charge in Coulomb.

17.5.1 Constructor & Destructor Documentation

17.5.1.1 AisConstantPowerElement() [1/2]

```
AisConstantPowerElement::AisConstantPowerElement (
    double power,
    double duration,
    double samplingInterval) [explicit]
```

Parameters

<i>power</i>	the value set for the power in watts.
<i>duration</i>	the maximum duration for the experiment in seconds.
<i>samplingInterval</i>	the data sampling interval value in seconds.

17.5.1.2 AisConstantPowerElement() [2/2]

```
AisConstantPowerElement::AisConstantPowerElement (
    bool isCharge,
    double power,
    double duration,
    double samplingInterval) [explicit]
```

Parameters

<i>isCharge</i>	true to set the experiment simulate charge and false to simulate discharge.
<i>power</i>	the value set for the power in watts.
<i>duration</i>	the maximum duration for the experiment in seconds.
<i>samplingInterval</i>	the data sampling interval value in seconds.

Attention

Deprecation Warning: the `isCharge` parameter will be deprecated in a future version. Using the alternative constructor is highly recommended to avoid compilation errors in a future version. If this constructor is used, the sign of the power will be ignored and `isCharge` state will be used to determine it instead.

17.5.2 Member Function Documentation

17.5.2.1 getCategory()

```
QStringList AisConstantPowerElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Energy Storage", "Charge/Discharge").

17.5.2.2 getMaxCapacity()

```
double AisConstantPowerElement::getMaxCapacity () const
```

Returns

the value set for the maximum capacity in Coulomb.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity.

17.5.2.3 getMaxCurrent()

```
double AisConstantPowerElement::getMaxCurrent () const
```

Returns

the value set for the maximum current in amps.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity.

17.5.2.4 getMaxDuration()

```
double AisConstantPowerElement::getMaxDuration () const
```

Returns

the maximum duration for the experiment in seconds.

17.5.2.5 getMaxVoltage()

```
double AisConstantPowerElement::getMaxVoltage () const
```

Returns

the value set for the maximum voltage.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity

17.5.2.6 getMinCurrent()

```
double AisConstantPowerElement::getMinCurrent () const
```

Returns

the value set for the minimum current in amps.

Note

this is an optional parameter. If no value has been set, the default value is 0.

17.5.2.7 getMinVoltage()

```
double AisConstantPowerElement::getMinVoltage () const
```

Returns

the minimum value set for the voltage in volts.

Note

this is an optional parameter. If no value has been set, the default value is negative infinity

17.5.2.8 getName()

```
QString AisConstantPowerElement::getName () const [override]
```

Returns

The name of the element: "Constant Power Charge/Discharge".

17.5.2.9 getPower()

```
double AisConstantPowerElement::getPower () const
```

Returns

the value set for the power in watts.

17.5.2.10 getSamplingInterval()

```
double AisConstantPowerElement::getSamplingInterval () const
```

Returns

the data sampling interval value in seconds.

17.5.2.11 isCharge()

```
bool AisConstantPowerElement::isCharge () const
```

Returns

true if the experiment is set to simulate charge and false if it is set to simulate discharge.

17.5.2.12 isMaximumVoltageVsOCP()

```
bool AisConstantPowerElement::isMaximumVoltageVsOCP () const
```

Returns

true if the specified maximum voltage is set against the open-circuit voltage and false if it is set against the reference terminal.

See also

setVsOcp

17.5.2.13 isMinimumVoltageVsOCP()

```
bool AisConstantPowerElement::isMinimumVoltageVsOCP () const
```

Returns

true if the specified voltage is set against the open-circuit minimum voltage and false if it is set against the reference terminal.

See also

setVsOcp

17.5.2.14 setCharge()

```
void AisConstantPowerElement::setCharge (
    bool isCharge)
```

Parameters

<i>isCharge</i>	if the given argument is true, the experiment will simulate charge and discharge if given false.
-----------------	--

Attention

Deprecation Warning: setCharge will be deprecated in a future version. Avoid using this function, and instead set the power to a positive or negative value. If [AisConstantPowerElement\(bool, double, double, double\)](#) is used, you must use this function to set the charge/discharge state.

17.5.2.15 setMaxCapacity()

```
void AisConstantPowerElement::setMaxCapacity (
    double maxCapacity)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit cumulative charge value. If a maximum capacity is set, the experiment will continue to run as long as the cumulative charge is below that value.

Parameters

<i>maxCapacity</i>	the value to set for the cell maximum capacity.
--------------------	---

17.5.2.16 setMaxCurrent()

```
void AisConstantPowerElement::setMaxCurrent (
    double maxCurrent)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit Current value. If a maximum current is set, the experiment will continue to run as long as the measured current is above that value.

Parameters

<i>maxCurrent</i>	the maximum current value in amps at which the experiment will stop.
-------------------	--

17.5.2.17 setMaxDuration()

```
void AisConstantPowerElement::setMaxDuration (
    double maxDuration)
```

The experiment will continue to run as long as the passed time is less than that the set duration value.

Parameters

<i>maxDuration</i>	the maximum duration for the experiment in seconds.
--------------------	---

17.5.2.18 setMaximumVoltageVsOCP()

```
void AisConstantPowerElement::setMaximumVoltageVsOCP (
    bool vsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>vsOCP</i>	true to set the specified maximum voltage to reference the open-circuit voltage and false to set against the reference terminal.
--------------	--

17.5.2.19 setMaxVoltage()

```
void AisConstantPowerElement::setMaxVoltage (
    double maxVoltage)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit voltage value. If a maximum voltage is set, the experiment will continue to run as long as the measured voltage is below that value.

Parameters

<i>maxVoltage</i>	the maximum voltage value in volts at which the experiment will stop.
-------------------	---

17.5.2.20 setMinCurrent()

```
void AisConstantPowerElement::setMinCurrent (
    double maxCurrent)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an lower-limit Current value. If a minimum current is set, the experiment will continue to run as long as the measured current is below that value.

Parameters

<i>maxCurrent</i>	the minimum current value in amps at which the experiment will stop.
-------------------	--

17.5.2.21 setMinimumVoltageVsOCP()

```
void AisConstantPowerElement::setMinimumVoltageVsOCP (
    bool vsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>vsOCP</i>	true to set the specified minimum voltage to reference the open-circuit voltage and false to set against the reference terminal.
--------------	--

17.5.2.22 setMinVoltage()

```
void AisConstantPowerElement::setMinVoltage (
    double minVoltage)
```

Parameters

<i>minVoltage</i>	the value for the voltage in volts.
-------------------	-------------------------------------

Note

this is an optional parameter. If no value has been set, the default value is negative infinity.

17.5.2.23 setPower()

```
void AisConstantPowerElement::setPower (
    double power)
```

Parameters

<i>power</i>	the value set for the power in watts.
--------------	---------------------------------------

17.5.2.24 setSamplingInterval()

```
void AisConstantPowerElement::setSamplingInterval (
    double samplingInterval)
```

Parameters

<code>samplingInterval</code>	the data sampling interval value in seconds.
-------------------------------	--

The documentation for this class was generated from the following file:

- `AisConstantPowerElement.h`

17.6 AisConstantResistanceElement Class Reference

This element/experiment simulates a constant resistance load.

```
#include <AisConstantResistanceElement.h>
```

```
#include <AisConstantResistanceElement.h>
```

Public Member Functions

- [AisConstantResistanceElement](#) (double resistance, double duration, double samplingInterval)
the constant resistance element constructor.
- **AisConstantResistanceElement** (const [AisConstantResistanceElement](#) &)
copy constructor for the [AisConstantResistanceElement](#) object.
- [AisConstantResistanceElement](#) & **operator=** (const [AisConstantResistanceElement](#) &)
overload equal to operator for the [AisConstantResistanceElement](#) object.
- QString [getName](#) () const override
get the name of the element.
- QStringList [getCategory](#) () const override
get a list of applicable categories of the element.
- double [getResistance](#) () const
get the value set for the resistance as a load.
- void [setResistance](#) (double resistance)
set the value for the resistance as a load
- double [getSamplingInterval](#) () const
get how frequently we are sampling the data.
- void [setSamplingInterval](#) (double samplingInterval)
set how frequently we are sampling the data.
- double [getMinVoltage](#) () const
get the value set minimum for the voltage in volts.
- void [setMinVoltage](#) (double minVoltage)
set a minimum voltage to stop the experiment.
- bool [isMinimumVoltageVsOCP](#) () const
tells whether the specified minimum voltage is set against the open-circuit voltage or the reference terminal.
- void [setMinimumVoltageVsOCP](#) (bool vsOCP)
set whether to reference the specified minimum voltage against the open-circuit voltage or the reference terminal.
- double [getMaxVoltage](#) () const
get the value set maximum for the voltage in volts.
- void [setMaxVoltage](#) (double maxVoltage)

- set a maximum voltage to stop the experiment.*

 - bool `isMaximumVoltageVsOCP` () const
 - tells whether the specified maximum voltage is set against the open-circuit voltage or the reference terminal.*
 - void `setMaximumVoltageVsOCP` (bool vsOCP)
 - set whether to reference the specified maximum voltage against the open-circuit voltage or the reference terminal.*
 - double `getMaxCurrent` () const
 - get the value set maximum for the current in amps.*
 - void `setMaxCurrent` (double maxCurrent)
 - set a maximum current to stop the experiment.*
 - double `getMinCurrent` () const
 - get the value set minimum for the current in amps.*
 - void `setMinCurrent` (double maxCurrent)
 - set a minimum current to stop the experiment.*
 - double `getMaxDuration` () const
 - get the maximum duration set for the experiment. The experiment will end when the duration of the experiment reaches this value.*
 - void `setMaxDuration` (double maxDuration)
 - set the maximum duration for the experiment.*
 - double `getMaxCapacity` () const
 - get the value set for the maximum capacity / cumulative charge.*
 - void `setMaxCapacity` (double maxCapacity)
 - set the value for the maximum capacity / cumulative charge in Coulomb.*

17.6.1 Constructor & Destructor Documentation

17.6.1.1 AisConstantResistanceElement()

```
AisConstantResistanceElement::AisConstantResistanceElement (
    double resistance,
    double duration,
    double samplingInterval) [explicit]
```

Parameters

<i>resistance</i>	the value in ohm of the load resistance
<i>duration</i>	the maximum duration for the experiment in seconds.
<i>samplingInterval</i>	the data sampling interval value in seconds.

17.6.2 Member Function Documentation

17.6.2.1 getCategory()

```
QStringList AisConstantResistanceElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Energy Storage", "Charge/Discharge").

17.6.2.2 getMaxCapacity()

```
double AisConstantResistanceElement::getMaxCapacity () const
```

Returns

the value set for the maximum capacity in Coulomb.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity.

17.6.2.3 getMaxCurrent()

```
double AisConstantResistanceElement::getMaxCurrent () const
```

Returns

the value set for the maximum current in amps.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity.

17.6.2.4 getMaxDuration()

```
double AisConstantResistanceElement::getMaxDuration () const
```

Returns

the maximum duration for the experiment in seconds.

17.6.2.5 getMaxVoltage()

```
double AisConstantResistanceElement::getMaxVoltage () const
```

Returns

the value set for the maximum voltage in volts.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity

17.6.2.6 getMinCurrent()

```
double AisConstantResistanceElement::getMinCurrent () const
```

Returns

the value set for the minimum current in amps.

Note

this is an optional parameter. If no value has been set, the default value is 0.

17.6.2.7 getMinVoltage()

```
double AisConstantResistanceElement::getMinVoltage () const
```

Returns

the value set for the minimum voltage in volts.

Note

this is an optional parameter. If no value has been set, the default value is negative infinity

17.6.2.8 getName()

```
QString AisConstantResistanceElement::getName () const [override]
```

Returns

The name of the element: "Constant Resistance".

17.6.2.9 getResistance()

```
double AisConstantResistanceElement::getResistance () const
```

Returns

the value in ohm of the load resistance.

17.6.2.10 getSamplingInterval()

```
double AisConstantResistanceElement::getSamplingInterval () const
```

Returns

the data sampling interval value in seconds.

17.6.2.11 isMaximumVoltageVsOCP()

```
bool AisConstantResistanceElement::isMaximumVoltageVsOCP () const
```

Returns

true if the specified maximum voltage is set against the open-circuit voltage and false if it is set against the reference terminal.

See also

setVsOcp

17.6.2.12 isMinimumVoltageVsOCP()

```
bool AisConstantResistanceElement::isMinimumVoltageVsOCP () const
```

Returns

true if the specified voltage is set against the open-circuit minimum voltage and false if it is set against the reference terminal.

See also

setVsOcp

17.6.2.13 setMaxCapacity()

```
void AisConstantResistanceElement::setMaxCapacity (
    double maxCapacity)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit cumulative charge value. If a maximum capacity is set, the experiment will continue to run as long as the cumulative charge is below that value.

Parameters

<i>maxCapacity</i>	the value to set for the cell maximum capacity.
--------------------	---

17.6.2.14 setMaxCurrent()

```
void AisConstantResistanceElement::setMaxCurrent (
    double maxCurrent)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit Current value. If a maximum current is set, the experiment will continue to run as long as the measured current is above that value.

Parameters

<i>maxCurrent</i>	the maximum current value in amps at which the experiment will stop.
-------------------	--

17.6.2.15 setMaxDuration()

```
void AisConstantResistanceElement::setMaxDuration (
    double maxDuration)
```

The experiment will continue to run as long as the passed time is less than that the set duration value.

Parameters

<i>maxDuration</i>	the maximum duration for the experiment in seconds.
--------------------	---

17.6.2.16 setMaximumVoltageVsOCP()

```
void AisConstantResistanceElement::setMaximumVoltageVsOCP (
    bool vsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>vsOCP</i>	true to set the specified maximum voltage to reference the open-circuit voltage and false to set against the reference terminal.
--------------	--

17.6.2.17 setMaxVoltage()

```
void AisConstantResistanceElement::setMaxVoltage (
    double maxVoltage)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit voltage value. If a maximum voltage is set, the experiment will continue to run as long as the measured voltage is above that value.

Parameters

<i>maxVoltage</i>	the maximum voltage value in volts at which the experiment will stop.
-------------------	---

17.6.2.18 setMinCurrent()

```
void AisConstantResistanceElement::setMinCurrent (
    double maxCurrent)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an lower-limit Current value. If a minimum current is set, the experiment will continue to run as long as the measured current is below that value.

Parameters

<i>maxCurrent</i>	the minimum current value in amps at which the experiment will stop.
-------------------	--

17.6.2.19 setMinimumVoltageVsOCP()

```
void AisConstantResistanceElement::setMinimumVoltageVsOCP (
    bool vsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>vsOCP</i>	true to set the specified minimum voltage to reference the open-circuit voltage and false to set against the reference terminal.
--------------	--

17.6.2.20 setMinVoltage()

```
void AisConstantResistanceElement::setMinVoltage (
    double minVoltage)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an lower-limit voltage value. If a maximum voltage is set, the experiment will continue to run as long as the measured voltage is above that value.

Parameters

<i>minVoltage</i>	the minimum voltage value in volts at which the experiment will stop.
-------------------	---

17.6.2.21 setResistance()

```
void AisConstantResistanceElement::setResistance (
    double resistance)
```

Parameters

<i>resistance</i>	the value in ohm of the load resistance.
-------------------	--

17.6.2.22 setSamplingInterval()

```
void AisConstantResistanceElement::setSamplingInterval (
    double samplingInterval)
```

Parameters

<i>samplingInterval</i>	the data sampling interval value in seconds.
-------------------------	--

The documentation for this class was generated from the following file:

- AisConstantResistanceElement.h

17.7 AisCumulativeCapacityData Struct Reference

A structure containing DC cumulative capacity data collected from the instrument.

```
#include <AisDataPoints.h>
```

```
#include <AisDataPoints.h>
```

Public Attributes

- double **absolute**
absolute cumulative capacity over the entire experiment.
- double **net**
net cumulative capacity over the entire experiment.

The documentation for this struct was generated from the following file:

- AisDataPoints.h

17.8 AisCyclicVoltammetryElement Class Reference

This experiment sweeps the potential of the working electrode back and forth between the **first voltage-limit** and the **second voltage-limit** at a constant **scan rate (dE/dt)** for a specified number of **cycles**.

```
#include <AisCyclicVoltammetryElement.h>
```

```
#include <AisCyclicVoltammetryElement.h>
```

Public Member Functions

- [AisCyclicVoltammetryElement](#) (double startVoltage, double firstVoltageLimit, double secondVoltageLimit, double endVoltage, double dEdt, double samplingInterval)
constructor of the cyclic voltammetry element.
- **AisCyclicVoltammetryElement** (const [AisCyclicVoltammetryElement](#) &)
copy constructor for the [AisCyclicVoltammetryElement](#) object.
- [AisCyclicVoltammetryElement](#) & **operator=** (const [AisCyclicVoltammetryElement](#) &)
overload equal to operator for the [AisCyclicVoltammetryElement](#) object.
- QString [getName](#) () const override
get the name of the element.
- QStringList [getCategory](#) () const override
get a list of applicable categories of the element.
- double [getQuietTime](#) () const
Gets the quiet time duration.
- void [setQuietTime](#) (double quietTime)
Sets the quiet time duration.
- double [getQuietTimeSamplingInterval](#) () const
gets the potential sampling interval.
- void [setQuietTimeSamplingInterval](#) (double quietTimeSamplingInterval)
Sets the quiet time sampling interval.
- double [getStartVoltage](#) () const
get the value set for the start voltage
- void [setStartVoltage](#) (double startVoltage)
set the value for the start voltage.
- bool [isStartVoltageVsOCP](#) () const
tells whether the start voltage is set with respect to the open circuit voltage or not.
- void [setStartVoltageVsOCP](#) (bool startVoltageVsOCP)
set whether to reference the start voltage against the open-circuit voltage or the reference terminal.
- double [getFirstVoltageLimit](#) () const
get the value set for the first voltage-limit.
- void [setFirstVoltageLimit](#) (double v1)
set the first voltage-limit
- bool [isFirstVoltageLimitVsOCP](#) () const
tells whether the first voltage-limit is set with respect to the open circuit voltage or not.
- void [setFirstVoltageLimitVsOCP](#) (bool firstVoltageLimitVsOCP)
set whether to reference the first voltage-limit against the open-circuit voltage or not.
- double [getSecondVoltageLimit](#) () const
get the value set for the second voltage-limit
- void [setSecondVoltageLimit](#) (double v2)
set the second voltage-limit
- bool [isSecondVoltageLimitVsOCP](#) () const
tells whether the second voltage-limit is set with respect to the open circuit voltage or not.
- void [setSecondVoltageLimitVsOCP](#) (bool secondVoltageLimitVsOCP)
set whether to reference the second voltage-limit against the open-circuit voltage or not.
- unsigned int [getNumberOfCycles](#) ()
get the value set for the number of cycles
- void [setNumberOfCycles](#) (unsigned int cycles)
set the number of cycles to oscillate between the first voltage-limit and the second voltage-limit.
- double [getEndVoltage](#) () const
get the value set for the ending potential value.

- void [setEndVoltage](#) (double endVoltage)
set the ending potential value.
- bool [isEndVoltageVsOCP](#) () const
tells whether the end voltage is set with respect to the open circuit voltage or the reference terminal.
- void [setEndVoltageVsOCP](#) (bool endVoltageVsOCP)
set whether to reference the end voltage against the open-circuit voltage or the reference terminal.
- double [getdEdt](#) () const
get the value set for the constant scan rate dE/dt .
- void [setdEdt](#) (double dEdt)
set the value for the constant scan rate dE/dt .
- double [getSamplingInterval](#) () const
get how frequently we are sampling the data.
- void [setSamplingInterval](#) (double samplInterval)
set how frequently we are sampling the data.
- bool [isAutoRange](#) () const
tells whether the current range is set to auto-select or not.
- void [setAutoRange](#) ()
set to auto-select the current range.
- double [getApproxMaxCurrent](#) () const
get the value set for the expected maximum current.
- void [setApproxMaxCurrent](#) (double approxMaxCurrent)
set maximum current expected, for manual current range selection.
- double [getAlphaFactor](#) () const
Get the value set for the alpha factor.
- void [setAlphaFactor](#) (double alphaFactor)
alpha factor controls the percentage of data sampled during a given interval. Data will be averaged over the last n% of the sampling interval.

17.8.1 Constructor & Destructor Documentation

17.8.1.1 AisCyclicVoltammetryElement()

```
AisCyclicVoltammetryElement::AisCyclicVoltammetryElement (
    double startVoltage,
    double firstVoltageLimit,
    double secondVoltageLimit,
    double endVoltage,
    double dEdt,
    double samplingInterval) [explicit]
```

Parameters

<i>startVoltage</i>	the value of the start voltage in volts
<i>firstVoltageLimit</i>	the value of the first voltage-limit in volts
<i>secondVoltageLimit</i>	the value of the second voltage-limit in volts
<i>endVoltage</i>	the value of the end voltage in volts
<i>dEdt</i>	the constant scan rate dE/dt in V/s.
<i>samplingInterval</i>	the data sampling interval value in seconds.

17.8.2 Member Function Documentation

17.8.2.1 getAlphaFactor()

```
double AisCyclicVoltammetryElement::getAlphaFactor () const
```

Returns

The value for the alpha factor is represented as a percent between 0 and 100.

Note

If nothing is set, this function will return a default value of 75.

17.8.2.2 getApproxMaxCurrent()

```
double AisCyclicVoltammetryElement::getApproxMaxCurrent () const
```

Returns

the value set for the expected maximum current in Amps.

Note

if nothing was manually set, the device will auto-select the current range and the return value will be positive infinity.

17.8.2.3 getCategory()

```
QStringList AisCyclicVoltammetryElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Potentiostatic Control", "Basic Experiments").

17.8.2.4 getdEdt()

```
double AisCyclicVoltammetryElement::getdEdt () const
```

Returns

the value set for the constant scan rate dE/dt in V/s.

17.8.2.5 getEndVoltage()

```
double AisCyclicVoltammetryElement::getEndVoltage () const
```

This is the value of the voltage at which the experiment will stop. After the last cycle, the experiment will do one last sweep towards this value.

Returns

the value set for the ending voltage in volts.

17.8.2.6 getFirstVoltageLimit()

```
double AisCyclicVoltammetryElement::getFirstVoltageLimit () const
```

After the starting voltage, the scan will go to the first voltage-limit. This could result in either upward scan first if the first voltage-limit is higher than the start voltage or downward scan first if the first voltage-limit is lower than the start voltage.

Returns

the first voltage-limit value in volts.

17.8.2.7 getName()

```
QString AisCyclicVoltammetryElement::getName () const [override]
```

Returns

The name of the element: "Cyclic Voltammetry".

17.8.2.8 getNumberOfCycles()

```
unsigned int AisCyclicVoltammetryElement::getNumberOfCycles ()
```

Returns

the number of cycles set.

17.8.2.9 getQuietTime()

```
double AisCyclicVoltammetryElement::getQuietTime () const
```

Returns

The quiet time duration in seconds.

17.8.2.10 getQuietTimeSamplingInterval()

```
double AisCyclicVoltammetryElement::getQuietTimeSamplingInterval () const
```

Returns

samplingInterval The quiet time sampling interval to set in seconds.

17.8.2.11 getSamplingInterval()

```
double AisCyclicVoltammetryElement::getSamplingInterval () const
```

Returns

the data sampling interval value in seconds.

17.8.2.12 getSecondVoltageLimit()

```
double AisCyclicVoltammetryElement::getSecondVoltageLimit () const
```

After starting from the start-voltage and reaching the first voltage-limit, the scan will go to the second voltage limit. The scan will continue to oscillate between the first and second voltage-limits according to the number of cycles.

Returns

the second voltage-limit value in volts.

17.8.2.13 getStartVoltage()

```
double AisCyclicVoltammetryElement::getStartVoltage () const
```

Returns

the value of the start voltage in volts

17.8.2.14 isAutoRange()

```
bool AisCyclicVoltammetryElement::isAutoRange () const
```

Returns

true if the current range is set to auto-select and false if a rage has been selected.

17.8.2.15 isEndVoltageVsOCP()

```
bool AisCyclicVoltammetryElement::isEndVoltageVsOCP () const
```

Returns

true if the end voltage is set with respect to the open-circuit voltage and false if set against the reference terminal.

Note

if no value was set, the default is false

17.8.2.16 isFirstVoltageLimitVsOCP()

```
bool AisCyclicVoltammetryElement::isFirstVoltageLimitVsOCP () const
```

Returns

true if the first voltage-limit is set with respect to the open-circuit voltage and false if not.

Note

if no value was set, the default is false.

17.8.2.17 isSecondVoltageLimitVsOCP()

```
bool AisCyclicVoltammetryElement::isSecondVoltageLimitVsOCP () const
```

Returns

true if the second voltage-limit is set with respect to the open-circuit voltage and false if not.

Note

if no value was set, the default is false.

17.8.2.18 isStartVoltageVsOCP()

```
bool AisCyclicVoltammetryElement::isStartVoltageVsOCP () const
```

Returns

true if the start voltage is set with respect to the open-circuit voltage and false if not.

17.8.2.19 setAlphaFactor()

```
void AisCyclicVoltammetryElement::setAlphaFactor (  
    double alphaFactor)
```

This is an **optional** parameter. If nothing is set, the device will use the default value of 75.

Parameters

<i>alphaFactor</i>	the value for the alphaFactor ranges from 0 to 100.
--------------------	---

17.8.2.20 setApproxMaxCurrent()

```
void AisCyclicVoltammetryElement::setApproxMaxCurrent (
    double approxMaxCurrent)
```

This is an **optional** parameter. If nothing is set, the device will auto-select the current range.

Parameters

<i>approxMaxCurrent</i>	the value for the maximum current expected in Amps.
-------------------------	---

17.8.2.21 setAutoRange()

```
void AisCyclicVoltammetryElement::setAutoRange ()
```

This option is set by default. There is no need to call this function to auto-select if the range was not manually set.

17.8.2.22 setdEdt()

```
void AisCyclicVoltammetryElement::setdEdt (
    double dEdt)
```

Parameters

<i>dEdt</i>	the value set for the constant scan rate dE/dt in V/s.
-------------	--

17.8.2.23 setEndVoltage()

```
void AisCyclicVoltammetryElement::setEndVoltage (
    double endVoltage)
```

This is the value of the voltage at which the experiment will stop. After the last cycle, the experiment will do one last sweep towards this value.

Parameters

<i>endVoltage</i>	the value to set for the ending potential in volts.
-------------------	---

17.8.2.24 setEndVoltageVsOCP()

```
void AisCyclicVoltammetryElement::setEndVoltageVsOCP (
    bool endVoltageVsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>endVoltageVsOCP</i>	true to set the end voltage to be referenced against the open-circuit voltage and false if set against the reference terminal.
------------------------	--

17.8.2.25 setFirstVoltageLimit()

```
void AisCyclicVoltammetryElement::setFirstVoltageLimit (
    double v1)
```

After the starting voltage, the scan will go to the first voltage-limit. This could result in either upward scan first if the first voltage-limit is higher than the start voltage or downward scan first if the first voltage-limit is lower than the start voltage.

Parameters

<i>v1</i>	first voltage-limit value in volts
-----------	------------------------------------

17.8.2.26 setFirstVoltageLimitVsOCP()

```
void AisCyclicVoltammetryElement::setFirstVoltageLimitVsOCP (
    bool firstVoltageLimitVsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>firstVoltageLimitVsOCP</i>	true to set the upper voltage to be referenced against the open-circuit voltage and false otherwise.
-------------------------------	--

17.8.2.27 setNumberOfCycles()

```
void AisCyclicVoltammetryElement::setNumberOfCycles (
    unsigned int cycles)
```

Parameters

<i>cycles</i>	the number of cycles to set
---------------	-----------------------------

17.8.2.28 setQuietTime()

```
void AisCyclicVoltammetryElement::setQuietTime (
    double quietTime)
```

Parameters

<i>quietTime</i>	The quiet time duration to set in seconds.
------------------	--

17.8.2.29 setQuietTimeSamplingInterval()

```
void AisCyclicVoltammetryElement::setQuietTimeSamplingInterval (
    double quietTimeSamplingInterval)
```

Parameters

<i>quietTimeSamplingInterval</i>	The quiet time sampling interval to set in seconds.
----------------------------------	---

17.8.2.30 setSamplingInterval()

```
void AisCyclicVoltammetryElement::setSamplingInterval (
    double sampInterval)
```

Parameters

<i>sampInterval</i>	the data sampling interval value in seconds.
---------------------	--

17.8.2.31 setSecondVoltageLimit()

```
void AisCyclicVoltammetryElement::setSecondVoltageLimit (
    double v2)
```

After starting from the start-voltage and reaching the first voltage-limit, the scan will go to the second voltage limit. The scan will continue to oscillate between the first and second voltage-limits according to the number of cycles.

Parameters

<i>v2</i>	the second voltage-limit value in volts
-----------	---

17.8.2.32 setSecondVoltageLimitVsOCP()

```
void AisCyclicVoltammetryElement::setSecondVoltageLimitVsOCP (
    bool secondVoltageLimitVsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>secondVoltageLimitVsOCP</i>	true to set the second voltage-limit to be referenced against the open-circuit voltage and false otherwise.
--------------------------------	---

17.8.2.33 setStartVoltage()

```
void AisCyclicVoltammetryElement::setStartVoltage (
    double startVoltage)
```

Parameters

<code>startVoltage</code>	the value of the start voltage in volts
---------------------------	---

17.8.2.34 setStartVoltageVsOCP()

```
void AisCyclicVoltammetryElement::setStartVoltageVsOCP (
    bool startVoltageVsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<code>startVoltageVsOCP</code>	true to if the start voltage is set to reference the open-circuit voltage and false if set against the reference terminal.
--------------------------------	--

The documentation for this class was generated from the following file:

- AisCyclicVoltammetryElement.h

17.9 AisDataManipulator Class Reference

This class offers advanced control over pulse data collection and manipulation. It provides methods to manipulate AIS primary data for all three pulse voltammetry experiments types, namely Differential Pulse Voltammetry (DPV), Square Wave Voltammetry (SWV), and Normal Pulse Voltammetry (NPV).

```
#include <AisDataManipulator.h>
```

```
#include <AisDataManipulator.h>
```

Public Member Functions

- **AisDataManipulator ()**
Default constructor for [AisDataManipulator](#) class.
- **AisDataManipulator (const [AisDiffPulseVoltammetryElement](#) &other)**
[AisDiffPulseVoltammetryElement](#) constructor for [AisDataManipulator](#) class.
- **AisDataManipulator (const [AisNormalPulseVoltammetryElement](#) &other)**
[AisNormalPulseVoltammetryElement](#) constructor for [AisDataManipulator](#) class.
- **AisDataManipulator (const [AisSquareWaveVoltammetryElement](#) &other)**
[AisSquareWavePulseVoltammetryElement](#) constructor for [AisDataManipulator](#) class.
- **[AisErrorCode](#) setPulseType ([AisPulseType](#) type, double pulseWidth, double pulsePeriod)**
Set pulse type with pulse width and pulse period.
- **[AisErrorCode](#) setPulseType ([AisPulseType](#) type, double frequency)**
Set pulse type with frequency.
- double **getPulseWidth () const**
Get the pulse width.
- double **getPulsePeriod () const**

- Get the pulse period.*

 - double [getFrequency](#) () const

Get the pulse frequency.
- bool [isPulseCompleted](#) () const

Check if the pulse is completed.
- double [getBaseCurrent](#) () const

Get the base current.
- double [getPulseCurrent](#) () const

Get the pulse current.
- double [getBaseVoltage](#) () const

Get the base voltage.
- double [getPulseVoltage](#) () const

Get the pulse voltage.
- void [loadPrimaryData](#) (const [AisDCData](#) &data)

Load primary data from [AisDCData](#) object.

17.9.1 Constructor & Destructor Documentation

17.9.1.1 AisDataManipulator() [1/3]

```
AisDataManipulator::AisDataManipulator (
    const AisDiffPulseVoltammetryElement & other)
```

Parameters

<i>other</i>	The AisDiffPulseVoltammetryElement tile to be executed
--------------	--

17.9.1.2 AisDataManipulator() [2/3]

```
AisDataManipulator::AisDataManipulator (
    const AisNormalPulseVoltammetryElement & other)
```

Parameters

<i>other</i>	the AisNormalPulseVoltammetryElement tile to be executed
--------------	--

17.9.1.3 AisDataManipulator() [3/3]

```
AisDataManipulator::AisDataManipulator (
    const AisSquareWaveVoltammetryElement & other)
```

Parameters

<i>other</i>	AisSquareWaveVoltammetryElement tile to be executed
--------------	---

17.9.2 Member Function Documentation

17.9.2.1 `getBaseCurrent()`

```
double AisDataManipulator::getBaseCurrent () const
```

Returns

The base current.

17.9.2.2 `getBaseVoltage()`

```
double AisDataManipulator::getBaseVoltage () const
```

Returns

The base voltage.

17.9.2.3 `getFrequency()`

```
double AisDataManipulator::getFrequency () const
```

Returns

The pulse frequency.

17.9.2.4 `getPulseCurrent()`

```
double AisDataManipulator::getPulseCurrent () const
```

Returns

The pulse current.

17.9.2.5 `getPulsePeriod()`

```
double AisDataManipulator::getPulsePeriod () const
```

Returns

The pulse period.

17.9.2.6 `getPulseVoltage()`

```
double AisDataManipulator::getPulseVoltage () const
```

Returns

The pulse voltage.

17.9.2.7 `getPulseWidth()`

```
double AisDataManipulator::getPulseWidth () const
```

Returns

The pulse width.

17.9.2.8 `isPulseCompleted()`

```
bool AisDataManipulator::isPulseCompleted () const
```

Returns

True if the pulse is completed, false otherwise.

17.9.2.9 `loadPrimaryData()`

```
void AisDataManipulator::loadPrimaryData (  
    const AisDCData & data)
```

Parameters

<i>data</i>	The AisDCData object containing primary data.
-------------	---

17.9.2.10 `setPulseType()` [1/2]

```
AisErrorCode AisDataManipulator::setPulseType (  
    AisPulseType type,  
    double frequency)
```

Note

This function is usefull only for SquarewavePulse.

Parameters

<i>type</i>	The type of pulse.
<i>frequency</i>	The frequency of the pulse.

Returns

[AisErrorCode::Success](#) if pulse setting was successful. If not successful, possible returned errors are:

- [AisErrorCode::FailedRequest](#)

17.9.2.11 setPulseType() [2/2]

```
AisErrorCode AisDataManipulator::setPulseType (
    AisPulseType type,
    double pulseWidth,
    double pulsePeriod)
```

Note

This function is usefull only for DifferentialPulse and NormalPulse.

Parameters

<i>type</i>	The type of pulse.
<i>pulseWidth</i>	The width of the pulse.
<i>pulsePeriod</i>	The period of the pulse.

Returns

[AisErrorCode::Success](#) if pulse setting was successful. If not successful, possible returned errors are:

- [AisErrorCode::FailedRequest](#)

The documentation for this class was generated from the following file:

- [AisDataManipulator.h](#)

17.10 AisDCCurrentSweepElement Class Reference

this experiment performs a DC current sweep from the **starting current** to the **ending current** which progresses linearly according to the **scan rate**.

```
#include <AisDCCurrentSweepElement.h>
```

```
#include <AisDCCurrentSweepElement.h>
```

Public Member Functions

- [AisDCCurrentSweepElement](#) (double startCurrent, double endCurrent, double scanRate, double samplingInterval)
the DC current sweep element.
- [AisDCCurrentSweepElement](#) (const [AisDCCurrentSweepElement](#) &)
copy constructor for the [AisDCCurrentSweepElement](#) object.
- [AisDCCurrentSweepElement](#) & **operator=** (const [AisDCCurrentSweepElement](#) &)
overload equal to operator for the [AisDCCurrentSweepElement](#) object.
- QString [getName](#) () const override
get the name of the element.
- QStringList [getCategory](#) () const override
get a list of applicable categories of the element.
- double [getQuietTime](#) () const
Gets the quiet time duration.
- void [setQuietTime](#) (double quietTime)
Sets the quiet time duration.
- double [getQuietTimeSamplingInterval](#) () const
gets the quiet time sampling interval.
- void [setQuietTimeSamplingInterval](#) (double quietTimeSamplingInterval)
Sets the quiet time sampling interval.
- double [getStartingCurrent](#) () const
get the value set for the starting current.
- void [setStartingCurrent](#) (double startingCurrent)
set the value for the starting current.
- double [getEndingCurrent](#) () const
get the value set for the ending current.
- void [setEndingCurrent](#) (double endingCurrent)
set the value for the ending current.
- double [getScanRate](#) () const
get the value set for the scan rate.
- void [setScanRate](#) (double scanRate)
set the value for the current scan rate.
- double [getSamplingInterval](#) () const
get how frequently we are sampling the data.
- void [setSamplingInterval](#) (double samplingInterval)
set how frequently we are sampling the data.
- double [getMaxVoltage](#) () const
get the value set for the maximum voltage. The experiment will end when it reaches this value.
- void [setMaxVoltage](#) (double maxVoltage)
set a maximum voltage to stop the experiment.
- double [getMinVoltage](#) () const
get the value set minimum for the voltage in volts.
- void [setMinVoltage](#) (double minVoltage)
set a minimum voltage to stop the experiment.
- double [getAlphaFactor](#) () const
Get the value set for the alpha factor.
- void [setAlphaFactor](#) (double alphaFactor)
alpha factor controls the percentage of data sampled during a given interval. Data will be averaged over the last n% of the sampling interval.

17.10.1 Constructor & Destructor Documentation

17.10.1.1 AisDCCurrentSweepElement()

```
AisDCCurrentSweepElement::AisDCCurrentSweepElement (
    double startCurrent,
    double endCurrent,
    double scanRate,
    double samplingInterval) [explicit]
```

Parameters

<i>startCurrent</i>	the value for the starting current in Amps.
<i>endCurrent</i>	the value for the ending current in Amps.
<i>scanRate</i>	the value for the current scan rate in A/s.
<i>samplingInterval</i>	how frequently we are sampling the data.

17.10.2 Member Function Documentation

17.10.2.1 getAlphaFactor()

```
double AisDCCurrentSweepElement::getAlphaFactor () const
```

Returns

The value for the alpha factor is represented as a percent between 0 and 100.

Note

If nothing is set, this function will return a default value of 75.

17.10.2.2 getCategory()

```
QStringList AisDCCurrentSweepElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Galvanostatic Control", "Basic Voltammetry").

17.10.2.3 getEndingCurrent()

```
double AisDCCurrentSweepElement::getEndingCurrent () const
```

Returns

the value for the ending current in Amps.

17.10.2.4 getMaxVoltage()

```
double AisDCCurrentSweepElement::getMaxVoltage () const
```

Returns

the value set for the maximum voltage.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity

17.10.2.5 getMinVoltage()

```
double AisDCCurrentSweepElement::getMinVoltage () const
```

Returns

the value set for the minimum voltage in volts.

Note

this is an optional parameter. If no value has been set, the default value is negative infinity

17.10.2.6 getName()

```
QString AisDCCurrentSweepElement::getName () const [override]
```

Returns

The name of the element: "DC Current Linear Sweep".

17.10.2.7 getQuietTime()

```
double AisDCCurrentSweepElement::getQuietTime () const
```

Returns

The quiet time duration in seconds.

17.10.2.8 getQuietTimeSamplingInterval()

```
double AisDCCurrentSweepElement::getQuietTimeSamplingInterval () const
```

Returns

samplingInterval The quiet time sampling interval to set in seconds.

17.10.2.9 getSamplingInterval()

```
double AisDCCurrentSweepElement::getSamplingInterval () const
```

Returns

the data sampling interval value in seconds.

17.10.2.10 getScanRate()

```
double AisDCCurrentSweepElement::getScanRate () const
```

Returns

the value set for the scan rate in A/s.

See also

[setScanRate](#)

17.10.2.11 getStartingCurrent()

```
double AisDCCurrentSweepElement::getStartingCurrent () const
```

Returns

the value set for the constant current in Amps.

17.10.2.12 setAlphaFactor()

```
void AisDCCurrentSweepElement::setAlphaFactor (  
    double alphaFactor)
```

This is an **optional** parameter. If nothing is set, the device will use the default value of 75.

Parameters

<i>alphaFactor</i>	the value for the alphaFactor ranges from 0 to 100.
--------------------	---

17.10.2.13 setEndingCurrent()

```
void AisDCCurrentSweepElement::setEndingCurrent (  
    double endingCurrent)
```

Parameters

<i>endingCurrent</i>	the value for the ending current in Amps
----------------------	--

17.10.2.14 setMaxVoltage()

```
void AisDCCurrentSweepElement::setMaxVoltage (
    double maxVoltage)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit voltage value. If a maximum voltage is set, the experiment will continue to run as long as the measured voltage is below that value.

Parameters

<i>maxVoltage</i>	the maximum voltage value in volts at which the experiment will stop.
-------------------	---

17.10.2.15 setMinVoltage()

```
void AisDCCurrentSweepElement::setMinVoltage (
    double minVoltage)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an lower-limit voltage value. If a maximum voltage is set, the experiment will continue to run as long as the measured voltage is above that value.

Parameters

<i>minVoltage</i>	the minimum voltage value in volts at which the experiment will stop.
-------------------	---

17.10.2.16 setQuietTime()

```
void AisDCCurrentSweepElement::setQuietTime (
    double quietTime)
```

Parameters

<i>quietTime</i>	The quiet time duration to set in seconds.
------------------	--

17.10.2.17 setQuietTimeSamplingInterval()

```
void AisDCCurrentSweepElement::setQuietTimeSamplingInterval (
    double quietTimeSamplingInterval)
```


Parameters

<i>quietTimeSamplingInterval</i>	The quiet time sampling interval to set in seconds.
----------------------------------	---

17.10.2.18 setSamplingInterval()

```
void AisDCCurrentSweepElement::setSamplingInterval (
    double samplingInterval)
```

Parameters

<i>samplingInterval</i>	the data sampling interval value in seconds.
-------------------------	--

17.10.2.19 setScanRate()

```
void AisDCCurrentSweepElement::setScanRate (
    double scanRate)
```

The scan rate represents the value of the discrete current step size in one second in the linear sweep.

Parameters

<i>scanRate</i>	the value to set for the scan rate.
-----------------	-------------------------------------

17.10.2.20 setStartingCurrent()

```
void AisDCCurrentSweepElement::setStartingCurrent (
    double startingCurrent)
```

Parameters

<i>startingCurrent</i>	the value to set for the starting current in Amps
------------------------	---

The documentation for this class was generated from the following file:

- AisDCCurrentSweepElement.h

17.11 AisDCData Struct Reference

A structure containing DC data collected from the instrument.

```
#include <AisDataPoints.h>
```

```
#include <AisDataPoints.h>
```

Public Attributes

- double **timestamp**
the time at which the DC data arrived.
- double **workingElectrodeVoltage**
the measured working electrode voltage in volts.
- double **counterElectrodeVoltage**
the measured counter electrode voltage in volts.
- double **current**
the measured electric current value in Amps
- double **temperature**
the measured temperature in Celsius.
- [AisCumulativeCapacityData](#) **cumulativeCapacity**
the cumulative capacity data

The documentation for this struct was generated from the following file:

- AisDataPoints.h

17.12 AisDCPotentialSweepElement Class Reference

this experiment performs a DC potential sweep from the **starting current** to the **ending current** which progresses linearly according to the **scan rate**.

```
#include <AisDCPotentialSweepElement.h>
```

```
#include <AisDCPotentialSweepElement.h>
```

Public Member Functions

- [AisDCPotentialSweepElement](#) (double startPotential, double endPotential, double scanRate, double samplingInterval)
the potential sweep element constructor.
- **AisDCPotentialSweepElement** (const [AisDCPotentialSweepElement](#) &)
copy constructor for the [AisDCPotentialSweepElement](#) object.
- [AisDCPotentialSweepElement](#) & **operator=** (const [AisDCPotentialSweepElement](#) &)
overload equal to operator for the [AisDCPotentialSweepElement](#) object.
- QString **getName** () const override
get the name of the element.
- QStringList **getCategory** () const override
get a list of applicable categories of the element.
- double **getQuietTime** () const
Gets the quiet time duration.
- void **setQuietTime** (double quietTime)
Sets the quiet time duration.
- double **getQuietTimeSamplingInterval** () const
gets the quiet time sampling interval.
- void **setQuietTimeSamplingInterval** (double quietTimeSamplingInterval)
Sets the quiet time sampling interval.

- double [getStartingPot](#) () const
get the value set for the starting potential.
- void [setStartingPot](#) (double startingPotential)
set the value for the starting potential.
- bool [isStartVoltageVsOCP](#) () const
tells whether the starting potential is set against the open-circuit voltage or the reference terminal.
- void [setStartVoltageVsOCP](#) (bool startVoltageVsOCP)
set whether to reference the starting potential against the open-circuit voltage or the reference terminal.
- double [getEndingPot](#) () const
get the value set for the ending potential value.
- void [setEndingPot](#) (double endingPotential)
set the ending potential value.
- bool [isEndVoltageVsOCP](#) () const
tells whether the end voltage is set with respect to the open circuit voltage or the reference terminal.
- void [setEndVoltageVsOCP](#) (bool endVoltageVsOCP)
set whether to reference the end voltage against the open-circuit voltage or the reference terminal.
- double [getScanRate](#) () const
get the value set for the voltage scan rate.
- void [setScanRate](#) (double scanRate)
set the value for the voltage scan rate.
- double [getSamplingInterval](#) () const
get how frequently we are sampling the data.
- void [setSamplingInterval](#) (double samplingInterval)
set how frequently we are sampling the data.
- bool [isAutoRange](#) () const
tells whether the current range is set to auto-select or not.
- void [setAutoRange](#) ()
set to auto-select the current range.
- double [getApproxMaxCurrent](#) () const
get the value set for the expected maximum current.
- void [setApproxMaxCurrent](#) (double approxMaxCurrent)
set maximum current expected, for manual current range selection.
- double [getMaxAbsoluteCurrent](#) () const
get the value set for the maximum Current. The experiment will end when it reaches this value.
- void [setMaxAbsoluteCurrent](#) (double maxCurrent)
set a maximum Current to stop the experiment.
- double [getMinAbsoluteCurrent](#) () const
get the value set minimum for the Current in amps.
- void [setMinAbsoluteCurrent](#) (double minCurrent)
set a minimum Current to stop the experiment.
- double [getAlphaFactor](#) () const
Get the value set for the alpha factor.
- void [setAlphaFactor](#) (double alphaFactor)
alpha factor controls the percentage of data sampled during a given interval. Data will be averaged over the last n% of the sampling interval.

17.12.1 Constructor & Destructor Documentation

17.12.1.1 AisDCPotentialSweepElement()

```
AisDCPotentialSweepElement::AisDCPotentialSweepElement (
    double startPotential,
    double endPotential,
    double scanRate,
    double samplingInterval) [explicit]
```

Parameters

<i>startPotential</i>	the value of the starting potential in volts
<i>endPotential</i>	the value of the ending potential in volts
<i>scanRate</i>	the voltage scan rate in V/s
<i>samplingInterval</i>	how frequently we are sampling the data.

17.12.2 Member Function Documentation

17.12.2.1 getAlphaFactor()

```
double AisDCPotentialSweepElement::getAlphaFactor () const
```

Returns

The value for the alpha factor is represented as a percent between 0 and 100.

Note

If nothing is set, this function will return a default value of 75.

17.12.2.2 getApproxMaxCurrent()

```
double AisDCPotentialSweepElement::getApproxMaxCurrent () const
```

Returns

the value set for the expected maximum current in Amps.

Note

if nothing was manually set, the device will auto-select the current range and the return value will be positive infinity.

17.12.2.3 getCategory()

```
QStringList AisDCPotentialSweepElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Potentiostatic Control", "Basic Experiments").

17.12.2.4 getEndingPot()

```
double AisDCPotentialSweepElement::getEndingPot () const
```

This is the value of the voltage at which the experiment will stop.

Returns

the value set for the ending voltage in volts.

17.12.2.5 getMaxAbsoluteCurrent()

```
double AisDCPotentialSweepElement::getMaxAbsoluteCurrent () const
```

Returns

the value set for the maximum Current.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity

17.12.2.6 getMinAbsoluteCurrent()

```
double AisDCPotentialSweepElement::getMinAbsoluteCurrent () const
```

Returns

the value set for the minimum Current in amps.

Note

this is an optional parameter. If no value has been set, the default value is negative infinity

17.12.2.7 getName()

```
QString AisDCPotentialSweepElement::getName () const [override]
```

Returns

The name of the element: "DC Potential Linear Sweep".

17.12.2.8 getQuietTime()

```
double AisDCPotentialSweepElement::getQuietTime () const
```

Returns

The quiet time duration in seconds.

17.12.2.9 getQuietTimeSamplingInterval()

```
double AisDCPotentialSweepElement::getQuietTimeSamplingInterval () const
```

Returns

samplingInterval The quiet time sampling interval to set in seconds.

17.12.2.10 getSamplingInterval()

```
double AisDCPotentialSweepElement::getSamplingInterval () const
```

Returns

the data sampling interval value in seconds.

17.12.2.11 getScanRate()

```
double AisDCPotentialSweepElement::getScanRate () const
```

Returns

the value set for the voltage scan rate in V/s

See also

[setScanRate](#)

17.12.2.12 getStartingPot()

```
double AisDCPotentialSweepElement::getStartingPot () const
```

Returns

the value of the starting potential in volts.

17.12.2.13 isAutoRange()

```
bool AisDCPotentialSweepElement::isAutoRange () const
```

Returns

true if the current range is set to auto-select and false if a range has been selected.

17.12.2.14 isEndVoltageVsOCP()

```
bool AisDCPotentialSweepElement::isEndVoltageVsOCP () const
```

Returns

true if the end voltage is set with respect to the open-circuit voltage and false if set against the reference terminal.

See also

[setEndVoltageVsOCP](#)

17.12.2.15 isStartVoltageVsOCP()

```
bool AisDCPotentialSweepElement::isStartVoltageVsOCP () const
```

Returns

true if the starting potential is set against the open-circuit voltage and false if it is set against the reference terminal.

See also

[setStartVoltageVsOCP](#)

17.12.2.16 setAlphaFactor()

```
void AisDCPotentialSweepElement::setAlphaFactor (
    double alphaFactor)
```

This is an **optional** parameter. If nothing is set, the device will use the default value of 75.

Parameters

<i>alphaFactor</i>	the value for the alphaFactor ranges from 0 to 100.
--------------------	---

17.12.2.17 setApproxMaxCurrent()

```
void AisDCPotentialSweepElement::setApproxMaxCurrent (
    double approxMaxCurrent)
```

This is an **optional** parameter. If nothing is set, the device will auto-select the current range.

Parameters

<i>approxMaxCurrent</i>	the value for the maximum current expected in Amps.
-------------------------	---

17.12.2.18 setAutoRange()

```
void AisDCPotentialSweepElement::setAutoRange ()
```

This option is set by default. There is no need to call this function to auto-select if the range was not manually set.

17.12.2.19 setEndingPot()

```
void AisDCPotentialSweepElement::setEndingPot (
    double endingPotential)
```

This is the value of the voltage at which the experiment will stop.

Parameters

<i>endingPotential</i>	the value to set for the ending potential in volts.
------------------------	---

17.12.2.20 setEndVoltageVsOCP()

```
void AisDCPotentialSweepElement::setEndVoltageVsOCP (
    bool endVoltageVsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>endVoltageVsOCP</i>	true to set the end voltage to be referenced against the open-circuit voltage and false if set against the reference terminal.
------------------------	--

Note

by default, this is set to false.

17.12.2.21 setMaxAbsoluteCurrent()

```
void AisDCPotentialSweepElement::setMaxAbsoluteCurrent (
    double maxCurrent)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit Current value. If a maximum Current is set, the experiment will continue to run as long as the measured Current is below that value with the hardware current limitation.

Parameters

<i>maxCurrent</i>	the maximum Current value in volts at which the experiment will stop.
-------------------	---

17.12.2.22 setMinAbsoluteCurrent()

```
void AisDCPotentialSweepElement::setMinAbsoluteCurrent (
    double minCurrent)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an lower-limit Current value. If a maximum Current is set, the experiment will continue to run as long as the measured voltage is above that value.

Parameters

<i>minCurrent</i>	the minimum Current value in volts at which the experiment will stop.
-------------------	---

17.12.2.23 setQuietTime()

```
void AisDCPotentialSweepElement::setQuietTime (
    double quietTime)
```

Parameters

<i>quietTime</i>	The quiet time duration to set in seconds.
------------------	--

17.12.2.24 setQuietTimeSamplingInterval()

```
void AisDCPotentialSweepElement::setQuietTimeSamplingInterval (
    double quietTimeSamplingInterval)
```

Parameters

<i>quietTimeSamplingInterval</i>	The quiet time sampling interval to set in seconds.
----------------------------------	---

17.12.2.25 setSamplingInterval()

```
void AisDCPotentialSweepElement::setSamplingInterval (
    double samplingInterval)
```

Parameters

<i>samplingInterval</i>	the data sampling interval value in seconds.
-------------------------	--

17.12.2.26 setScanRate()

```
void AisDCPotentialSweepElement::setScanRate (
    double scanRate)
```

The scan rate represents the value of the discrete voltage step size in one second in the linear sweep.

Parameters

<i>scanRate</i>	the value to set for the scan rate.
-----------------	-------------------------------------

17.12.2.27 setStartingPot()

```
void AisDCPotentialSweepElement::setStartingPot (
    double startingPotential)
```

Parameters

<i>startingPotential</i>	the value of the starting potential in volts
--------------------------	--

17.12.2.28 setStartVoltageVsOCP()

```
void AisDCPotentialSweepElement::setStartVoltageVsOCP (
    bool startVoltageVsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>startVoltageVsOCP</i>	true to if the starting potential is set to reference the open-circuit voltage and false if set against the reference terminal.
--------------------------	---

Note

by default, this is set to false.

The documentation for this class was generated from the following file:

- AisDCPotentialSweepElement.h

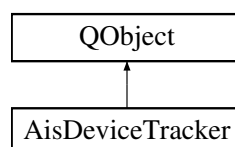
17.13 AisDeviceTracker Class Reference

This class is used track device connections to the computer. It also provides instrument handlers specific to each connected device which provide control of the relevant device.

```
#include <AisDeviceTracker.h>
```

```
#include <AisDeviceTracker.h>
```

Inheritance diagram for AisDeviceTracker:



Signals

- void [newDeviceConnected](#) (const QString &deviceName)
a signal to be emitted whenever a new connection has been successfully established with a device.
- void [deviceDisconnected](#) (const QString &deviceName)
a signal to be emitted whenever a device has been disconnected.
- void [firmwareUpdateNotification](#) (const QString &message)
a signal which is emitted regularly during a firmware update, providing information about the progress of the update.
- void [firmwareUpdateFinished](#) (const QString &deviceName, bool success)
Emitted when a device finishes the firmware update process.

Public Member Functions

- [AisErrorCode connectToDevice](#) (const QString &deviceName)
Establish a connection with a device connected on a USB port via its name.
- [AisErrorCode connectToDeviceOnComPort](#) (const QString &comPort)
establish a connection with a device connected on a USB port.
- const [AisInstrumentHandler](#) & [getInstrumentHandler](#) (const QString &deviceName) const
get an instrument handler to control a specific device.
- const std::list< QString > [getConnectedDevices](#) () const
get a list of all the connected devices.
- int [connectAllPluggedInDevices](#) ()
connect all devices physically plugged to the computer.
- [AisErrorCode updateFirmwareOnComPort](#) (const QString &comport) const
update firmware on connected device at USB port.
- int [updateFirmwareOnAllAvailableDevices](#) ()
request firmware update for all available devices.
- void [saveLogToFile](#) (bool save)
Allow to collect device error message in file for debugging purpose.
- void [setLogFilePath](#) (const QString &path)
This will help to change the log file directory.
- void [purgeStaleDevices](#) ()
Calling this function will remove all disconnected devices from memory. Any pointers and references to these devices will be invalidated.

Static Public Member Functions

- static [AisDeviceTracker](#) * [Instance](#) ()
get the instance of the device tracker.

17.13.1 Member Function Documentation

17.13.1.1 connectAllPluggedInDevices()

```
int AisDeviceTracker::connectAllPluggedInDevices ()
```

This will automatically detect all the communication ports that have devices plugged in and establish a connection with each.

Returns

the number of *new* devices that have successfully established a connection with the computer. If a device has already been connected before calling this function, it will not be counted in the return value.

Note

emits [newDeviceConnected\(\)](#) signal with the device name for each successful connection.

17.13.1.2 connectToDevice()

```
AisErrorCode AisDeviceTracker::connectToDevice (
    const QString & deviceName)
```

Move a device handler to the specified thread. All signals from that handler will then be emitted from that thread.

Parameters

<i>deviceName</i>	The name of the device to connect. (case sensitive)
-------------------	---

Return values

AisErrorCode::Success	if the device was connected, or is currently active. No AisDeviceTracker::newDeviceConnected signal will be emitted if the device is already connected.
AisErrorCode::DeviceNotFound	if the device was not found on any comport or it is open in another application

Note

This function will briefly open all available Squidstat serial connections and then close them. It may conflict with other applications attempting to initialize Squidstats.

emits [newDeviceConnected\(\)](#) signal with the device name if establishing the connection was successful.

17.13.1.3 connectToDeviceOnComPort()

```
AisErrorCode AisDeviceTracker::connectToDeviceOnComPort (
    const QString & comPort)
```

Parameters

<i>comPort</i>	the communication port to connect through.
----------------	--

Returns

[AisErrorCode::Success](#) if a connection was established with the device through the given communication port. If not successful, possible returned errors are:

- [AisErrorCode::Unknown](#)
- [AisErrorCode::FirmwareNotSupported](#)
- [AisErrorCode::ConnectionFailed](#)

Note

emits [newDeviceConnected\(\)](#) signal with the device name if establishing the connection was successful.

You need to specify the communication port specific to your computer. For example, on PC, you may find your port number through the 'device manager'. An example would be "COM15".

17.13.1.4 deviceDisconnected

```
void AisDeviceTracker::deviceDisconnected (
    const QString & deviceName) [signal]
```

Parameters

<i>deviceName</i>	the name of the newly disconnected device.
-------------------	--

Examples

[advancedControlFlow.cpp](#), [basicExperiment.cpp](#), and [pulseData.cpp](#).

17.13.1.5 firmwareUpdateFinished

```
void AisDeviceTracker::firmwareUpdateFinished (  
    const QString & deviceName,  
    bool success) [signal]
```

Parameters

<i>deviceName</i>	The name of the device which is finished updating.
<i>success</i>	Indicates whether the firmware updated successfully.

Note

If the device is in reset mode when the firmware is updated, the device name will be "Not Available" for all firmware update messaging.

17.13.1.6 firmwareUpdateNotification

```
void AisDeviceTracker::firmwareUpdateNotification (  
    const QString & message) [signal]
```

Parameters

<i>message</i>	a string containing the progress percentage message.
----------------	--

Note

If the device is in reset mode when the firmware is updated, the device name will be "Not Available" for all firmware update messaging.

Examples

[firmwareUpdate.cpp](#).

17.13.1.7 getConnectedDevices()

```
const std::list< QString > AisDeviceTracker::getConnectedDevices () const
```

Returns

a list of all the connected devices.

17.13.1.8 getInstrumentHandler()

```
const AisInstrumentHandler & AisDeviceTracker::getInstrumentHandler (  
    const QString & deviceName) const
```

Parameters

<i>deviceName</i>	the name of the connected device to get the instrument handler for (case sensitive).
-------------------	--

Returns

The instrument handler that controls the specified device. If the device has not been connected, it will return a stale device.

Note

You may get a list of the connected devices using [getConnectedDevices\(\)](#). Also, whenever a device has been connected by calling [connectToDeviceOnComPort\(\)](#), a signal is emitted with the device name.

Device handlers will become stale if a device disconnects, and you will need to acquire new instances on reconnection.

See also

[AisInstrumentHandler](#)

[AisDeviceTracker::connectToDeviceOnComPort\(\)](#)

[AisDeviceTracker::getConnectedDevices\(\)](#)

17.13.1.9 Instance()

```
static AisDeviceTracker * AisDeviceTracker::Instance () [static]
```

Returns

the static instance of the [AisDeviceTracker](#)

Examples

[advancedControlFlow.cpp](#), [advancedExperiment.cpp](#), [basicExperiment.cpp](#), [dataOutput.cpp](#), [firmwareUpdate.cpp](#), [linkedChannels.cpp](#), [manualExperiment.cpp](#), [nonblockingExperiment.cpp](#), and [pulseData.cpp](#).

17.13.1.10 newDeviceConnected

```
void AisDeviceTracker::newDeviceConnected (
    const QString & deviceName) [signal]
```

Parameters

<i>deviceName</i>	the name of the newly connected device.
-------------------	---

Note

this signal will be emitted for each newly connected device whenever either [connectToDeviceOnComPort\(\)](#) or [connectAllPluggedInDevices\(\)](#) successfully established connections.

Examples

[advancedControlFlow.cpp](#), [advancedExperiment.cpp](#), [basicExperiment.cpp](#), [dataOutput.cpp](#), [linkedChannels.cpp](#), [manualExperiment.cpp](#), [nonblockingExperiment.cpp](#), and [pulseData.cpp](#).

17.13.1.11 purgeStaleDevices()

```
void AisDeviceTracker::purgeStaleDevices ()
```

Note

This function should only be called after handling a disconnection event and removing all references to the disconnected device.

This is a temporary patch function to fix memory leaks over long periods of time with many disconnection events. In a future implementation this function will no longer be needed.

17.13.1.12 saveLogToFile()

```
void AisDeviceTracker::saveLogToFile (
    bool save)
```

Note

by default it will be true.

Parameters

<i>save</i>	When set to 'false,' it will not write logs to the file. When set to 'true,' it will begin writing device error logs to the file.
-------------	---

See also

[setLogFilePath](#)

17.13.1.13 setLogFilePath()

```
void AisDeviceTracker::setLogFilePath (
    const QString & path)
```

Note

by default it will be Document/Admiral Instrument/API

Parameters

<i>path</i>	Set the path value at which you want to save the log file.
-------------	--

Note

If you set 'false' for 'saveLogToFile,' it will not generate the log file. It is recommended to set it to 'true' or leave the permission as the default setting.

See also

[saveLogToFile](#)

17.13.1.14 `updateFirmwareOnAllAvailableDevices()`

```
int AisDeviceTracker::updateFirmwareOnAllAvailableDevices ()
```

This will automatically detect devices not currently in use and update firmware if necessary.

Returns

the number of devices that have successfully requested for firmware update. If a device has already been updated firmware before calling this function, it will not be counted in the return value. If any error is generated while requesting firmware update, it will not be counted in the return value.

Note

emits [firmwareUpdateNotification\(\)](#) signal will provide notification regarding firmware update of all devices. You can update firmware when you reset the device physically through reset button.

See also

[updateFirmwareOnComPort](#)

17.13.1.15 `updateFirmwareOnComPort()`

```
AisErrorCode AisDeviceTracker::updateFirmwareOnComPort (
    const QString & comport) const
```

Parameters

<i>comport</i>	the communication port to connect through.
----------------	--

Returns

[AisErrorCode::Success](#) if firmware update successfully initiated through the given communication port. If not successful, possible returned errors are:

- [AisErrorCode::FirmwareUptodate](#)
- [AisErrorCode::ConnectionFailed](#)

Note

emits [firmwareUpdateNotification\(\)](#) signal to provide firmware update progress.

You need to specify the communication port specific to your computer. For example, on PC, you may find your port number through the 'device manager'. An example would be "COM15".

The documentation for this class was generated from the following file:

- `AisDeviceTracker.h`

17.14 AisDiffPulseVoltammetryElement Class Reference

In this experiment, the working electrode holds at a **starting potential** during the **quiet time**. Then it applies a train of pulses superimposed on a staircase waveform, with a uniform **potential step** size. The potential continues to step until the **final potential** is reached.

```
#include <AisDiffPulseVoltammetryElement.h>
```

```
#include <AisDiffPulseVoltammetryElement.h>
```

Public Member Functions

- [AisDiffPulseVoltammetryElement](#) (double startVoltage, double endVoltage, double voltageStep, double pulseHeight, double pulseWidth, double pulsePeriod, double approxMaxCurrent)
the differential pulse element constructor.
- [AisDiffPulseVoltammetryElement](#) (double startVoltage, double endVoltage, double voltageStep, double pulseHeight, double pulseWidth, double pulsePeriod)
the differential pulse element constructor.
- [AisDiffPulseVoltammetryElement](#) (const [AisDiffPulseVoltammetryElement](#) &)
copy constructor for the [AisDiffPulseVoltammetryElement](#) object.
- [AisDiffPulseVoltammetryElement](#) & **operator=** (const [AisDiffPulseVoltammetryElement](#) &)
overload equal to operator for the [AisDiffPulseVoltammetryElement](#) object.
- QString [getName](#) () const override
get the name of the element.
- QStringList [getCategory](#) () const override
get a list of applicable categories of the element.
- double [getQuietTime](#) () const
Gets the quiet time duration.
- void [setQuietTime](#) (double quietTime)
Sets the quiet time duration.
- double [getQuietTimeSamplingInterval](#) () const
gets the quiet time sampling interval.
- void [setQuietTimeSamplingInterval](#) (double quietTimeSamplingInterval)
Sets the quiet time sampling interval.
- double [getStartVoltage](#) () const
get the value set for the start voltage.
- void [setStartVoltage](#) (double startVoltage)
set the value for the start voltage.
- bool [isStartVoltageVsOCP](#) () const
tells whether the starting potential is set against the open-circuit voltage or the reference terminal.
- void [setStartVoltageVsOCP](#) (bool startVoltageVsOCP)
set whether to reference the starting potential against the open-circuit voltage or the reference terminal.
- double [getEndVoltage](#) () const
get the value set for the ending potential value.
- void [setEndVoltage](#) (double endVoltage)
set the ending potential value.
- bool [isEndVoltageVsOCP](#) () const
tells whether the end voltage is set with respect to the open circuit voltage or the reference terminal.
- void [setEndVoltageVsOCP](#) (bool endVoltageVsOCP)
set whether to reference the end voltage against the open-circuit voltage or the reference terminal.
- double [getVStep](#) () const

- get the value set for the potential step.*
- void [setVStep](#) (double vStep)
set the value for the potential step.
- double [getPulseHeight](#) () const
get the value set for the pulse height.
- void [setPulseHeight](#) (double pulseHeight)
set the value for the pulse height.
- double [getPulseWidth](#) () const
get the value set for the pulse width.
- void [setPulseWidth](#) (double pulseWidth)
set the value for the pulse width.
- double [getPulsePeriod](#) () const
get the value set for the pulse period.
- void [setPulsePeriod](#) (double pulsePeriod)
set the value for the pulse period.
- bool [isAutoRange](#) () const
tells whether the current range is set to auto-select or not.
- void [setAutoRange](#) ()
set to auto-select the current range.
- double [getApproxMaxCurrent](#) () const
get the value set for the expected maximum current.
- void [setApproxMaxCurrent](#) (double approxMaxCurrent)
set maximum current expected, for manual current range selection.
- double [getAlphaFactor](#) () const
Get the value set for the alpha factor.
- void [setAlphaFactor](#) (double alphaFactor)
alpha factor controls the percentage of data sampled during a given interval. Data will be averaged over the last n% of the sampling interval.

17.14.1 Constructor & Destructor Documentation

17.14.1.1 AisDiffPulseVoltammetryElement() [1/2]

```
AisDiffPulseVoltammetryElement::AisDiffPulseVoltammetryElement (
    double startVoltage,
    double endVoltage,
    double voltageStep,
    double pulseHeight,
    double pulseWidth,
    double pulsePeriod,
    double approxMaxCurrent) [explicit]
```

Parameters

<i>startVoltage</i>	the value of the starting potential in volts
<i>endVoltage</i>	the value of the ending potential in volts
<i>voltageStep</i>	the value set for the voltage step in volts.
<i>pulseHeight</i>	the value for the pulse height in volts.
<i>pulseWidth</i>	the value for the pulse width in seconds.
<i>pulsePeriod</i>	the value for the pulse period in seconds.
<i>approxMaxCurrent</i>	the value for the approximate maximum current in amperes.

17.14.1.2 AisDiffPulseVoltammetryElement() [2/2]

```
AisDiffPulseVoltammetryElement::AisDiffPulseVoltammetryElement (
    double startVoltage,
    double endVoltage,
    double voltageStep,
    double pulseHeight,
    double pulseWidth,
    double pulsePeriod) [explicit]
```

Parameters

<i>startVoltage</i>	the value of the starting potential in volts
<i>endVoltage</i>	the value of the ending potential in volts
<i>voltageStep</i>	the value set for the voltage step in volts.
<i>pulseHeight</i>	the value for the pulse height in volts.
<i>pulseWidth</i>	the value for the pulse width in seconds.
<i>pulsePeriod</i>	the value for the pulse period in seconds.

17.14.2 Member Function Documentation

17.14.2.1 getAlphaFactor()

```
double AisDiffPulseVoltammetryElement::getAlphaFactor () const
```

Returns

The value for the alpha factor is represented as a percent between 0 and 100.

Note

If nothing is set, this function will return a default value of 75.

17.14.2.2 getApproxMaxCurrent()

```
double AisDiffPulseVoltammetryElement::getApproxMaxCurrent () const
```

Returns

the value set for the expected maximum current in Amps.

Note

if nothing was manually set, the device will auto-select the current range and the return value will be positive infinity.

17.14.2.3 getCategory()

```
QStringList AisDiffPulseVoltammetryElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Potentiostatic Control", "Basic Voltammetry", "Pulse Voltammetry").

17.14.2.4 getEndVoltage()

```
double AisDiffPulseVoltammetryElement::getEndVoltage () const
```

This is the value of the voltage at which the experiment will stop.

Returns

the value set for the ending voltage in volts.

17.14.2.5 getName()

```
QString AisDiffPulseVoltammetryElement::getName () const [override]
```

Returns

The name of the element: "Differential Pulse Potential Voltammetry".

17.14.2.6 getPulseHeight()

```
double AisDiffPulseVoltammetryElement::getPulseHeight () const
```

Returns

the value set for the pulse height in volts.

See also

[setPulseHeight](#)

17.14.2.7 getPulsePeriod()

```
double AisDiffPulseVoltammetryElement::getPulsePeriod () const
```

Returns

the value set for the pulse period in seconds.

See also

[setPulsePeriod](#)

17.14.2.8 getPulseWidth()

```
double AisDiffPulseVoltammetryElement::getPulseWidth () const
```

Returns

the value set for the pulse width in seconds.

See also

[setPulseWidth](#)

17.14.2.9 getQuietTime()

```
double AisDiffPulseVoltammetryElement::getQuietTime () const
```

Returns

The quiet time duration in seconds.

17.14.2.10 getQuietTimeSamplingInterval()

```
double AisDiffPulseVoltammetryElement::getQuietTimeSamplingInterval () const
```

Returns

samplingInterval The quiet time sampling interval to set in seconds.

17.14.2.11 getStartVoltage()

```
double AisDiffPulseVoltammetryElement::getStartVoltage () const
```

Returns

the value of the start voltage in volts.

17.14.2.12 getVStep()

```
double AisDiffPulseVoltammetryElement::getVStep () const
```

Returns

the value set for the potential step in volts.

See also

[setVStep](#)

17.14.2.13 isAutoRange()

```
bool AisDiffPulseVoltammetryElement::isAutoRange () const
```

Returns

true if the current range is set to auto-select and false if a range has been selected.

Deprecated This function is deprecated and no longer supports auto range for this element. Specify the current using [setApproxMaxCurrent\(\)](#). The device will determine the appropriate range based on the current value.

17.14.2.14 isEndVoltageVsOCP()

```
bool AisDiffPulseVoltammetryElement::isEndVoltageVsOCP () const
```

Returns

true if the end voltage is set with respect to the open-circuit voltage and false if set against the reference terminal.

See also

[setEndVoltageVsOCP](#)

17.14.2.15 isStartVoltageVsOCP()

```
bool AisDiffPulseVoltammetryElement::isStartVoltageVsOCP () const
```

Returns

true if the starting potential is set against the open-circuit voltage and false if it is set against the reference terminal.

See also

[setStartVoltageVsOCP](#)

17.14.2.16 setAlphaFactor()

```
void AisDiffPulseVoltammetryElement::setAlphaFactor (
    double alphaFactor)
```

This is an **optional** parameter. If nothing is set, the device will use the default value of 75.

Parameters

<i>alphaFactor</i>	the value for the alphaFactor ranges from 0 to 100.
--------------------	---

17.14.2.17 setApproxMaxCurrent()

```
void AisDiffPulseVoltammetryElement::setApproxMaxCurrent (
    double approxMaxCurrent)
```

This is an **optional** parameter. If nothing is set, the device will auto-select the current range.

Parameters

<i>approxMaxCurrent</i>	the value for the maximum current expected in Amps.
-------------------------	---

17.14.2.18 setAutoRange()

```
void AisDiffPulseVoltammetryElement::setAutoRange ()
```

This option is set by default. There is no need to call this function to auto-select if the range was not manually set.

Deprecated This function is deprecated. Use [setApproxMaxCurrent\(\)](#) to specify the current range instead.

17.14.2.19 setEndVoltage()

```
void AisDiffPulseVoltammetryElement::setEndVoltage (
    double endVoltage)
```

This is the value of the voltage at which the experiment will stop.

Parameters

<i>endVoltage</i>	the value to set for the ending voltage in volts.
-------------------	---

17.14.2.20 setEndVoltageVsOCP()

```
void AisDiffPulseVoltammetryElement::setEndVoltageVsOCP (
    bool endVoltageVsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>endVoltageVsOCP</i>	true to set the end voltage to be referenced against the open-circuit voltage and false if set against the reference terminal.
------------------------	--

Note

by default, this is set to false.

Examples

[pulseData.cpp](#).

17.14.2.21 setPulseHeight()

```
void AisDiffPulseVoltammetryElement::setPulseHeight (
    double pulseHeight)
```

For the first pulse, the pulse height is added to the starting potential. For the next pulse, the pulse height is added to the potential voltage and the potential step. In general, the pulse height is added to the potential step and the starting voltage of the last pulse.

Parameters

<i>pulseHeight</i>	the value to set for the pulse height in volts.
--------------------	---

17.14.2.22 setPulsePeriod()

```
void AisDiffPulseVoltammetryElement::setPulsePeriod (
    double pulsePeriod)
```

The pulse period is the time spent between the starts of two consecutive pulses.

Parameters

<i>pulsePeriod</i>	the value to set for the pulse period in seconds.
--------------------	---

17.14.2.23 setPulseWidth()

```
void AisDiffPulseVoltammetryElement::setPulseWidth (
    double pulseWidth)
```

The pulse width is the value in seconds for the time spent at the same voltage set for the pulse height.

Parameters

<i>pulseWidth</i>	the value to set for the pulse width in seconds.
-------------------	--

See also

[setPulseHeight](#)

17.14.2.24 setQuietTime()

```
void AisDiffPulseVoltammetryElement::setQuietTime (
    double quietTime)
```


Parameters

<i>quietTime</i>	The quiet time duration to set in seconds.
------------------	--

17.14.2.25 setQuietTimeSamplingInterval()

```
void AisDiffPulseVoltammetryElement::setQuietTimeSamplingInterval (
    double quietTimeSamplingInterval)
```

Parameters

<i>quietTimeSamplingInterval</i>	The quiet time sampling interval to set in seconds.
----------------------------------	---

17.14.2.26 setStartVoltage()

```
void AisDiffPulseVoltammetryElement::setStartVoltage (
    double startVoltage)
```

Parameters

<i>startVoltage</i>	the value of the start voltage in volts
---------------------	---

17.14.2.27 setStartVoltageVsOCP()

```
void AisDiffPulseVoltammetryElement::setStartVoltageVsOCP (
    bool startVoltageVsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>startVoltageVsOCP</i>	true to if the starting potential is set to reference the open-circuit voltage and false if set against the reference terminal.
--------------------------	---

Note

by default, this is set to false.

Examples

[pulseData.cpp](#).

17.14.2.28 setVStep()

```
void AisDiffPulseVoltammetryElement::setVStep (
    double vStep)
```

The potential step is the difference between the starting potential of two consecutive pulses.

Parameters

<code>vStep</code>	the value to set for the potential step in volts.
--------------------	---

Note

Regardless of `vStep`'s sign, the device will determine the step direction based on the start and end voltage.

The documentation for this class was generated from the following file:

- `AisDiffPulseVoltammetryElement.h`

17.15 AisEISGalvanostaticElement Class Reference

This experiment records the complex impedance of the experimental cell in galvanostatic mode, starting from the **start frequency** and sweeping through towards the **end frequency**, with a fixed number of frequency **steps per decade**.

```
#include <AisEISGalvanostaticElement.h>
```

```
#include <AisEISGalvanostaticElement.h>
```

Public Member Functions

- [AisEISGalvanostaticElement](#) (double startFrequency, double endFrequency, double stepsPerDecade, double currentBias, double currentAmplitude)
the EIS galvanostatic element constructor.
- [AisEISGalvanostaticElement](#) (const [AisEISGalvanostaticElement](#) &)
copy constructor for the [AisEISGalvanostaticElement](#) object.
- [AisEISGalvanostaticElement](#) & **operator=** (const [AisEISGalvanostaticElement](#) &)
overload equal to operator for the [AisEISGalvanostaticElement](#) object.
- `QString getName ()` const override
get the name of the element.
- `QStringList getCategory ()` const override
get a list of applicable categories of the element.
- `double getQuietTime ()` const
Gets the quiet time duration.
- `void setQuietTime` (double quietTime)
Sets the quiet time duration.
- `double getQuietTimeSamplingInterval ()` const
gets the quiet time sampling interval.
- `void setQuietTimeSamplingInterval` (double quietTimeSamplingInterval)
Sets the quiet time sampling interval.
- `double getStartFreq ()` const
get the value set for the current starting frequency
- `void setStartFreq` (double startFreq)
set the value for the current starting frequency.
- `double getEndFreq ()` const

- the value set for the current ending frequency.*
- void [setEndFreq](#) (double endFreq)
set the value for the current end frequency.
- double [getStepsPerDecade](#) () const
get the value set for the current frequency steps per decade.
- void [setStepsPerDecade](#) (double stepsPerDecade)
set the number of the current frequency steps per decade.
- double [getBiasCurrent](#) () const
get the value set for the DC bias (DC offset).
- void [setBiasCurrent](#) (double biasCurrent)
set the value for the DC bias (DC offset).
- double [getAmplitude](#) () const
the value to set for the AC current amplitude.
- void [setAmplitude](#) (double amplitude)
set the value for the AC current amplitude.
- unsigned int [getMinimumCycles](#) () const
get the minimum number of periods of applied sinusoidal current to sample at each frequency.
- void [setMinimumCycles](#) (unsigned int numberOfCycle)
set the minimum number of periods of applied sinusoidal current to sample at each frequency.

17.15.1 Constructor & Destructor Documentation

17.15.1.1 AisEISGalvanostaticElement()

```
AisEISGalvanostaticElement::AisEISGalvanostaticElement (
    double startFrequency,
    double endFrequency,
    double stepsPerDecade,
    double currentBias,
    double currentAmplitude) [explicit]
```

Parameters

<i>startFrequency</i>	the value for the current starting frequency
<i>endFrequency</i>	the value for the current ending frequency
<i>stepsPerDecade</i>	the value for the current frequency steps per decade.
<i>currentBias</i>	the value for the DC bias (DC offset).
<i>currentAmplitude</i>	the AC current amplitude.

17.15.2 Member Function Documentation

17.15.2.1 getAmplitude()

```
double AisEISGalvanostaticElement::getAmplitude () const
```

Returns

the value set for the AC current amplitude in Amps.

17.15.2.2 `getBiasCurrent()`

```
double AisEISGalvanostaticElement::getBiasCurrent () const
```

Returns

the value set for the DC bias in Amps.

17.15.2.3 `getCategory()`

```
QStringList AisEISGalvanostaticElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Galvanostatic Control", "Impedance Methods", "Basic Experiments").

17.15.2.4 `getEndFreq()`

```
double AisEISGalvanostaticElement::getEndFreq () const
```

Returns

the value set for the current end frequency in Hz

17.15.2.5 `getMinimumCycles()`

```
unsigned int AisEISGalvanostaticElement::getMinimumCycles () const
```

Returns

get number of cycles to sample at each frequency.

17.15.2.6 `getName()`

```
QString AisEISGalvanostaticElement::getName () const [override]
```

Returns

The name of the element: "Galvanostatic EIS".

17.15.2.7 `getQuietTime()`

```
double AisEISGalvanostaticElement::getQuietTime () const
```

Returns

The quiet time duration in seconds.

17.15.2.8 getQuietTimeSamplingInterval()

```
double AisEISGalvanostaticElement::getQuietTimeSamplingInterval () const
```

Returns

samplingInterval The quiet time sampling interval to set in seconds.

17.15.2.9 getStartFreq()

```
double AisEISGalvanostaticElement::getStartFreq () const
```

Returns

the value set for the current start frequency in Hz?

17.15.2.10 getStepsPerDecade()

```
double AisEISGalvanostaticElement::getStepsPerDecade () const
```

Returns

the value set for the current frequency steps per decade. This is unit-less.

17.15.2.11 setAmplitude()

```
void AisEISGalvanostaticElement::setAmplitude (
    double amplitude)
```

Parameters

<i>amplitude</i>	the value to set for the AC current amplitude in Amps.
------------------	--

17.15.2.12 setBiasCurrent()

```
void AisEISGalvanostaticElement::setBiasCurrent (
    double biasCurrent)
```

Parameters

<i>biasCurrent</i>	the value to set for the DC bias in Amps.
--------------------	---

17.15.2.13 setEndFreq()

```
void AisEISGalvanostaticElement::setEndFreq (
    double endFreq)
```

Parameters

<i>endFreq</i>	the value to set for the current end frequency in Hz
----------------	--

17.15.2.14 setMinimumCycles()

```
void AisEISGalvanostaticElement::setMinimumCycles (  
    unsigned int numberOfCycle)
```

Parameters

<i>numberOfCycle</i>	number of cycles to sample at each frequency.
----------------------	---

17.15.2.15 setQuietTime()

```
void AisEISGalvanostaticElement::setQuietTime (  
    double quietTime)
```

Parameters

<i>quietTime</i>	The quiet time duration to set in seconds.
------------------	--

17.15.2.16 setQuietTimeSamplingInterval()

```
void AisEISGalvanostaticElement::setQuietTimeSamplingInterval (  
    double quietTimeSamplingInterval)
```

Parameters

<i>quietTimeSamplingInterval</i>	The quiet time sampling interval to set in seconds.
----------------------------------	---

17.15.2.17 setStartFreq()

```
void AisEISGalvanostaticElement::setStartFreq (  
    double startFreq)
```

Parameters

<i>startFreq</i>	the value to set the current starting frequency in Hz
------------------	---

17.15.2.18 setStepsPerDecade()

```
void AisEISGalvanostaticElement::setStepsPerDecade (  
    double stepsPerDecade)
```

Parameters

<code>stepsPerDecade</code>	the value to set for the number of steps per decade.
-----------------------------	--

The documentation for this class was generated from the following file:

- AisEISGalvanostaticElement.h

17.16 AisEISPotentiostaticElement Class Reference

This experiment records the complex impedance of the experimental cell in potentiostatic mode, starting from the **start frequency** and sweeping through towards the **end frequency**, with a fixed number of frequency **steps per decade**.

```
#include <AisEISPotentiostaticElement.h>
```

```
#include <AisEISPotentiostaticElement.h>
```

Public Member Functions

- [AisEISPotentiostaticElement](#) (double startFrequency, double endFrequency, double stepsPerDecade, double voltageBias, double voltageAmplitude)
the EIS potentiostatic element
- [AisEISPotentiostaticElement](#) (const [AisEISPotentiostaticElement](#) &)
copy constructor for the [AisEISPotentiostaticElement](#) object.
- [AisEISPotentiostaticElement](#) & **operator=** (const [AisEISPotentiostaticElement](#) &)
overload equal to operator for the [AisEISPotentiostaticElement](#) object.
- QString [getName](#) () const override
get the name of the element.
- QStringList [getCategory](#) () const override
get a list of applicable categories of the element.
- double [getQuietTime](#) () const
Gets the quiet time duration.
- void [setQuietTime](#) (double quietTime)
Sets the quiet time duration.
- double [getQuietTimeSamplingInterval](#) () const
gets the quiet time sampling interval.
- void [setQuietTimeSamplingInterval](#) (double quietTimeSamplingInterval)
Sets the quiet time sampling interval.
- double [getStartFreq](#) () const
get the value set for the voltage starting frequency
- void [setStartFreq](#) (double startFreq)
set the value for the voltage starting frequency.
- double [getEndFreq](#) () const
the value set for the voltage ending frequency.
- void [setEndFreq](#) (double endFreq)
set the value for the voltage end frequency.
- double [getStepsPerDecade](#) () const

- get the value set for the voltage frequency steps per decade.*
- void [setStepsPerDecade](#) (double stepsPerDecade)
set the number of the voltage frequency steps per decade.
- double [getBiasVoltage](#) () const
get the value set for the DC bias (DC offset).
- void [setBiasVoltage](#) (double biasVoltage)
set the value for the DC bias (DC offset).
- bool [isBiasVoltageVsOCP](#) () const
tells whether the DC-bias voltage is referenced against the open-circuit voltage or the reference cable.
- void [setBiasVoltageVsOCP](#) (bool biasVsOCP)
set whether to reference the DC-bias voltage against the open-circuit voltage or the reference terminal.
- double [getAmplitude](#) () const
the value to set for the AC voltage amplitude.
- void [setAmplitude](#) (double amplitude)
set the value for the AC voltage amplitude.
- unsigned int [getMinimumCycles](#) () const
get the minimum number of periods of applied sinusoidal voltage to sample at each frequency.
- void [setMinimumCycles](#) (unsigned int numberOfCycle)
set the minimum number of periods of applied sinusoidal voltage to sample at each frequency.

17.16.1 Constructor & Destructor Documentation

17.16.1.1 AisEISPotentiostaticElement()

```
AisEISPotentiostaticElement::AisEISPotentiostaticElement (
    double startFrequency,
    double endFrequency,
    double stepsPerDecade,
    double voltageBias,
    double voltageAmplitude) [explicit]
```

Parameters

<i>startFrequency</i>	the value for the voltage starting frequency
<i>endFrequency</i>	the value for the voltage ending frequency
<i>stepsPerDecade</i>	the value for the voltage frequency steps per decade.
<i>voltageBias</i>	the value for the DC bias (DC offset).
<i>voltageAmplitude</i>	the AC voltage amplitude.

17.16.2 Member Function Documentation

17.16.2.1 getAmplitude()

```
double AisEISPotentiostaticElement::getAmplitude () const
```

Returns

the value set for the AC voltage amplitude in volts.

17.16.2.2 getBiasVoltage()

```
double AisEISPotentiostaticElement::getBiasVoltage () const
```

Returns

the value set for the DC bias in volts.

17.16.2.3 getCategory()

```
QStringList AisEISPotentiostaticElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Potentiostatic Control", "Impedance Methods", "Basic Experiments").

17.16.2.4 getEndFreq()

```
double AisEISPotentiostaticElement::getEndFreq () const
```

Returns

the value set for the voltage end frequency in Hz

17.16.2.5 getMinimumCycles()

```
unsigned int AisEISPotentiostaticElement::getMinimumCycles () const
```

Returns

get number of cycles to sample at each frequency.

17.16.2.6 getName()

```
QString AisEISPotentiostaticElement::getName () const [override]
```

Returns

The name of the element: "Potentiostatic EIS".

17.16.2.7 getQuietTime()

```
double AisEISPotentiostaticElement::getQuietTime () const
```

Returns

The quiet time duration in seconds.

17.16.2.8 getQuietTimeSamplingInterval()

```
double AisEISPotentiostaticElement::getQuietTimeSamplingInterval () const
```

Returns

samplingInterval The quiet time sampling interval to set in seconds.

17.16.2.9 getStartFreq()

```
double AisEISPotentiostaticElement::getStartFreq () const
```

Returns

the value set for the start frequency in Hz

17.16.2.10 getStepsPerDecade()

```
double AisEISPotentiostaticElement::getStepsPerDecade () const
```

Returns

the value set for the frequency steps per decade. This is unit-less.

17.16.2.11 isBiasVoltageVsOCP()

```
bool AisEISPotentiostaticElement::isBiasVoltageVsOCP () const
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Returns

true if the DC-bias voltage is set against the open-circuit voltage and false if it is set against the reference terminal.

17.16.2.12 setAmplitude()

```
void AisEISPotentiostaticElement::setAmplitude (  
    double amplitude)
```

Parameters

<i>amplitude</i>	the value to set for the AC voltage amplitude in volts.
------------------	---

17.16.2.13 setBiasVoltage()

```
void AisEISPotentiostaticElement::setBiasVoltage (  
    double biasVoltage)
```

Parameters

<i>biasVoltage</i>	the value to set for the DC bias in volts.
--------------------	--

17.16.2.14 setBiasVoltageVsOCP()

```
void AisEISPotentiostaticElement::setBiasVoltageVsOCP (
    bool biasVsOCP)
```

Parameters

<i>biasVsOCP</i>	true to if the DC-bias voltage is set to reference the open-circuit voltage and false if set against the reference terminal.
------------------	--

17.16.2.15 setEndFreq()

```
void AisEISPotentiostaticElement::setEndFreq (
    double endFreq)
```

Parameters

<i>endFreq</i>	the value to set for the voltage end frequency in Hz
----------------	--

17.16.2.16 setMinimumCycles()

```
void AisEISPotentiostaticElement::setMinimumCycles (
    unsigned int numberOfCycle)
```

Parameters

<i>numberOfCycle</i>	number of cycles to sample at each frequency.
----------------------	---

17.16.2.17 setQuietTime()

```
void AisEISPotentiostaticElement::setQuietTime (
    double quietTime)
```

Parameters

<i>quietTime</i>	The quiet time duration to set in seconds.
------------------	--

17.16.2.18 setQuietTimeSamplingInterval()

```
void AisEISPotentiostaticElement::setQuietTimeSamplingInterval (
    double quietTimeSamplingInterval)
```

Parameters

<i>quietTimeSamplingInterval</i>	The quiet time sampling interval to set in seconds.
----------------------------------	---

17.16.2.19 setStartFreq()

```
void AisEISPotentiostaticElement::setStartFreq (  
    double startFreq)
```

Parameters

<i>startFreq</i>	the value to set the starting frequency Hz
------------------	--

17.16.2.20 setStepsPerDecade()

```
void AisEISPotentiostaticElement::setStepsPerDecade (  
    double stepsPerDecade)
```

Parameters

<i>stepsPerDecade</i>	the value to set for the number of steps per decade.
-----------------------	--

The documentation for this class was generated from the following file:

- AisEISPotentiostaticElement.h

17.17 AisErrorCode Class Reference

This class contains the possible error codes returned to the user when working with the API. Error codes can help diagnose issues such as invalid parameters, communication failures, or device malfunctions. By handling errors properly, you can ensure reliable operation of your experiments.

```
#include <AisErrorCode.h>
```

```
#include <AisErrorCode.h>
```

Public Types

- enum `ErrorCode` : `uint8_t` {
`Unknown` = 255 , `Success` = 0 , `ConnectionFailed` = 1 , `FirmwareNotSupported` = 2 ,
`FirmwareFileNotFound` = 3 , `FirmwareUptodate` = 4 , `InvalidChannel` = 10 , `BusyChannel` = 11 ,
`DeviceNotFound` = 13 , `FeatureNotSupported` = 14 , `ManualExperimentNotRunning` = 51 , `ExperimentNotUploaded`
= 52 ,
`ExperimentIsEmpty` = 53 , `InvalidParameters` = 54 , `ChannelNotBusy` = 55 , `ExperimentUploaded` = 56 ,
`DeviceCommunicationFailed` = 100 , `FailedToSetManualModeCurrentRange` = 101 , `FailedToSetManualModeConstantVoltage`
= 102 , `FailedToPauseExperiment` = 103 ,
`FailedToResumeExperiment` = 104 , `FailedToStopExperiment` = 105 , `FailedToUploadExperiment` = 106 ,
`ExperimentAlreadyPaused` = 107 ,
`ExperimentAlreadyRun` = 108 , `FailedToSetManualModeVoltageRange` = 109 , `FailedToSetManualModeConstantCurrent`
= 110 , `FailedToSetManualModeInOCP` = 111 ,
`FailedToSetManualModeSamplingInterval` = 112 , `FailedToSetIRComp` = 113 , `FailedToSetCompRange` =
114 , `FailedToSetChannelMaximumVoltage` = 115 ,
`FailedToSetChannelMinimumVoltage` = 116 , `FailedToSetChannelMaximumCurrent` = 117 , `FailedToSetChannelMinimumCurrent`
= 118 , `FailedToSetChannelMinimumTemperature` = 119 ,
`FailedRequest` = 254 }

The possible error codes that can be returned to the user.

Public Member Functions

- QString `message` () const
a function to get a message explaining the error.
- int `value` () const
a function to get the error code.

17.17.1 Member Enumeration Documentation

17.17.1.1 ErrorCode

```
enum AisErrorCode::ErrorCode : uint8_t
```

Enumerator

Unknown	indicates that the command failed for an unknown reason.
Success	indicates success.
ConnectionFailed	indicates failure connecting the plugged in device when calling AisDeviceTracker::connectToDeviceOnComPort .
FirmwareNotSupported	indicates failure connecting the plugged in device when calling AisDeviceTracker::connectToDeviceOnComPort because firmware update require.
InvalidChannel	indicates that the given channel number is not valid.
BusyChannel	indicates that the failure was due to the channel being busy.
DeviceNotFound	indicates that no device was detected to be connected.
FeatureNotSupported	indicates that the feature is not available on the device.
ManualExperimentNotRunning	indicates that the given command applies when there is a manual experiment running on the channel but there is none.
ExperimentNotUploaded	indicates that the given command applies when an experiment has already been uploaded to the channel but there is none.
ExperimentIsEmpty	indicates that the given experiment has no elements. It need to contain at least one.

Enumerator

InvalidParameters	indicates that a given parameter is invalid. For example, it is out of the allowed range.
ChannelNotBusy	indicates that the given command applies when there is an experiment running or paused on the channel but there is none.
ExperimentUploaded	indicates that the given command could not be completed because an experiment is already uploaded to the channel.
DeviceCommunicationFailed	indicates that there was failure in communication with the device.
FailedToSetManualModeCurrentRange	indicates failure to set manual mode current range due to a possible communication failure with the device.
FailedToSetManualModeConstantVoltage	indicates failure to set manual mode constant voltage due to a possible communication failure with the device
FailedToPauseExperiment	indicates that pausing the experiment failed because either there is no active experiment or due to a possible communication failure with the device.
FailedToResumeExperiment	indicates that resuming the experiment failed because either there is no paused experiment or due to a possible communication failure with the device.
FailedToStopExperiment	indicates that stopping the experiment failed because either there is no experiment running, the experiment is paused, or due to a possible communication failure with the device.
FailedToUploadExperiment	indicates failure to communicate with the device to upload the experiment.
ExperimentAlreadyPaused	indicates that pausing the experiment failed because the experiment is already paused.
ExperimentAlreadyRun	indicates that resuming the experiment failed because an experiment is already running.
FailedToSetManualModeConstantCurrent	indicates failure to set manual mode constant current due to a possible communication failure with the device.
FailedToSetManualModeInOCP	indicates failure of setting manual mode in open circuit mode for possible communication failure with the device.
FailedToSetManualModeSamplingInterval	indicates failure of setting manual mode sampling interval. possible communication failure with the device.
FailedToSetIRComp	indicates failure of setting IR Compensation. Possible communication failure with the device.
FailedToSetCompRange	indicates failure of setting Compensation Range. Possible communication failure with the device.
FailedToSetChannelMaximumVoltage	indicates failure of setting Channel Maximum Voltage. Possible communication failure with the device.
FailedToSetChannelMinimumVoltage	indicates failure of setting Channel Minimum Voltage. Possible communication failure with the device.
FailedToSetChannelMaximumCurrent	indicates failure of setting Channel Maximum Current. Possible communication failure with the device.
FailedToSetChannelMinimumCurrent	indicates failure of setting Channel Minimum Current. Possible communication failure with the device.
FailedToSetChannelMinimumTemperature	indicates failure of setting Channel Maximum Temperature. Possible communication failure with the device.

17.17.2 Member Function Documentation

17.17.2.1 message()

```
QString AisErrorCode::message () const
```

Returns

a message that explains the error.

17.17.2.2 value()

```
int AisErrorCode::value () const
```

Returns

the error code

The documentation for this class was generated from the following file:

- AisErrorCode.h

17.18 AisExperiment Class Reference

this class is used to create custom experiments. A custom experiment contains one or more elements. Once you create elements and set their parameters, you can add them to the container.

```
#include <AisExperiment.h>
```

```
#include <AisExperiment.h>
```

Public Member Functions

- **AisExperiment ()**
this is the default constructor for the custom experiment.
- **AisExperiment (const AisExperiment &exp)**
this is the copy constructor for the custom experiment.
- void **operator=** (const AisExperiment &exp)
the assignment operator for the custom experiment.
- QString **getExperimentName** () const
get the name of the custom experiment.
- QString **getDescription** () const
get a brief description of the custom experiment.
- QStringList **getCategory** () const
get the category for the custom experiment.
- void **setExperimentName** (QString name)
set a name for the custom experiment.
- void **setDescription** (QString description)
set a description for the experiment.
- bool **appendElement** (AisAbstractElement &element, unsigned int repeat=1)
Append an element to this experiment.
- bool **appendSubExperiment** (const AisExperiment &subExp, unsigned int repeat=1)
Append a sub experiment to this experiment.

Friends

- class **AisInstrumentHandler**

17.18.1 Constructor & Destructor Documentation

17.18.1.1 AisExperiment()

```
AisExperiment::AisExperiment (
    const AisExperiment & exp) [explicit]
```

Parameters

<i>exp</i>	the custom experiment to copy from.
------------	-------------------------------------

17.18.2 Member Function Documentation

17.18.2.1 appendElement()

```
bool AisExperiment::appendElement (
    AisAbstractElement & element,
    unsigned int repeat = 1)
```

Parameters

<i>element</i>	The experiment element to be appended to this experiment.
<i>repeat</i>	The number of times this element will be repeated. This is an optional parameter with a default value of 1. The minimum value is 1. If smaller the function <u>will not append</u> the node to the experiment. The maximum value is 65535. The function will clamp any value greater than this to 65535, and it <u>will append</u> the node to the experiment.

Returns

true if the element was appended to the experiment and **false** otherwise.

Note

Although an element is an experiment, in the context of custom experiments, it is referred to as an element to make a distinction between the two. In other contexts where such distinction is not needed, an element may still be referred to as an experiment.

17.18.2.2 appendSubExperiment()

```
bool AisExperiment::appendSubExperiment (
    const AisExperiment & subExp,
    unsigned int repeat = 1)
```


Parameters

<i>subExp</i>	A sub experiment to be appended to this experiment.
<i>repeat</i>	The number of times this sub experiment will be repeated. This is an optional parameter with a default value of 1. The minimum value is 1. If smaller, the function will not append the sub experiment to the experiment The maximum value is 65535. The function will clamp any value greater than this to 65535, and it <u>will</u> <u>append</u> the sub experiment to the experiment.

Returns

true if the sub experiment was appended to the experiment and **false** otherwise.

17.18.2.3 getCategory()

```
QStringList AisExperiment::getCategory () const
```

Returns

the category set for the custom experiment. If no category has been set, the default category returned is ("Custom").

17.18.2.4 getDescription()

```
QString AisExperiment::getDescription () const
```

Returns

the description set for the custom experiment. If no description has been set, the default description returned is "Not Defined".

17.18.2.5 getExperimentName()

```
QString AisExperiment::getExperimentName () const
```

Returns

the name set for the custom experiment. If no name has been set, the default name returned is "Custom Experiment"

17.18.2.6 operator=()

```
void AisExperiment::operator= (
    const AisExperiment & exp)
```

Parameters

<i>exp</i>	the custom experiment to copy from.
------------	-------------------------------------

17.18.2.7 setDescription()

```
void AisExperiment::setDescription (
    QString description)
```

Parameters

<i>description</i>	the description to be set for the custom experiment.
--------------------	--

17.18.2.8 setExperimentName()

```
void AisExperiment::setExperimentName (
    QString name)
```

Parameters

<i>name</i>	the name to be set for the custom experiment.
-------------	---

The documentation for this class was generated from the following file:

- AisExperiment.h

17.19 AisExperimentNode Struct Reference

A structure containing some information regarding the running element.

```
#include <AisDataPoints.h>
```

```
#include <AisDataPoints.h>
```

Public Attributes

- QString **stepName**
This is the name of the current element running.
- int **stepNumber**
this number is the order of the element within the custom experiment.
- int **substepNumber**
this number is the order of the step within the element.
- int **cycle**
this number is cycle within the element.

The documentation for this struct was generated from the following file:

- AisDataPoints.h

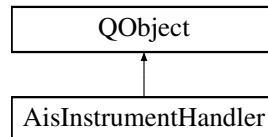
17.20 AisInstrumentHandler Class Reference

this class provides control of the device including starting, pausing, resuming and stopping an experiment on a channel as well as reading the data and other controls of the device.

```
#include <AisInstrumentHandler.h>
```

```
#include <AisInstrumentHandler.h>
```

Inheritance diagram for AisInstrumentHandler:



Signals

- void **deviceDisconnected** ()
a signal that is emitted if the device associated with this handler has been disconnected.
- void **groundFloatStateChanged** (bool grounded)
a signal that is emitted when the floating ground connection state has changed.
- void **experimentStarting** (uint8_t channel)
a signal that is emitted whenever an experiment begins
- void **experimentNewElementStarting** (uint8_t channel, const **AisExperimentNode** &stepInfo)
a signal that is emitted whenever a new elemental experiment has started.
- void **activeDCDataReady** (uint8_t channel, const **AisDCData** &DCData)
a signal that is emitted whenever new DC data for an active experiment are ready.
- void **idleDCDataReady** (uint8_t channel, const **AisDCData** &DCData)
a signal that is emitted whenever new DC data are ready when the device is in an idle state.
- void **recoveryDCDataReady** (uint8_t channel, const **AisDCData** &DCData)
a signal that is emitted whenever new DC recovery data are ready.
- void **activeACDataReady** (uint8_t channel, const **AisACData** &ACData)
a signal that is emitted whenever new AC data for an active experiment are ready.
- void **recoveryACDataReady** (uint8_t channel, const **AisACData** &ACData)
a signal that is emitted whenever new AC recovery data are ready.
- void **experimentStopped** (uint8_t channel, const QString &reason)
a signal that is emitted whenever an experiment was stopped manually or has completed.
- void **experimentPaused** (uint8_t channel)
a signal that is emitted whenever an experiment was paused.
- void **experimentResumed** (uint8_t channel)
a signal that is emitted whenever an experiment was resumed.
- void **recoverDataErased** (bool successful)
a signal that is emitted whenever data erase process is completed.
- void **deviceError** (uint8_t channel, const QString &error)
a signal that is emitted whenever device send any critical error.

Public Member Functions

- [AisErrorCode uploadExperimentToChannel](#) (uint8_t channel, std::shared_ptr< [AisExperiment](#) > experiment) const
upload an already created custom experiment to a specific channel on the device.
- [AisErrorCode uploadExperimentToChannel](#) (uint8_t channel, const [AisExperiment](#) &experiment) const
upload an already created custom experiment to a specific channel on the device.
- [AisErrorCode startUploadedExperiment](#) (uint8_t channel) const
start the previously uploaded experiment on the specific channel.
- [AisErrorCode startIdleSampling](#) (uint8_t channel) const
start idle sampling when an experiment is neither uploaded nor running on the specified channel.
- [AisErrorCode skipExperimentStep](#) (uint8_t channel) const
skip the current experiment step and proceed to the next.
- [AisErrorCode pauseExperiment](#) (uint8_t channel) const
pause a running experiment on the channel.
- [AisErrorCode resumeExperiment](#) (uint8_t channel) const
resume a paused experiment on the channel.
- [AisErrorCode stopExperiment](#) (uint8_t channel) const
stop a running or a paused experiment on the channel.
- double [getExperimentUTCStartTime](#) (uint8_t channel) const
get UTC time for the start of the experiment in seconds.
- [AisErrorCode setIRComp](#) (uint8_t channel, double uncompensatedResistance, double compensationLevel) const
set IR compensation.
- [AisErrorCode setCompRange](#) (uint8_t channel, const [AisCompRange](#) &compRange) const
set a compensation range with stability factor and bandwidth index.
- int8_t [setLinkedChannels](#) (std::vector< uint8_t > channels) const
connect several channels together in parallel mode.
- int8_t [setBipolarLinkedChannels](#) (std::vector< uint8_t > channels) const
connect two channels together in bipolar mode.
- bool [hasBipolarMode](#) (uint8_t channel) const
tells whether the given channel is bipolar mode
- std::vector< uint8_t > [getLinkedChannels](#) (uint8_t channel) const
get a list of channels linked to the given channel.
- [AisErrorCode setFanSpeedMaximum](#) () const
Set the fan speed to maximum when an experiment is active. Only available on some devices.
- [AisErrorCode setFanSpeedVariable](#) () const
Set the fan speed to always adjust automatically based on the internal temperature of the instrument. Only available on some devices.
- bool [isChannelBusy](#) (uint8_t channel) const
tells whether the given channel is busy or not.
- bool [isChannelPaused](#) (uint8_t channel) const
tells whether the given channel has a paused experiment or not.
- std::vector< uint8_t > [getFreeChannels](#) () const
get a list of the currently free channels.
- int [getNumberOfChannels](#) () const
get the number of all the channels on this device.
- [AisErrorCode eraseRecoverData](#) () const
delete the recover data from device.
- [AisErrorCode startManualExperiment](#) (uint8_t channel) const
start a manual experiment.

- [AisErrorCode setManualModeSamplingInterval](#) (uint8_t channel, double value) const
set an interval for sampling the data.
- [AisErrorCode setManualModeOCP](#) (uint8_t channel) const
set open-circuit potential mode.
- [AisErrorCode setManualModeConstantVoltage](#) (uint8_t channel, double value) const
set constant voltage for the manual experiment.
- [AisErrorCode setManualModeConstantVoltage](#) (uint8_t channel, double value, int currentRangeIndex) const
set constant voltage for the manual experiment and also set a manual current range.
- [AisErrorCode setManualModeCurrentRange](#) (uint8_t channel, int currentRangeIndex) const
set the current range for the manual experiment. Once a range is set, autoranging capability is turned off. That means that during potentiostatic control, the current range may range up if necessary, but it will not drop below the user-set range. During galvanostatic control, the lowest current range that contains the designated setpoint will be chosen, provided it is not lower than the user-set range.
- [AisErrorCode setManualModeCurrentAutorange](#) (uint8_t channel) const
enable current autoranging for the manual experiment.
- [AisErrorCode setManualModeVoltageRange](#) (uint8_t channel, int voltageRangeIndex) const
set the voltage range for the manual experiment. Once a range is set, autoranging capability is turned off. That means that during galvanostatic control, the voltage range may range up if necessary, but it will not drop below the user-set range. During potentiostatic control, the lowest voltage range that contains the designated setpoint will be chosen, provided it is not lower than the user-set range.
- [AisErrorCode setManualModeVoltageAutorange](#) (uint8_t channel) const
enable voltage autoranging for the manual experiment.
- [AisErrorCode setManualModeConstantCurrent](#) (uint8_t channel, double value) const
set constant current for the manual experiment.
- `std::vector< std::pair< double, double > >` [getManualModeCurrentRangeList](#) (uint8_t channel) const
get a list of the applicable current ranges to the given channel specific to your device.
- `std::vector< std::pair< double, double > >` [getManualModeVoltageRangeList](#) (uint8_t channel) const
get a list of the applicable voltage ranges to the given channel specific to your device.
- [AisErrorCode setChannelMaximumVoltage](#) (uint8_t channel, double Vmax) const
Sets the maximum allowable voltage for a channel. The experiment will stop if the measured voltage exceeds this limit.
- [AisErrorCode setChannelMinimumVoltage](#) (uint8_t channel, double Vmin) const
Sets the minimum allowable voltage for a channel. The experiment will stop if the measured voltage falls below this limit.
- [AisErrorCode setChannelMaximumCurrent](#) (uint8_t channel, double Imax) const
Sets the maximum allowable current for a channel. The experiment will stop if the measured current exceeds this limit.
- [AisErrorCode setChannelMinimumCurrent](#) (uint8_t channel, double Imin) const
Sets the minimum allowable current for a channel. The experiment will stop if the measured current falls below this limit.
- [AisErrorCode setChannelMaximumTemperature](#) (uint8_t channel, double MaxTemperature) const
Sets the maximum allowable temperature for a channel. The experiment will stop if the measured temperature exceeds this limit.
- [AisErrorCode resetChannelLimits](#) (uint8_t channel) const
Resets all limits for the specified channel.
- [AisErrorCode setLoadImpedance](#) (uint8_t channel, const float &loadImpedance)
Sets the expected load impedance for a specific channel.
- `float` [getLoadImpedance](#) (uint8_t channel)
Retrieves the currently configured load impedance for a specific channel.

17.20.1 Member Function Documentation

17.20.1.1 activeACDataReady

```
void AisInstrumentHandler::activeACDataReady (
    uint8_t channel,
    const AisACData & ACData) [signal]
```

Parameters

<i>channel</i>	the channel number from which the AC data arrived.
<i>ACData</i>	the AC data that just arrived.

Examples

[basicExperiment.cpp](#), [dataOutput.cpp](#), [nonblockingExperiment.cpp](#), and [pulseData.cpp](#).

17.20.1.2 activeDCDataReady

```
void AisInstrumentHandler::activeDCDataReady (
    uint8_t channel,
    const AisDCData & DCData) [signal]
```

Parameters

<i>channel</i>	the channel number from which the DC data arrived.
<i>DCData</i>	the DC data that just arrived.

Examples

[advancedExperiment.cpp](#), [basicExperiment.cpp](#), [dataOutput.cpp](#), [linkedChannels.cpp](#), [manualExperiment.cpp](#), [nonblockingExperiment.cpp](#), and [pulseData.cpp](#).

17.20.1.3 deviceError

```
void AisInstrumentHandler::deviceError (
    uint8_t channel,
    const QString & error) [signal]
```

Parameters

<i>channel</i>	the channel number at which error rise.
<i>error</i>	information about error message.

Note

stop experiment command will automatically send on channel.

Examples

[advancedExperiment.cpp](#), [basicExperiment.cpp](#), [linkedChannels.cpp](#), and [manualExperiment.cpp](#).

17.20.1.4 eraseRecoverData()

```
AisErrorCode AisInstrumentHandler::eraseRecoverData () const
```

Returns

[AisErrorCode::Success](#) if request is successfully send for delete the data. If not successful, possible returned errors are:

- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::DeviceCommunicationFailed](#)

17.20.1.5 experimentNewElementStarting

```
void AisInstrumentHandler::experimentNewElementStarting (  
    uint8_t channel,  
    const AisExperimentNode & stepInfo) [signal]
```

Parameters

<i>channel</i>	the channel number on which the experiment was started.
<i>stepInfo</i>	information regarding the current step.

See also

[AisExperimentNode](#)

Examples

[advancedExperiment.cpp](#), [basicExperiment.cpp](#), [dataOutput.cpp](#), [nonblockingExperiment.cpp](#), and [pulseData.cpp](#).

17.20.1.6 experimentPaused

```
void AisInstrumentHandler::experimentPaused (  
    uint8_t channel) [signal]
```

Parameters

<i>channel</i>	the channel on which the experiment was paused.
----------------	---

Examples

[pulseData.cpp](#).

17.20.1.7 experimentResumed

```
void AisInstrumentHandler::experimentResumed (  
    uint8_t channel) [signal]
```

Parameters

<i>channel</i>	the channel on which the experiment was resumed.
----------------	--

Examples

[pulseData.cpp](#).

17.20.1.8 experimentStarting

```
void AisInstrumentHandler::experimentStarting (
    uint8_t channel) [signal]
```

Parameters

<i>channel</i>	the channel number on which the experiment has begun
----------------	--

17.20.1.9 experimentStopped

```
void AisInstrumentHandler::experimentStopped (
    uint8_t channel,
    const QString & reason) [signal]
```

Parameters

<i>channel</i>	the channel on which the experiment has stopped.
<i>reason</i>	the reason why the experiment has stopped.

Examples

[advancedControlFlow.cpp](#), [advancedExperiment.cpp](#), [basicExperiment.cpp](#), [dataOutput.cpp](#), [manualExperiment.cpp](#), [nonblockingExperiment.cpp](#), and [pulseData.cpp](#).

17.20.1.10 getExperimentUTCStartTime()

```
double AisInstrumentHandler::getExperimentUTCStartTime (
    uint8_t channel) const
```

This will give the time in seconds between the origin of UTC time and the start of the experiment aka Unix Epoch.

Parameters

<i>channel</i>	the channel for which to get the start time of the experiment.
----------------	--

Returns

the Unix Epoch up to the start of the experiment in seconds.

17.20.1.11 getFreeChannels()

```
std::vector< uint8_t > AisInstrumentHandler::getFreeChannels () const
```

Returns

a list of the currently free channels. If all channels are busy, an empty list is returned.

17.20.1.12 getLinkedChannels()

```
std::vector< uint8_t > AisInstrumentHandler::getLinkedChannels (
    uint8_t channel) const
```

Parameters

<i>channel</i>	a valid channel number to find which other channels are linked to it.
----------------	---

Returns

a list of channels linked to the channel parameter.

17.20.1.13 getLoadImpedance()

```
float AisInstrumentHandler::getLoadImpedance (
    uint8_t channel)
```

This allows the user to query the load impedance previously set for a channel.

Parameters

<i>channel</i>	The channel number for which the load impedance is requested.
----------------	---

Returns

The load impedance value in ohms for the specified channel.

17.20.1.14 getManualModeCurrentRangeList()

```
std::vector< std::pair< double, double > > AisInstrumentHandler::getManualModeCurrentRange↵
List (
    uint8_t channel) const
```

The list is indexed, with each index containing a range with minimum and maximum current for the range. You can pass the index of the desired current range to setManualModeConstantVoltage or setManualModeConstantCurrent.

Parameters

<i>channel</i>	a valid channel number for which to check the current range.
----------------	--

Returns

a list of the of the applicable current ranges to the given channel specific to your device.

17.20.1.15 getManualModeVoltageRangeList()

```
std::vector< std::pair< double, double > > AisInstrumentHandler::getManualModeVoltageRange↵
List (
    uint8_t channel) const
```

The list is indexed, with each index containing a range with minimum and maximum voltage for the range. You can pass the index of the desired current range to setManualModeConstantVoltage or setManualModeConstantCurrent.

Parameters

<i>channel</i>	a valid channel number for which to check the current range.
----------------	--

Returns

a list of the of the applicable current ranges to the given channel specific to your device.

17.20.1.16 getNumberOfChannels()

```
int AisInstrumentHandler::getNumberOfChannels () const
```

Returns

the number of channels on the connected device. If no device found, -1 will be returned.

17.20.1.17 groundFloatStateChanged

```
void AisInstrumentHandler::groundFloatStateChanged (
    bool grounded) [signal]
```

Parameters

<i>grounded</i>	true if there is a connection to ground and false if the ground has disconnected.
-----------------	---

17.20.1.18 hasBipolarMode()

```
bool AisInstrumentHandler::hasBipolarMode (
    uint8_t channel) const
```

Parameters

<i>channel</i>	the channel number to check if it is bipolar mode
----------------	---

Returns

true only if given a valid channel number that has bipolar mode.

17.20.1.19 idleDCDataReady

```
void AisInstrumentHandler::idleDCDataReady (
    uint8_t channel,
    const AisDCData & DCData) [signal]
```

A manual experiment displays real time values. These values are displayed even if the channel does not have an experiment running on it.

Parameters

<i>channel</i>	the channel number from which the DC data arrived.
<i>DCData</i>	the DC data that just arrived.

17.20.1.20 isChannelBusy()

```
bool AisInstrumentHandler::isChannelBusy (
    uint8_t channel) const
```

Parameters

<i>channel</i>	the channel number to check if it is busy or not.
----------------	---

Returns

true only if given a valid channel number that has either a running or a paused experiment.

17.20.1.21 isChannelPaused()

```
bool AisInstrumentHandler::isChannelPaused (
    uint8_t channel) const
```

Parameters

<i>channel</i>	the channel number to check if it has a paused experiment.
----------------	--

Returns

true only if given a valid channel number that has an experiment that has been paused.

17.20.1.22 pauseExperiment()

```
AisErrorCode AisInstrumentHandler::pauseExperiment (
    uint8_t channel) const
```

Parameters

<i>channel</i>	the channel number to pause the experiment on.
----------------	--

Returns

true if an experiment was successfully paused on the channel and false otherwise. If not successful, possible returned errors are:

- [AisErrorCode::FailedToPauseExperiment](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)
- [AisErrorCode::ChannelNotBusy](#)

This will return [AisErrorCode::Success](#) only if there is currently a running experiment on a valid channel on a connected device.

17.20.1.23 recoverDataErased

```
void AisInstrumentHandler::recoverDataErased (
    bool successful) [signal]
```

Parameters

<i>successful</i>	is true on erased correctly, and false on data is not erased.
-------------------	---

17.20.1.24 recoveryACDataReady

```
void AisInstrumentHandler::recoveryACDataReady (
    uint8_t channel,
    const AisACData & ACData) [signal]
```

Parameters

<i>channel</i>	the channel number from which the AC data are recovered from.
<i>ACData</i>	the AC data that just arrived.

17.20.1.25 recoveryDCDataReady

```
void AisInstrumentHandler::recoveryDCDataReady (
    uint8_t channel,
    const AisDCData & DCData) [signal]
```

Parameters

<i>channel</i>	the channel number from which the DC data are recovered from.
<i>DCData</i>	the DC data that just arrived.

17.20.1.26 resetChannelLimits()

```
AisErrorCode AisInstrumentHandler::resetChannelLimits (  
    uint8_t channel) const
```

This function removes all configured voltage, current, and temperature limits for a specific channel, restoring it to its default state.

Parameters

<i>channel</i>	The channel number for which the limits are to be reset.
----------------	--

Returns

[AisErrorCode](#) indicating success or failure of the operation.

17.20.1.27 resumeExperiment()

```
AisErrorCode AisInstrumentHandler::resumeExperiment (
    uint8_t channel) const
```

Parameters

<i>channel</i>	the channel number to resume the experiment on.
----------------	---

Returns

[AisErrorCode::Success](#) if an experiment was successfully resumed on the channel. If not successful, possible returned errors are:

- [AisErrorCode::FailedToResumeExperiment](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)
- [AisErrorCode::ChannelNotBusy](#)

This will return [AisErrorCode::Success](#) only if there is currently a paused experiment on a valid channel on a connected device.

17.20.1.28 setBipolarLinkedChannels()

```
int8_t AisInstrumentHandler::setBipolarLinkedChannels (
    std::vector< uint8_t > channels) const
```

You may combine two channels to expand the voltage range to include negative voltages. Note that this is only applicable to the cyclor model. For 4 channel Cyclor models, you can combine channels 1 and 2 or channels 3 and 4. You cannot use any other channel combinations.

Parameters

<i>channels</i>	a list of two channels to be operate in bipolar mode.
-----------------	---

Returns

the master channel out of the given list of two channels. The master channel is your interface to upload an experiment to and then control it. If not successful set in bipolar mode, possible returned errors as -1.

Note

this functionality is only applicable to the cyclor model.

17.20.1.29 setChannelMaximumCurrent()

```
AisErrorCode AisInstrumentHandler::setChannelMaximumCurrent (
    uint8_t channel,
    double Imax) const
```

Sets the maximum allowable current for a channel. The experiment will stop if the measured current exceeds this limit.

Parameters

<i>channel</i>	The channel number for which the current limit is to be set.
<i>I_{max}</i>	The maximum allowable current in amperes.

Returns

[AisErrorCode](#) indicating success or failure of the operation.

Note

- For Squidstat Cyclers, this limit will not apply to AC elements.
- For all other devices, it is not recommended to use this for AC elements.

17.20.1.30 setChannelMaximumTemperature()

```
AisErrorCode AisInstrumentHandler::setChannelMaximumTemperature (
    uint8_t channel,
    double MaxTemperature) const
```

Sets the maximum allowable temperature for a channel. The experiment will stop if the measured temperature exceeds this limit.

Parameters

<i>channel</i>	The channel number for which the temperature limit is to be set.
<i>MaxTemperature</i>	The maximum allowable temperature in degrees Celsius.

Returns

[AisErrorCode](#) indicating success or failure of the operation.

Note

- For Squidstat Cyclers, this limit will not apply to AC elements.

17.20.1.31 setChannelMaximumVoltage()

```
AisErrorCode AisInstrumentHandler::setChannelMaximumVoltage (
    uint8_t channel,
    double Vmax) const
```

Sets the maximum allowable voltage for a channel. The experiment will stop if the measured voltage exceeds this limit.

Parameters

<i>channel</i>	The channel number for which the voltage limit is to be set.
<i>Vmax</i>	The maximum allowable voltage in volts.

Returns

[AisErrorCode](#) indicating success or failure of the operation.

Note

- For Squidstat Cyclers, this limit will not apply to AC elements.
- For all other devices, it is not recommended to use this for AC elements.

17.20.1.32 setChannelMinimumCurrent()

```
AisErrorCode AisInstrumentHandler::setChannelMinimumCurrent (
    uint8_t channel,
    double Imin) const
```

Sets the minimum allowable current for a channel. The experiment will stop if the measured current falls below this limit.

Parameters

<i>channel</i>	The channel number for which the current limit is to be set.
<i>Imin</i>	The minimum allowable current in amperes.

Returns

[AisErrorCode](#) indicating success or failure of the operation.

Note

- For Squidstat Cyclers, this limit will not apply to AC elements.
- For all other devices, it is not recommended to use this for AC elements.

17.20.1.33 setChannelMinimumVoltage()

```
AisErrorCode AisInstrumentHandler::setChannelMinimumVoltage (
    uint8_t channel,
    double Vmin) const
```

Sets the minimum allowable voltage for a channel. The experiment will stop if the measured voltage falls below this limit.

Parameters

<i>channel</i>	The channel number for which the voltage limit is to be set.
<i>Vmin</i>	The minimum allowable voltage in volts.

Returns

[AisErrorCode](#) indicating success or failure of the operation.

Note

- For Squidstat Cyclers, this limit will not apply to AC elements.
- For all other devices, it is not recommended to use this for AC elements.

17.20.1.34 setCompRange()

```
AisErrorCode AisInstrumentHandler::setCompRange (  
    uint8_t channel,  
    const AisCompRange & compRange) const
```

Parameters

<i>channel</i>	the channel for which to set the compensation range.
<i>compRange</i>	an object of type <code>compRange</code> that is initialized with a stability factor (0-10) and a bandwidth index (0-10).

Returns

[AisErrorCode::Success](#) if setting the IR compensation was successful. If not successful, possible returned errors are:

- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)
- [AisErrorCode::InvalidParameters](#)

See also

[AisCompRange](#)

17.20.1.35 setFanSpeedMaximum()

```
AisErrorCode AisInstrumentHandler::setFanSpeedMaximum () const
```

Note

Setting this mode may remove small variations in data caused by fan cycling in some electrochemical systems.

Set the fan speed to maximum when an experiment is active. Only available on some devices.

Model-specific fan behavior:

- Squidstat Cycler: Not available.
- Squidstat Ace, Prime, Solo, and Squidstat Plus (serial numbers below 1700): Always operate in this mode. Fans will turn off when no experiment is running.
- Squidstat Plus (serial numbers 1700 or higher), Squidstat Penta, Decka, and Venta: Default operating mode. Fans will adjust based on the internal temperature of the instrument when no experiment is running.

Return values

<i>AisErrorCode::Success</i>	
<i>AisErrorCode::DeviceNotFound</i>	
<i>AisErrorCode::FeatureNotSupported</i>	
<i>AisErrorCode::DeviceCommunicationFailed</i>	

See also

[setFanSpeedVariable](#)

17.20.1.36 setFanSpeedVariable()

```
AisErrorCode AisInstrumentHandler::setFanSpeedVariable () const
```

Set the fan speed to always adjust automatically based on the internal temperature of the instrument. Only available on some devices.

Model-specific fan behavior:

- Squidstat Cycler: Always operates in this mode; no data variances occur from fan cycling.
- Squidstat Ace, Prime, Solo, and Squidstat Plus (serial numbers below 1700): Not available.
- Squidstat Plus (serial numbers 1700 or higher), Squidstat Penta, Decka, and Venta: Available on these models.

Return values

<i>AisErrorCode::Success</i>	
<i>AisErrorCode::DeviceNotFound</i>	
<i>AisErrorCode::FeatureNotSupported</i>	
<i>AisErrorCode::DeviceCommunicationFailed</i>	

See also

[setFanSpeedMaximum](#)

17.20.1.37 setIRComp()

```
AisErrorCode AisInstrumentHandler::setIRComp (
    uint8_t channel,
    double uncompensatedResistance,
    double compensationLevel) const
```

Parameters

<i>channel</i>	the channel for which to set the IR compensation.
<i>uncompensatedResistance</i>	the value of the uncompensated resistance in Ohms.
<i>compensationLevel</i>	the compensation percentage (0%-100%). This is unit-less.

Returns

[AisErrorCode::Success](#) if setting the IR compensation was successful. If not successful, possible returned errors are:

- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)
- [AisErrorCode::InvalidParameters](#)

17.20.1.38 setLinkedChannels()

```
int8_t AisInstrumentHandler::setLinkedChannels (
    std::vector< uint8_t > channels) const
```

You may connect a list of channels so you can get a higher combined output current of all channels. Note that this is only applicable to the cyclor model.

Parameters

<i>channels</i>	a list of channels to be linked.
-----------------	----------------------------------

Returns

the master channel out of the given list of channels. The master channel is your interface to upload an experiment to and then control it.

Note

this functionality is only applicable to the cyclor model.

17.20.1.39 setLoadImpedance()

```
AisErrorCode AisInstrumentHandler::setLoadImpedance (
    uint8_t channel,
    const float & loadImpedance)
```

By providing the load impedance, the Cyclor can calculate and apply an initial current that will result in a voltage close to the desired setpoint at the start of a potentiostatic experiment. This helps prevent immediately hitting lower current safety limits when the experiment begins.

Parameters

<i>channel</i>	The channel number for which the load impedance is being set.
<i>loadImpedance</i>	The load impedance value in ohms to be applied to the channel.

Returns

[AisErrorCode](#) indicating success or failure of the operation.

17.20.1.40 setManualModeConstantCurrent()

```
AisErrorCode AisInstrumentHandler::setManualModeConstantCurrent (
    uint8_t channel,
    double value) const
```

Parameters

<i>channel</i>	a valid channel number to set a constant voltage for.
<i>value</i>	the value to set the constant current in Amps.

Returns

[AisErrorCode::Success](#) if setting the constant current was successful. If not successful, possible returned errors are:

- [AisErrorCode::ManualExperimentNotRunning](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)
- [AisErrorCode::DeviceCommunicationFailed](#)
- [AisErrorCode::FailedToSetManualModeConstantCurrent](#)

17.20.1.41 setManualModeConstantVoltage() [1/2]

```
AisErrorCode AisInstrumentHandler::setManualModeConstantVoltage (
    uint8_t channel,
    double value) const
```

Parameters

<i>channel</i>	a valid channel number to set a constant voltage for.
<i>value</i>	the value to set the constant voltage in volts.

Returns

[AisErrorCode::Success](#) if setting the constant voltage was successful. If not successful, possible returned errors are:

- [AisErrorCode::FailedToSetManualModeConstantVoltage](#)
- [AisErrorCode::ManualExperimentNotRunning](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)

17.20.1.42 setManualModeConstantVoltage() [2/2]

```
AisErrorCode AisInstrumentHandler::setManualModeConstantVoltage (
    uint8_t channel,
    double value,
    int currentRangeIndex) const
```

Parameters

<i>channel</i>	a valid channel number to set a constant voltage for.
<i>value</i>	the value to set the constant voltage in volts.
<i>currentRangeIndex</i>	the index of the desired current range.

Returns

[AisErrorCode::Success](#) if setting the constant voltage was successful. You can get a list of the available ranges for your model by calling [getManualModeCurrentRangeList](#). If not successful, possible returned errors are:

- [AisErrorCode::FailedToSetManualModeConstantVoltage](#)
- [AisErrorCode::FailedToSetManualModeCurrentRange](#)
- [AisErrorCode::ManualExperimentNotRunning](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)

17.20.1.43 setManualModeCurrentAutorange()

```
AisErrorCode AisInstrumentHandler::setManualModeCurrentAutorange (
    uint8_t channel) const
```

Parameters

<i>channel</i>	a valid channel number to enable current autoranging for.
----------------	---

Returns

[AisErrorCode::Success](#) if enabling current autoranging successful. If not successful, possible returned errors are:

- [AisErrorCode::FailedToSetManualModeCurrentRange](#)
- [AisErrorCode::ManualExperimentNotRunning](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)

17.20.1.44 setManualModeCurrentRange()

```
AisErrorCode AisInstrumentHandler::setManualModeCurrentRange (
    uint8_t channel,
    int currentRangeIndex) const
```

Parameters

<i>channel</i>	a valid channel number to set the current range for.
<i>currentRangeIndex</i>	the index of the desired current range.

Returns

[AisErrorCode::Success](#) if setting the current range was successful. You can get a list of the available ranges for your model by calling [getManualModeCurrentRangeList](#). If not successful, possible returned errors are:

- [AisErrorCode::FailedToSetManualModeCurrentRange](#)
- [AisErrorCode::ManualExperimentNotRunning](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)

17.20.1.45 setManualModeOCP()

```
AisErrorCode AisInstrumentHandler::setManualModeOCP (
    uint8_t channel) const
```

To apply the set potential or current, leave the open circuit potential mode off. This operation is reversed automatically when calling either [setManualModeConstantVoltage\(\)](#) or [setManualModeConstantCurrent\(\)](#)

Parameters

<i>channel</i>	a valid channel number to set open circuit mode on.
----------------	---

Returns

[AisErrorCode::Success](#) if turning on the open circuit mode was successful. If not successful, possible returned errors are:

- [AisErrorCode::ManualExperimentNotRunning](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)
- [AisErrorCode::DeviceCommunicationFailed](#)

17.20.1.46 setManualModeSamplingInterval()

```
AisErrorCode AisInstrumentHandler::setManualModeSamplingInterval (
    uint8_t channel,
    double value) const
```

Parameters

<i>channel</i>	the channel to set the sampling interval for.
<i>value</i>	the value for the sampling interval in seconds.

Returns

[AisErrorCode::Success](#) if the operation was set successfully. If not successful, possible returned errors are:

- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::Unknown](#)
- [AisErrorCode::InvalidChannel](#)

17.20.1.47 setManualModeVoltageAutorange()

```
AisErrorCode AisInstrumentHandler::setManualModeVoltageAutorange (
    uint8_t channel) const
```

Parameters

<i>channel</i>	a valid channel number to enable voltage autoranging for.
----------------	---

Returns

[AisErrorCode::Success](#) if enabling voltage autoranging successful. If not successful, possible returned errors are:

- [AisErrorCode::FailedToSetManualModeVoltageRange](#)
- [AisErrorCode::ManualExperimentNotRunning](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)

17.20.1.48 setManualModeVoltageRange()

```
AisErrorCode AisInstrumentHandler::setManualModeVoltageRange (
    uint8_t channel,
    int voltageRangeIndex) const
```

Parameters

<i>channel</i>	a valid channel number to set the voltage range for.
<i>voltageRangeIndex</i>	the index of the desired voltage range.

Returns

[AisErrorCode::Success](#) if setting the voltage range was successful. You can get a list of the available ranges for your model by calling [getManualModeVoltageRangeList](#). If not successful, possible returned errors are:

- [AisErrorCode::FailedToSetManualModeVoltageRange](#)
- [AisErrorCode::ManualExperimentNotRunning](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)

17.20.1.49 skipExperimentStep()

```
AisErrorCode AisInstrumentHandler::skipExperimentStep (
    uint8_t channel) const
```

When running an element that has several steps like going from CC to CV, then skipping the step goes to the next step within the element. When having several elements in the custom experiment and the current element has one step or we are at the last step within the element, then skipping the step results in going to the next element. If this is the final step of the final element, the experiment will stop.

Parameters

<i>channel</i>	a valid channel number with an experiment to skip the step.
----------------	---

Returns

[AisErrorCode::Success](#) the experiment step was successfully skipped. If not successful, possible returned errors are:

- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)
- [AisErrorCode::ChannelNotBusy](#)
- [AisErrorCode::DeviceCommunicationFailed](#)

17.20.1.50 startIdleSampling()

```
AisErrorCode AisInstrumentHandler::startIdleSampling (
    uint8_t channel) const
```

Parameters

<i>channel</i>	the channel number to start collecting idle data on.
----------------	--

Returns

[AisErrorCode::Success](#) if the request to start idle data was sent. If not successful, possible returned errors are:

- [AisErrorCode::ExperimentUploaded](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)
- [AisErrorCode::BusyChannel](#)

See also

[isChannelBusy](#)

17.20.1.51 startManualExperiment()

```
AisErrorCode AisInstrumentHandler::startManualExperiment (
    uint8_t channel) const
```

With manual experiments, users can turn on any connected channel and toggle between open circuit mode and voltage or current setpoints that can be changed in real-time and run for indefinite periods.

Parameters

<i>channel</i>	a valid channel number to run the manual experiment on.
----------------	---

Returns

[AisErrorCode::Success](#) if the manual experiment was successfully started. If not successful, possible returned errors are:

- [AisErrorCode::FailedManualModeStartExperiment](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)
- [AisErrorCode::BusyChannel](#)

17.20.1.52 startUploadedExperiment()

```
AisErrorCode AisInstrumentHandler::startUploadedExperiment (
    uint8_t channel) const
```

Parameters

<i>channel</i>	the channel number to start the experiment on.
----------------	--

Returns

[AisErrorCode::Success](#) if the experiment was successfully started on the channel. If not successful, possible returned errors are:

- [AisErrorCode::DeviceCommunicationFailed](#)
- [AisErrorCode::ExperimentNotUploaded](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)
- [AisErrorCode::BusyChannel](#)

See also

[uploadExperimentToChannel](#)
[isChannelBusy](#)

17.20.1.53 stopExperiment()

```
AisErrorCode AisInstrumentHandler::stopExperiment (
    uint8_t channel) const
```

Parameters

<i>channel</i>	the channel number to stop the experiment on.
----------------	---

Returns

[AisErrorCode::Success](#) if an experiment was successfully stopped on the channel. If not successful, possible returned errors are:

- [AisErrorCode::FailedToStopExperiment](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)

This will only return [AisErrorCode::Success](#) if there is currently a running or a paused experiment on a valid channel on a connected device.

17.20.1.54 uploadExperimentToChannel() [1/2]

```
AisErrorCode AisInstrumentHandler::uploadExperimentToChannel (
    uint8_t channel,
    const AisExperiment & experiment) const
```

Any running experiment is run on a specific device on a specific channel. This function uploads an experiment to a channel so that you may start, pause, resume and stop the experiment. All of these four control functionalities and others require a channel number to control the experiment. Therefore, if we have several channels, we need to keep track of which experiment is on which channel.

Parameters

<i>channel</i>	the channel number to upload the experiment to.
<i>experiment</i>	the custom experiment to be uploaded to the channel.

Returns

[AisErrorCode::Success](#) if the experiment was successfully uploaded to the channel. If not successful, possible returned errors are:

- [AisErrorCode::FailedToUploadExperiment](#)
- [AisErrorCode::ExperimentIsEmpty](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)
- [AisErrorCode::BusyChannel](#)
- [AisErrorCode::InvalidParameters](#)

This returns [AisErrorCode::Success](#) only when given a valid channel number that is not busy on a connected device.

See also

[isChannelBusy](#)

17.20.1.55 uploadExperimentToChannel() [2/2]

```
AisErrorCode AisInstrumentHandler::uploadExperimentToChannel (
    uint8_t channel,
    std::shared_ptr< AisExperiment > experiment) const
```

Any running experiment is run on a specific device on a specific channel. This function uploads an experiment to a channel so that you may start, pause, resume and stop the experiment. All of these four control functionalities and others require a channel number to control the experiment. Therefore, if we have several channels, we need to keep track of which experiment is on which channel.

Parameters

<i>channel</i>	the channel number to upload the experiment to.
<i>experiment</i>	the custom experiment to be uploaded to the channel.

Returns

[AisErrorCode::Success](#) if the experiment was successfully uploaded to the channel. If not successful, possible returned errors are:

- [AisErrorCode::FailedToUploadExperiment](#)
- [AisErrorCode::ExperimentIsEmpty](#)
- [AisErrorCode::DeviceNotFound](#)
- [AisErrorCode::InvalidChannel](#)
- [AisErrorCode::BusyChannel](#)
- [AisErrorCode::InvalidParameters](#)

This returns [AisErrorCode::Success](#) only when given a valid channel number that is not busy on a connected device.

See also

[isChannelBusy](#)

The documentation for this class was generated from the following file:

- AisInstrumentHandler.h

17.21 AisMottSchottkyElement Class Reference

This class performs Mott-Schottky analysis on the working electrode for a specified range of potentials.

```
#include <AisMottSchottkyElement.h>
```

```
#include <AisMottSchottkyElement.h>
```

Public Member Functions

- [AisMottSchottkyElement](#) (double startingPotential, double endingPotential, double voltageStep, double startFrequency, double endFrequency, double stepsPerDecade, double amplitude, unsigned int minCycles, double stepQuietTime)
Constructor for the Mott-Schottky experiment element.
- [AisMottSchottkyElement](#) (double startingPotential, double endingPotential, double voltageStep, double startFrequency, double endFrequency, double stepsPerDecade, double amplitude, unsigned int minCycles)
Constructor for the Mott-Schottky experiment element.
- [AisMottSchottkyElement](#) (const [AisMottSchottkyElement](#) &other)
Copy constructor for the [AisMottSchottkyElement](#) object.
- [AisMottSchottkyElement](#) & operator= (const [AisMottSchottkyElement](#) &other)
Assignment operator for the [AisMottSchottkyElement](#) object.
- ~[AisMottSchottkyElement](#) () override
Destructor for the [AisMottSchottkyElement](#) object.
- QString [getName](#) () const override
Get the name of the experiment element.
- QStringList [getCategory](#) () const override
Get a list of applicable categories of the experiment element.
- double [getStartingPotential](#) () const
Get the starting potential for the experiment.
- void [setStartingPotential](#) (double startingPotential)
Set the starting potential for the experiment.
- double [getEndingPotential](#) () const
Get the ending potential for the experiment.
- void [setEndingPotential](#) (double endingPotential)
Set the ending potential for the experiment.
- double [getVoltageStep](#) () const
Get the voltage step size between each potential.
- void [setVoltageStep](#) (double voltageStep)
Set the voltage step size between each potential.
- double [getStartFrequency](#) () const
Get the starting frequency for the EIS measurement.
- void [setStartFrequency](#) (double startFrequency)
Set the starting frequency for the EIS measurement.
- double [getEndFrequency](#) () const
Get the ending frequency for the EIS measurement.
- void [setEndFrequency](#) (double endFrequency)
Set the ending frequency for the EIS measurement.
- double [getStepsPerDecade](#) () const
Get the number of frequency steps per decade.
- void [setStepsPerDecade](#) (double stepsPerDecade)
Set the number of frequency steps per decade.

- double [getAmplitude](#) () const
Get the amplitude of the AC signal used in the EIS measurement.
- void [setAmplitude](#) (double amplitude)
Set the amplitude of the AC signal used in the EIS measurement.
- unsigned int [getMinCycles](#) () const
Get the minimum number of cycles per frequency step.
- void [setMinCycles](#) (unsigned int minCycles)
Set the minimum number of cycles per frequency step.
- double [getQuietTime](#) () const
Get the quiet time before starting the EIS measurement.
- void [setQuietTime](#) (double quietTime)
Set the quiet time before starting the EIS measurement.
- double [getQuietTimeSamplInterval](#) () const
Get the sampling interval during the quiet time.
- void [setQuietTimeSamplInterval](#) (double quietTimeSamplInterval)
Set the sampling interval during the quiet time.
- double [getStepQuietTime](#) () const
Get the quiet time after each potential step before starting the EIS measurement.
- void [setStepQuietTime](#) (double stepQuietTime)
Set the quiet time after each potential step before starting the EIS measurement.
- double [getStepQuietSamplInterval](#) () const
Get the sampling interval during the quiet time after each potential step.
- void [setStepQuietSamplInterval](#) (double stepQuietTimeSamplInterval)
Set the sampling interval during the quiet time after each potential step.
- bool [isStartVoltageVsOCP](#) () const
Check if the starting voltage is measured versus the open circuit potential (OCP).
- void [setStartVoltageVsOCP](#) (bool startVsOCP)
Set whether the starting voltage is measured versus the open circuit potential (OCP).
- bool [isEndVoltageVsOCP](#) () const
Check if the ending voltage is measured versus the open circuit potential (OCP).
- void [setEndVoltageVsOCP](#) (bool endVsOCP)
Set whether the ending voltage is measured versus the open circuit potential (OCP).

17.21.1 Constructor & Destructor Documentation

17.21.1.1 AisMottSchottkyElement() [1/3]

```
AisMottSchottkyElement::AisMottSchottkyElement (
    double startingPotential,
    double endingPotential,
    double voltageStep,
    double startFrequency,
    double endFrequency,
    double stepsPerDecade,
    double amplitude,
    unsigned int minCycles,
    double stepQuietTime) [explicit]
```

Parameters

<i>startingPotential</i>	The starting potential (voltage) for the experiment.
--------------------------	--

Parameters

<i>endingPotential</i>	The ending potential (voltage) for the experiment.
<i>voltageStep</i>	The voltage step size between each potential during the experiment.
<i>startFrequency</i>	The starting frequency for the EIS measurement.
<i>endFrequency</i>	The ending frequency for the EIS measurement.
<i>stepsPerDecade</i>	The number of frequency steps per decade.
<i>amplitude</i>	The amplitude of the AC signal used in the EIS measurement.
<i>minCycles</i>	The minimum number of cycles per frequency step during the EIS measurement.
<i>stepQuietTime</i>	The time spent at the starting potential.

17.21.1.2 AisMottSchottkyElement() [2/3]

```
AisMottSchottkyElement::AisMottSchottkyElement (
    double startingPotential,
    double endingPotential,
    double voltageStep,
    double startFrequency,
    double endFrequency,
    double stepsPerDecade,
    double amplitude,
    unsigned int minCycles) [explicit]
```

Parameters

<i>startingPotential</i>	The starting potential (voltage) for the experiment.
<i>endingPotential</i>	The ending potential (voltage) for the experiment.
<i>voltageStep</i>	The voltage step size between each potential during the experiment.
<i>startFrequency</i>	The starting frequency for the EIS measurement.
<i>endFrequency</i>	The ending frequency for the EIS measurement.
<i>stepsPerDecade</i>	The number of frequency steps per decade.
<i>amplitude</i>	The amplitude of the AC signal used in the EIS measurement.
<i>minCycles</i>	The minimum number of cycles per frequency step during the EIS measurement.

17.21.1.3 AisMottSchottkyElement() [3/3]

```
AisMottSchottkyElement::AisMottSchottkyElement (
    const AisMottSchottkyElement & other) [explicit]
```

Parameters

<i>other</i>	The object to copy from.
--------------	--------------------------

17.21.2 Member Function Documentation

17.21.2.1 getAmplitude()

```
double AisMottSchottkyElement::getAmplitude () const
```

Returns

The AC amplitude in volts.

17.21.2.2 getCategory()

```
QStringList AisMottSchottkyElement::getCategory () const [override]
```

Returns

A list of categories where the experiment is applicable, such as "Advanced Experiments".

17.21.2.3 getEndFrequency()

```
double AisMottSchottkyElement::getEndFrequency () const
```

Returns

The ending frequency in Hz.

17.21.2.4 getEndingPotential()

```
double AisMottSchottkyElement::getEndingPotential () const
```

Returns

The ending potential in volts.

17.21.2.5 getMinCycles()

```
unsigned int AisMottSchottkyElement::getMinCycles () const
```

Returns

The minimum number of cycles.

17.21.2.6 getName()

```
QString AisMottSchottkyElement::getName () const [override]
```

Returns

The name of the element, "Mott-Schottky".

17.21.2.7 getQuietTime()

```
double AisMottSchottkyElement::getQuietTime () const
```

Returns

The quiet time in seconds.

17.21.2.8 getQuietTimeSampInterval()

```
double AisMottSchottkyElement::getQuietTimeSampInterval () const
```

Returns

The sampling interval in seconds.

17.21.2.9 getStartFrequency()

```
double AisMottSchottkyElement::getStartFrequency () const
```

Returns

The starting frequency in Hz.

17.21.2.10 getStartingPotential()

```
double AisMottSchottkyElement::getStartingPotential () const
```

Returns

The starting potential in volts.

17.21.2.11 getStepQuietSampInterval()

```
double AisMottSchottkyElement::getStepQuietSampInterval () const
```

Returns

The sampling interval in seconds.

17.21.2.12 getStepQuietTime()

```
double AisMottSchottkyElement::getStepQuietTime () const
```

Returns

The quiet time after each potential step in seconds.

17.21.2.13 getStepsPerDecade()

```
double AisMottSchottkyElement::getStepsPerDecade () const
```

Returns

The number of steps per decade.

17.21.2.14 getVoltageStep()

```
double AisMottSchottkyElement::getVoltageStep () const
```

Returns

The voltage step size in volts.

17.21.2.15 isEndVoltageVsOCP()

```
bool AisMottSchottkyElement::isEndVoltageVsOCP () const
```

Returns

True if the ending voltage is measured versus OCP, false otherwise.

17.21.2.16 isStartVoltageVsOCP()

```
bool AisMottSchottkyElement::isStartVoltageVsOCP () const
```

Returns

True if the starting voltage is measured versus OCP, false otherwise.

17.21.2.17 operator=()

```
AisMottSchottkyElement & AisMottSchottkyElement::operator= (  
    const AisMottSchottkyElement & other)
```


Parameters

<i>other</i>	The object to assign from.
--------------	----------------------------

Returns

A reference to the assigned object.

17.21.2.18 setAmplitude()

```
void AisMottSchottkyElement::setAmplitude (  
    double amplitude)
```

Parameters

<i>amplitude</i>	The AC amplitude in volts.
------------------	----------------------------

17.21.2.19 setEndFrequency()

```
void AisMottSchottkyElement::setEndFrequency (  
    double endFrequency)
```

Parameters

<i>endFrequency</i>	The ending frequency in Hz.
---------------------	-----------------------------

17.21.2.20 setEndingPotential()

```
void AisMottSchottkyElement::setEndingPotential (  
    double endingPotential)
```

Parameters

<i>endingPotential</i>	The ending potential in volts.
------------------------	--------------------------------

17.21.2.21 setEndVoltageVsOCP()

```
void AisMottSchottkyElement::setEndVoltageVsOCP (  
    bool endVsOCP)
```

Parameters

<i>endVsOCP</i>	True if the ending voltage is measured versus OCP, false otherwise.
-----------------	---

17.21.2.22 setMinCycles()

```
void AisMottSchottkyElement::setMinCycles (  
    unsigned int minCycles)
```

Parameters

<i>minCycles</i>	The minimum number of cycles.
------------------	-------------------------------

17.21.2.23 setQuietTime()

```
void AisMottSchottkyElement::setQuietTime (
    double quietTime)
```

Parameters

<i>quietTime</i>	The quiet time in seconds.
------------------	----------------------------

17.21.2.24 setQuietTimeSampInterval()

```
void AisMottSchottkyElement::setQuietTimeSampInterval (
    double quietTimeSampInterval)
```

Parameters

<i>quietTimeSampInterval</i>	The sampling interval in seconds.
------------------------------	-----------------------------------

17.21.2.25 setStartFrequency()

```
void AisMottSchottkyElement::setStartFrequency (
    double startFrequency)
```

Parameters

<i>startFrequency</i>	The starting frequency in Hz.
-----------------------	-------------------------------

17.21.2.26 setStartingPotential()

```
void AisMottSchottkyElement::setStartingPotential (
    double startingPotential)
```

Parameters

<i>startingPotential</i>	The starting potential in volts.
--------------------------	----------------------------------

17.21.2.27 setStartVoltageVsOCP()

```
void AisMottSchottkyElement::setStartVoltageVsOCP (
    bool startVsOCP)
```

Parameters

<i>startVsOCP</i>	True if the starting voltage is measured versus OCP, false otherwise.
-------------------	---

17.21.2.28 setStepQuietSampInterval()

```
void AisMottSchottkyElement::setStepQuietSampInterval (
    double stepQuietTimeSampInterval)
```

Parameters

<i>stepQuietTimeSampInterval</i>	The sampling interval in seconds.
----------------------------------	-----------------------------------

17.21.2.29 setStepQuietTime()

```
void AisMottSchottkyElement::setStepQuietTime (
    double stepQuietTime)
```

Parameters

<i>stepQuietTime</i>	The quiet time after each potential step in seconds.
----------------------	--

17.21.2.30 setStepsPerDecade()

```
void AisMottSchottkyElement::setStepsPerDecade (
    double stepsPerDecade)
```

Parameters

<i>stepsPerDecade</i>	The number of steps per decade.
-----------------------	---------------------------------

17.21.2.31 setVoltageStep()

```
void AisMottSchottkyElement::setVoltageStep (
    double voltageStep)
```

Parameters

<i>voltageStep</i>	The voltage step size in volts.
--------------------	---------------------------------

The documentation for this class was generated from the following file:

- AisMottSchottkyElement.h

17.22 AisNormalPulseVoltammetryElement Class Reference

This experiment holds the working electrode at a **baseline potential** during the **quiet time**, then applies a train of pulses, which increase in amplitude until the **final potential** is reached.

```
#include <AisNormalPulseVoltammetryElement.h>
```

```
#include <AisNormalPulseVoltammetryElement.h>
```

Public Member Functions

- [AisNormalPulseVoltammetryElement](#) (double startVoltage, double endVoltage, double voltageStep, double pulseWidth, double pulsePeriod)
the normal-pulse-voltammetry element constructor
- [AisNormalPulseVoltammetryElement](#) (double startVoltage, double endVoltage, double voltageStep, double pulseWidth, double pulsePeriod, double approxMaxCurrent)
the normal-pulse-voltammetry element constructor
- **AisNormalPulseVoltammetryElement** (const [AisNormalPulseVoltammetryElement](#) &)
copy constructor for the [AisNormalPulseVoltammetryElement](#) object.
- [AisNormalPulseVoltammetryElement](#) & **operator=** (const [AisNormalPulseVoltammetryElement](#) &)
overload equal to operator for the [AisNormalPulseVoltammetryElement](#) object.
- QString [getName](#) () const override
get the name of the element.
- QStringList [getCategory](#) () const override
get a list of applicable categories of the element.
- double [getQuietTime](#) () const
Gets the quiet time duration.
- void [setQuietTime](#) (double quietTime)
Sets the quiet time duration.
- double [getQuietTimeSamplingInterval](#) () const
gets the quiet time sampling interval.
- void [setQuietTimeSamplingInterval](#) (double quietTimeSamplingInterval)
Sets the quiet time sampling interval.
- double [getStartVoltage](#) () const
get the value set for the start voltage.
- void [setStartVoltage](#) (double startVoltage)
set the value for the start voltage.
- bool [isStartVoltageVsOCP](#) () const
tells whether the start voltage is set against the open-circuit voltage or the reference terminal.
- void [setStartVoltageVsOCP](#) (bool startVoltageVsOCP)
set whether to reference the start voltage against the open-circuit voltage or the reference terminal.
- double [getEndVoltage](#) () const
get the value set for the ending potential value.
- void [setEndVoltage](#) (double endVoltage)
set the ending potential value.
- bool [isEndVoltageVsOCP](#) () const
tells whether the end voltage is set with respect to the open circuit voltage or the reference terminal.
- void [setEndVoltageVsOCP](#) (bool endVoltageVsOcp)
set whether to reference the end voltage against the open-circuit voltage or the reference terminal.
- double [getVStep](#) () const
get the value set for the voltage step.

- void [setVStep](#) (double vStep)
set the value for the voltage step.
- double [getPulseWidth](#) () const
get the value set for the pulse width
- void [setPulseWidth](#) (double pulseWidth)
set the value in seconds for pulse width.
- double [getPulsePeriod](#) () const
get the value set for the pulse period.
- void [setPulsePeriod](#) (double pulsePeriod)
set the value for the pulse period.
- bool [isAutoRange](#) () const
tells whether the current range is set to auto-select or not.
- void [setAutoRange](#) ()
set to auto-select the current range.
- double [getApproxMaxCurrent](#) () const
get the value set for the expected maximum current.
- void [setApproxMaxCurrent](#) (double approxMaxCurrent)
set maximum current expected, for manual current range selection.
- double [getAlphaFactor](#) () const
Get the value set for the alpha factor.
- void [setAlphaFactor](#) (double alphaFactor)
alpha factor controls the percentage of data sampled during a given interval. Data will be averaged over the last n% of the sampling interval.

17.22.1 Constructor & Destructor Documentation

17.22.1.1 AisNormalPulseVoltammetryElement() [1/2]

```
AisNormalPulseVoltammetryElement::AisNormalPulseVoltammetryElement (
    double startVoltage,
    double endVoltage,
    double voltageStep,
    double pulseWidth,
    double pulsePeriod) [explicit]
```

Parameters

<i>startVoltage</i>	the value of the starting potential in volts
<i>endVoltage</i>	the value of the ending potential in volts
<i>voltageStep</i>	the value set for the voltage step in volts.
<i>pulseWidth</i>	the value for the pulse width in seconds.
<i>pulsePeriod</i>	the value for the pulse period in seconds.

17.22.1.2 AisNormalPulseVoltammetryElement() [2/2]

```
AisNormalPulseVoltammetryElement::AisNormalPulseVoltammetryElement (
    double startVoltage,
    double endVoltage,
    double voltageStep,
    double pulseWidth,
    double pulsePeriod,
    double approxMaxCurrent) [explicit]
```

Parameters

<i>startVoltage</i>	the value of the starting potential in volts
<i>endVoltage</i>	the value of the ending potential in volts
<i>voltageStep</i>	the value set for the voltage step in volts.
<i>pulseWidth</i>	the value for the pulse width in seconds.
<i>pulsePeriod</i>	the value for the pulse period in seconds.
<i>approxMaxCurrent</i>	the value for the approximate maximum current in amperes.

17.22.2 Member Function Documentation**17.22.2.1 getAlphaFactor()**

```
double AisNormalPulseVoltammetryElement::getAlphaFactor () const
```

Returns

The value for the alpha factor is represented as a percent between 0 and 100.

Note

If nothing is set, this function will return a default value of 75.

17.22.2.2 getApproxMaxCurrent()

```
double AisNormalPulseVoltammetryElement::getApproxMaxCurrent () const
```

Returns

the value set for the expected maximum current in Amps.

Note

if nothing was manually set, the device will auto-select the current range and the return value will be positive infinity.

17.22.2.3 getCategory()

```
QStringList AisNormalPulseVoltammetryElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Potentiostatic Control", "Basic Voltammetry", "Pulse Voltammetry").

17.22.2.4 getEndVoltage()

```
double AisNormalPulseVoltammetryElement::getEndVoltage () const
```

This is the value of the voltage at which the experiment will stop.

Returns

the value set for the ending voltage in volts.

17.22.2.5 getName()

```
QString AisNormalPulseVoltammetryElement::getName () const [override]
```

Returns

The name of the element: "Normal Pulse Potential Voltammetry".

17.22.2.6 getPulsePeriod()

```
double AisNormalPulseVoltammetryElement::getPulsePeriod () const
```

Returns

the value for the pulse period in seconds.

See also

[setPulsePeriod](#)

17.22.2.7 getPulseWidth()

```
double AisNormalPulseVoltammetryElement::getPulseWidth () const
```

Returns

the value of the pulse width in seconds.

See also

[setPulseWidth](#)

17.22.2.8 getQuietTime()

```
double AisNormalPulseVoltammetryElement::getQuietTime () const
```

Returns

The quiet time duration in seconds.

17.22.2.9 getQuietTimeSamplingInterval()

```
double AisNormalPulseVoltammetryElement::getQuietTimeSamplingInterval () const
```

Returns

quiet time The quiet time sampling interval to set in seconds.

17.22.2.10 getStartVoltage()

```
double AisNormalPulseVoltammetryElement::getStartVoltage () const
```

Returns

the value of the start voltage in volts.

17.22.2.11 getVStep()

```
double AisNormalPulseVoltammetryElement::getVStep () const
```

Returns

the value set for the voltage step in volts.

See also

[setVStep](#)

17.22.2.12 isAutoRange()

```
bool AisNormalPulseVoltammetryElement::isAutoRange () const
```

Returns

true if the current range is set to auto-select and false if a range has been selected.

Deprecated This function is deprecated and no longer supports auto range for this element. Specify the current using [setApproxMaxCurrent\(\)](#). The device will determine the appropriate range based on the current value.

17.22.2.13 isEndVoltageVsOCP()

```
bool AisNormalPulseVoltammetryElement::isEndVoltageVsOCP () const
```

Returns

true if the end voltage is set with respect to the open-circuit voltage and false if set against the reference terminal.

Note

if nothing is set, the default is false.

17.22.2.14 isStartVoltageVsOCP()

```
bool AisNormalPulseVoltammetryElement::isStartVoltageVsOCP () const
```

Returns

true if the start voltage is set against the open-circuit voltage and false if it is set against the reference terminal.

Note

if nothing is set, the default is false.

See also

[setStartVoltageVsOCP](#)

17.22.2.15 setAlphaFactor()

```
void AisNormalPulseVoltammetryElement::setAlphaFactor (
    double alphaFactor)
```

This is an **optional** parameter. If nothing is set, the device will use the default value of 75.

Parameters

<i>alphaFactor</i>	the value for the alphaFactor ranges from 0 to 100.
--------------------	---

17.22.2.16 setApproxMaxCurrent()

```
void AisNormalPulseVoltammetryElement::setApproxMaxCurrent (
    double approxMaxCurrent)
```

This is an **optional** parameter. If nothing is set, the device will auto-select the current range.

Parameters

<i>approxMaxCurrent</i>	the value for the maximum current expected in Amps.
-------------------------	---

17.22.2.17 setAutoRange()

```
void AisNormalPulseVoltammetryElement::setAutoRange ()
```

This option is set by default. There is no need to call this function to auto-select if the range was not manually set.

Deprecated This function is deprecated. Use [setApproxMaxCurrent\(\)](#) to specify the current range instead.

17.22.2.18 setEndVoltage()

```
void AisNormalPulseVoltammetryElement::setEndVoltage (
    double endVoltage)
```

This is the value of the voltage at which the experiment will stop.

Parameters

<i>endVoltage</i>	the value to set for the ending potential in volts.
-------------------	---

17.22.2.19 setEndVoltageVsOCP()

```
void AisNormalPulseVoltammetryElement::setEndVoltageVsOCP (
    bool endVoltageVsOcp)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>endVoltageVsOcp</i>	true to set the end voltage to be referenced against the open-circuit voltage and false if set against the reference terminal.
------------------------	--

Note

by default, this is set to false.

17.22.2.20 setPulsePeriod()

```
void AisNormalPulseVoltammetryElement::setPulsePeriod (
    double pulsePeriod)
```

The pulse period is the time spent between the starts of two consecutive pulses.

Parameters

<i>pulsePeriod</i>	the value to set for the pulse period in seconds.
--------------------	---

17.22.2.21 setPulseWidth()

```
void AisNormalPulseVoltammetryElement::setPulseWidth (
    double pulseWidth)
```

The pulse width is the value in seconds for the time spent at the same voltage set for the pulse height.

Parameters

<i>pulseWidth</i>	the value to set for the pulse width in seconds.
-------------------	--

17.22.2.22 setQuietTime()

```
void AisNormalPulseVoltammetryElement::setQuietTime (
    double quietTime)
```

Parameters

<i>quietTime</i>	The quiet time duration to set in seconds.
------------------	--

17.22.2.23 setQuietTimeSamplingInterval()

```
void AisNormalPulseVoltammetryElement::setQuietTimeSamplingInterval (
    double quietTimeSamplingInterval)
```

Parameters

<i>quietTimeSamplingInterval</i>	The quiet time sampling interval to set in seconds.
----------------------------------	---

17.22.2.24 setStartVoltage()

```
void AisNormalPulseVoltammetryElement::setStartVoltage (
    double startVoltage)
```

Parameters

<i>startVoltage</i>	the value of the start voltage in volts
---------------------	---

17.22.2.25 setStartVoltageVsOCP()

```
void AisNormalPulseVoltammetryElement::setStartVoltageVsOCP (
    bool startVoltageVsOCP)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>startVoltageVsOCP</i>	true to if the start voltage is set to reference the open-circuit voltage and false if set against the reference terminal.
--------------------------	--

Note

by default, this is set to false.

17.22.2.26 setVStep()

```
void AisNormalPulseVoltammetryElement::setVStep (
    double vStep)
```

The voltage step is the voltage difference between the heights of two consecutive pulses.

Parameters

<code>vStep</code>	the value for the voltage step in volts.
--------------------	--

Note

Regardless of `vStep`'s sign, the device will determine the step direction based on the start and end voltage.

The documentation for this class was generated from the following file:

- `AisNormalPulseVoltammetryElement.h`

17.23 AisOpenCircuitElement Class Reference

This experiment observes the **open circuit potential** of the working electrode for a specific period of time.

```
#include <AisOpenCircuitElement.h>
```

```
#include <AisOpenCircuitElement.h>
```

Public Member Functions

- [AisOpenCircuitElement](#) (double duration, double samplingInterval)
the open-circuit element constructor.
- [AisOpenCircuitElement](#) (const [AisOpenCircuitElement](#) &)
copy constructor for the [AisOpenCircuitElement](#) object.
- [AisOpenCircuitElement](#) & **operator=** (const [AisOpenCircuitElement](#) &)
overload equal to operator for the [AisOpenCircuitElement](#) object.
- `QString getName ()` const override
get the name of the element.
- `QStringList getCategory ()` const override
get a list of applicable categories of the element.
- `double getSamplingInterval ()` const
get how frequently we are sampling the data.
- `void setSamplingInterval` (double samplingInterval)
set how frequently we are sampling the data.
- `double getMaxDuration ()` const
get the value set for the duration of the experiment.
- `void setMaxDuration` (double maxDuration)
set the value set for the duration of the experiment.
- `double getMaxVoltage ()` const
get the value set for the maximum voltage. The experiment will end when it reaches this value.
- `void setMaxVoltage` (double maxVoltage)
set a maximum voltage to stop the experiment.
- `double getMinVoltage ()` const
get the value set minimum for the voltage in volts.
- `void setMinVoltage` (double minVoltage)

- set a minimum voltage to stop the experiment.*
- double `getMindVdt ()` const
get the value set for the minimum voltage rate of change with respect to time (minimum dV/dt).
- void `setMindVdt` (double mindVdt)
set the minimum value for the voltage rate of change with respect to time (minimum dV/dt).
- bool `isAutoVoltageRange ()` const
tells whether the voltage range is set to auto-select or not.
- void `setAutoVoltageRange ()`
set to auto-select the voltage range.
- double `getApproxMaxVoltage ()` const
get the value set for the expected maximum voltage.
- void `setApproxMaxVoltage` (double approxMaxVoltage)
set maximum voltage expected, for manual voltage range selection.

17.23.1 Constructor & Destructor Documentation

17.23.1.1 AisOpenCircuitElement()

```
AisOpenCircuitElement::AisOpenCircuitElement (
    double duration,
    double samplingInterval) [explicit]
```

Parameters

<i>duration</i>	the maximum duration for the experiment in seconds.
<i>samplingInterval</i>	the data sampling interval value in seconds.

17.23.2 Member Function Documentation

17.23.2.1 getApproxMaxVoltage()

```
double AisOpenCircuitElement::getApproxMaxVoltage () const
```

Returns

the value set for the expected maximum Voltage in volt.

Note

if nothing was manually set, the device will auto-select the voltage range and the return value will be positive infinity.

17.23.2.2 getCategory()

```
QStringList AisOpenCircuitElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Basic Experiments").

17.23.2.3 getMaxDuration()

```
double AisOpenCircuitElement::getMaxDuration () const
```

Returns

the value set for the duration of the experiment in seconds.

17.23.2.4 getMaxVoltage()

```
double AisOpenCircuitElement::getMaxVoltage () const
```

Returns

the value set for the maximum voltage.

Note

this is an optional parameter. If no value has been set, the default value is positive infinity.

17.23.2.5 getMindVdt()

```
double AisOpenCircuitElement::getMindVdt () const
```

Returns

the value set for the minimum voltage rate of change with respect to time (minimum dV/dt).

Note

this is an optional parameter. If no value has been set, the default value is zero.

17.23.2.6 getMinVoltage()

```
double AisOpenCircuitElement::getMinVoltage () const
```

Returns

the value set for the minimum voltage in volts.

Note

this is an optional parameter. If no value has been set, the default value is negative infinity.

17.23.2.7 getName()

```
QString AisOpenCircuitElement::getName () const [override]
```

Returns

The name of the element: "Open Circuit Potential".

17.23.2.8 getSamplingInterval()

```
double AisOpenCircuitElement::getSamplingInterval () const
```

Returns

the data sampling interval value in seconds.

17.23.2.9 isAutoVoltageRange()

```
bool AisOpenCircuitElement::isAutoVoltageRange () const
```

Returns

true if the voltage range is set to auto-select and false if a range has been selected.

17.23.2.10 setApproxMaxVoltage()

```
void AisOpenCircuitElement::setApproxMaxVoltage (  
    double approxMaxVoltage)
```

This is an **optional** parameter. If nothing is set, the device will auto-select the voltage range.

Parameters

<i>approxMaxVoltage</i>	the value for the maximum voltage expected in V.
-------------------------	--

17.23.2.11 setAutoVoltageRange()

```
void AisOpenCircuitElement::setAutoVoltageRange ()
```

This option is set by default. There is no need to call this function to auto-select if the range was not manually set.

17.23.2.12 setMaxDuration()

```
void AisOpenCircuitElement::setMaxDuration (  
    double maxDuration)
```

Parameters

<i>maxDuration</i>	the value to set for the duration of the experiment in seconds.
--------------------	---

17.23.2.13 setMaxVoltage()

```
void AisOpenCircuitElement::setMaxVoltage (
    double maxVoltage)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an upper-limit voltage value. If a maximum voltage is set, the experiment will continue to run as long as the measured voltage is below that value.

Parameters

<i>maxVoltage</i>	the maximum voltage value in volts at which the experiment will stop.
-------------------	---

17.23.2.14 setMindVdt()

```
void AisOpenCircuitElement::setMindVdt (
    double mindVdt)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an lower-limit rate of change value. If a minimum value is set, the experiment will continue to run as long as the rage of change is above that value.

Parameters

<i>mindVdt</i>	the minimum value for the voltage rate of change with respect to time (minimum dV/dt).
----------------	--

17.23.2.15 setMinVoltage()

```
void AisOpenCircuitElement::setMinVoltage (
    double minVoltage)
```

This is an **optional** condition. If nothing is set, then the experiment will not stop based on an lower-limit voltage value. If a maximum voltage is set, the experiment will continue to run as long as the measured voltage is above that value.

Parameters

<i>minVoltage</i>	the minimum voltage value in volts at which the experiment will stop.
-------------------	---

17.23.2.16 setSamplingInterval()

```
void AisOpenCircuitElement::setSamplingInterval (
    double samplingInterval)
```


Parameters

<i>samplingInterval</i>	the data sampling interval value in seconds.
-------------------------	--

The documentation for this class was generated from the following file:

- AisOpenCircuitElement.h

17.24 AisSquareWaveVoltammetryElement Class Reference

This experiment holds the working electrode at the **starting potential** during the **quiet time**. Then it applies a train of square pulses superimposed on a staircase waveform with a uniform **potential step** magnitude.

```
#include <AisSquareWaveVoltammetryElement.h>
```

```
#include <AisSquareWaveVoltammetryElement.h>
```

Public Member Functions

- [AisSquareWaveVoltammetryElement](#) (double startVoltage, double endVoltage, double voltageStep, double pulseAmp, double pulseFrequency)
the square wave element constructor
- [AisSquareWaveVoltammetryElement](#) (double startVoltage, double endVoltage, double voltageStep, double pulseAmp, double pulseFrequency, double approxMaxCurrent)
the square wave element constructor
- [AisSquareWaveVoltammetryElement](#) (const [AisSquareWaveVoltammetryElement](#) &)
copy constructor for the [AisSquareWaveVoltammetryElement](#) object.
- [AisSquareWaveVoltammetryElement](#) & **operator=** (const [AisSquareWaveVoltammetryElement](#) &)
overload equal to operator for the [AisSquareWaveVoltammetryElement](#) object.
- QString [getName](#) () const override
get the name of the element.
- QStringList [getCategory](#) () const override
get a list of applicable categories of the element.
- double [getQuietTime](#) () const
Gets the quiet time duration.
- void [setQuietTime](#) (double quietTime)
Sets the quiet time duration.
- double [getQuietTimeSamplingInterval](#) () const
gets the quiet time sampling interval.
- void [setQuietTimeSamplingInterval](#) (double quietTimeSamplingInterval)
Sets the quiet time sampling interval.
- double [getStartVoltage](#) () const
get the value set for the start voltage.
- void [setStartVoltage](#) (double startVoltage)
set the value for the start voltage.
- bool [isStartVoltageVsOCP](#) () const
tells whether the start voltage is set against the open-circuit voltage or the reference terminal.
- void [setStartVoltageVsOCP](#) (bool startVoltageVsOcp)

- set whether to reference the start voltage against the open-circuit voltage or the reference terminal.*
- double [getEndVoltage](#) () const
get the value set for the ending potential value.
- void [setEndVoltage](#) (double endVoltage)
set the ending potential value.
- bool [isEndVoltageVsOCP](#) () const
tells whether the end voltage is set with respect to the open circuit voltage or the reference terminal.
- void [setEndVoltageVsOCP](#) (bool endVoltageVsOcp)
set whether to reference the end voltage against the open-circuit voltage or the reference terminal.
- double [getVStep](#) () const
get the value set for the voltage step.
- void [setVStep](#) (double vStep)
set the value for the voltage step. The voltage step is added to the value of the starting potential of the previous pulse to start the new pulse.
- double [getPulseAmp](#) () const
get the value set for the pulse amplitude.
- void [setPulseAmp](#) (double pulseAmp)
set the value for the pulse amplitude.
- double [getPulseFreq](#) () const
get the value set for the pulse frequency.
- void [setPulseFreq](#) (double pulseFreq)
set the value for the pulse frequency.
- bool [isAutoRange](#) () const
tells whether the current range is set to auto-select or not.
- void [setAutoRange](#) ()
set to auto-select the current range.
- double [getApproxMaxCurrent](#) () const
get the value set for the expected maximum current.
- void [setApproxMaxCurrent](#) (double approxMaxCurrent)
set maximum current expected, for manual current range selection.
- double [getAlphaFactor](#) () const
Get the value set for the alpha factor.
- void [setAlphaFactor](#) (double alphaFactor)
alpha factor controls the percentage of data sampled during a given interval. Data will be averaged over the last n% of the sampling interval.

17.24.1 Constructor & Destructor Documentation

17.24.1.1 AisSquareWaveVoltammetryElement() [1/2]

```
AisSquareWaveVoltammetryElement::AisSquareWaveVoltammetryElement (
    double startVoltage,
    double endVoltage,
    double voltageStep,
    double pulseAmp,
    double pulseFrequency) [explicit]
```

Parameters

<i>startVoltage</i>	the value of the starting potential in volts
<i>endVoltage</i>	the value of the ending potential in volts

<i>voltageStep</i>	the value set for the voltage step in volts.
<i>pulseAmp</i>	the value for the pulse amplitude in volts.
<i>pulseFrequency</i>	the value for the pulse frequency in Hz.

Deprecated Use the constructor with the `approxMaxCurrent` parameter instead.

17.24.1.2 AisSquareWaveVoltammetryElement() [2/2]

```
AisSquareWaveVoltammetryElement::AisSquareWaveVoltammetryElement (  
    double startVoltage,  
    double endVoltage,  
    double voltageStep,  
    double pulseAmp,  
    double pulseFrequency,  
    double approxMaxCurrent) [explicit]
```

Parameters

<i>startVoltage</i>	the value of the starting potential in volts
<i>endVoltage</i>	the value of the ending potential in volts
<i>voltageStep</i>	the value set for the voltage step in volts.
<i>pulseAmp</i>	the value for the pulse amplitude in volts.
<i>pulseFrequency</i>	the value for the pulse frequency in Hz.
<i>approxMaxCurrent</i>	the value for the approximate maximum current in amperes.

17.24.2 Member Function Documentation

17.24.2.1 getAlphaFactor()

```
double AisSquareWaveVoltammetryElement::getAlphaFactor () const
```

Returns

The value for the alpha factor is represented as a percent between 0 and 100.

Note

If nothing is set, this function will return a default value of 75.

17.24.2.2 getApproxMaxCurrent()

```
double AisSquareWaveVoltammetryElement::getApproxMaxCurrent () const
```

Returns

the value set for the expected maximum current in Amps.

Note

if nothing was manually set, the device will auto-select the current range and the return value will be positive infinity.

17.24.2.3 getCategory()

```
QStringList AisSquareWaveVoltammetryElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Potentiostatic Control", "Pulse Voltammetry").

17.24.2.4 getEndVoltage()

```
double AisSquareWaveVoltammetryElement::getEndVoltage () const
```

This is the value of the voltage at which the experiment will stop.

Returns

the value set for the ending voltage in volts.

17.24.2.5 getName()

```
QString AisSquareWaveVoltammetryElement::getName () const [override]
```

Returns

The name of the element: "Square Wave Potential Voltammetry".

17.24.2.6 getPulseAmp()

```
double AisSquareWaveVoltammetryElement::getPulseAmp () const
```

Returns

the value set for the pulse amplitude in volts.

See also

[setPulseAmp](#)

17.24.2.7 getPulseFreq()

```
double AisSquareWaveVoltammetryElement::getPulseFreq () const
```

Returns

the value set for the frequency in Hz.

17.24.2.8 getQuietTime()

```
double AisSquareWaveVoltammetryElement::getQuietTime () const
```

Returns

The quiet time duration in seconds.

17.24.2.9 getQuietTimeSamplingInterval()

```
double AisSquareWaveVoltammetryElement::getQuietTimeSamplingInterval () const
```

Returns

samplingInterval The quiet time sampling interval to set in seconds.

17.24.2.10 getStartVoltage()

```
double AisSquareWaveVoltammetryElement::getStartVoltage () const
```

Returns

the value of the start voltage in volts.

17.24.2.11 getVStep()

```
double AisSquareWaveVoltammetryElement::getVStep () const
```

Returns

the value set for the voltage step in volts.

See also

[setVStep](#)

17.24.2.12 isAutoRange()

```
bool AisSquareWaveVoltammetryElement::isAutoRange () const
```

Returns

true if the current range is set to auto-select and false if a range has been selected.

Deprecated This function is deprecated and no longer supports auto range for this element. Specify the current using [setApproxMaxCurrent\(\)](#). The device will determine the appropriate range based on the current value.

17.24.2.13 isEndVoltageVsOCP()

```
bool AisSquareWaveVoltammetryElement::isEndVoltageVsOCP () const
```

Returns

true if the end voltage is set with respect to the open-circuit voltage and false if set against the reference terminal.

Note

if nothing is set, the default is false.

17.24.2.14 isStartVoltageVsOCP()

```
bool AisSquareWaveVoltammetryElement::isStartVoltageVsOCP () const
```

Returns

true if the start voltage is set against the open-circuit voltage and false if it is set against the reference terminal.

Note

if nothing is set, the default is false.

See also

[setStartVoltageVsOCP](#)

17.24.2.15 setAlphaFactor()

```
void AisSquareWaveVoltammetryElement::setAlphaFactor (
    double alphaFactor)
```

This is an **optional** parameter. If nothing is set, the device will use the default value of 75.

Parameters

<i>alphaFactor</i>	the value for the alphaFactor ranges from 0 to 100.
--------------------	---

17.24.2.16 setApproxMaxCurrent()

```
void AisSquareWaveVoltammetryElement::setApproxMaxCurrent (
    double approxMaxCurrent)
```

This is an **optional** parameter. If nothing is set, the device will auto-select the current range.

Parameters

<i>approxMaxCurrent</i>	the value for the maximum current expected in Amps.
-------------------------	---

17.24.2.17 setAutoRange()

```
void AisSquareWaveVoltammetryElement::setAutoRange ()
```

This option is set by default. There is no need to call this function to auto-select if the range was not manually set.

Deprecated This function is deprecated. Use [setApproxMaxCurrent\(\)](#) to specify the current range instead.

17.24.2.18 setEndVoltage()

```
void AisSquareWaveVoltammetryElement::setEndVoltage (
    double endVoltage)
```

This is the value of the voltage at which the experiment will stop.

Parameters

<i>endVoltage</i>	the value to set for the ending potential in volts.
-------------------	---

17.24.2.19 setEndVoltageVsOCP()

```
void AisSquareWaveVoltammetryElement::setEndVoltageVsOCP (
    bool endVoltageVsOcp)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>endVoltageVsOcp</i>	true to set the end voltage to be referenced against the open-circuit voltage and false if set against the reference terminal.
------------------------	--

Note

by default, this is set to false.

17.24.2.20 setPulseAmp()

```
void AisSquareWaveVoltammetryElement::setPulseAmp (  
    double pulseAmp)
```

The voltage pulse goes up in height by the given amplitude in addition to the starting potential (of the previous pulse). It then goes back down twice the amplitude to end up one amplitude below the starting potential (of the previous pulse).

Parameters

<i>pulseAmp</i>	the value to set for the pulse amplitude in volts.
-----------------	--

17.24.2.21 setPulseFreq()

```
void AisSquareWaveVoltammetryElement::setPulseFreq (  
    double pulseFreq)
```

Parameters

<i>pulseFreq</i>	the value to set for the pulse frequency in Hz.
------------------	---

17.24.2.22 setQuietTime()

```
void AisSquareWaveVoltammetryElement::setQuietTime (  
    double quietTime)
```

Parameters

<i>quietTime</i>	The quiet time duration to set in seconds.
------------------	--

17.24.2.23 setQuietTimeSamplingInterval()

```
void AisSquareWaveVoltammetryElement::setQuietTimeSamplingInterval (  
    double quietTimeSamplingInterval)
```


Parameters

<i>quietTimeSamplingInterval</i>	The quiet time sampling interval to set in seconds.
----------------------------------	---

17.24.2.24 setStartVoltage()

```
void AisSquareWaveVoltammetryElement::setStartVoltage (  
    double startVoltage)
```

Parameters

<i>startVoltage</i>	the value of the start voltage in volts
---------------------	---

17.24.2.25 setStartVoltageVsOCP()

```
void AisSquareWaveVoltammetryElement::setStartVoltageVsOCP (  
    bool startVoltageVsOcp)
```

The reference terminal is for you to connect to any reference point you like. Connect it to the working electrode to reference ground.

Parameters

<i>startVoltageVsOcp</i>	true to if the start voltage is set to reference the open-circuit voltage and false if set against the reference terminal.
--------------------------	--

Note

by default, this is set to false.

17.24.2.26 setVStep()

```
void AisSquareWaveVoltammetryElement::setVStep (  
    double vStep)
```

Parameters

<i>vStep</i>	the value for the voltage step in volts.
--------------	--

Note

Regardless of vStep's sign, the device will determine the step direction based on the start and end voltage.

The documentation for this class was generated from the following file:

- AisSquareWaveVoltammetryElement.h

17.25 AisStaircasePotentialVoltammetryElement Class Reference

[AisStaircasePotentialVoltammetryElement](#) class represents an element for staircase potential voltammetry experiments. It inherits from [AisAbstractElement](#).

```
#include <AisStaircasePotentialVoltammetryElement.h>
```

```
#include <AisStaircasePotentialVoltammetryElement.h>
```

Public Member Functions

- [AisStaircasePotentialVoltammetryElement](#) (double startVoltage, double firstVoltageLimit, double secondVoltageLimit, double endVoltage, double stepSize, double stepDuration, double samplingInterval)
Constructs an [AisStaircasePotentialVoltammetryElement](#) with specified parameters.
- [AisStaircasePotentialVoltammetryElement](#) (const [AisStaircasePotentialVoltammetryElement](#) &other)
Copy constructor for [AisStaircasePotentialVoltammetryElement](#).
- [AisStaircasePotentialVoltammetryElement](#) & operator= (const [AisStaircasePotentialVoltammetryElement](#) &other)
Assignment operator for [AisStaircasePotentialVoltammetryElement](#).
- ~[AisStaircasePotentialVoltammetryElement](#) () override
Destructor for [AisStaircasePotentialVoltammetryElement](#).
- QString [getName](#) () const override
Gets the name of the element.
- QStringList [getCategory](#) () const override
Gets the category of the element.
- double [getQuietTime](#) () const
Gets the quiet time duration.
- void [setQuietTime](#) (double quietTime)
Sets the quiet time duration.
- double [getQuietTimeSamplingInterval](#) () const
gets the quiet time sampling interval.
- void [setQuietTimeSamplingInterval](#) (double quietTimeSamplingInterval)
Sets the quiet time sampling interval.
- double [getStartVoltage](#) () const
Gets the starting voltage.
- void [setStartVoltage](#) (double startVoltage)
Sets the starting voltage.
- bool [isStartVoltageVsOCP](#) () const
Checks if the starting voltage is with respect to the open circuit mode.
- void [setStartVoltageVsOCP](#) (bool startVsOCP)
Sets whether the starting voltage is with respect to the open circuit mode.
- double [getEndVoltage](#) () const
Gets the ending voltage.
- void [setEndVoltage](#) (double endVoltage)
Sets the ending voltage.
- bool [isEndVoltageVsOCP](#) () const
Checks if the ending voltage is with respect to the open circuit mode.
- void [setEndVoltageVsOCP](#) (bool endVsOCP)
Sets whether the ending voltage is with respect to the open circuit mode.
- double [getFirstVoltageLimit](#) () const
Gets the first voltage limit.

- void [setFirstVoltageLimit](#) (double firstVoltageLimit)
Sets the first voltage limit.
- bool [isFirstVoltageLimitVsOCP](#) () const
Checks if the first voltage limit is with respect to the open circuit mode.
- void [setFirstVoltageLimitVsOCP](#) (bool firstVoltageLimitVsOCP)
Sets whether the first voltage limit is with respect to the open circuit mode.
- double [getSecondVoltageLimit](#) () const
Gets the second voltage limit.
- void [setSecondVoltageLimit](#) (double secondVoltageLimit)
Sets the second voltage limit.
- bool [isSecondVoltageLimitVsOCP](#) () const
Checks if the second voltage limit is with respect to the open circuit mode.
- void [setSecondVoltageLimitVsOCP](#) (bool secondVoltageLimitVsOCP)
Sets whether the second voltage limit is with respect to the open circuit mode.
- double [getStepSize](#) () const
Gets the potential step size.
- void [setStepSize](#) (double stepSize)
Sets the potential step size.
- double [getStepDuration](#) () const
Gets the potential step duration.
- void [setStepDuration](#) (double stepDuration)
Sets the potential step duration.
- double [getSamplingInterval](#) () const
Gets the potential sampling interval.
- void [setSamplingInterval](#) (double samplingInterval)
Sets the potential sampling interval.
- bool [isAutorange](#) () const
Checks if the experiment should autorange the current.
- void [setAutorange](#) ()
Enables autorange for the experiment.
- double [getApproxMaxCurrent](#) () const
Gets the approximate maximum current.
- void [setApproxMaxCurrent](#) (double approxMaxCurrent)
Sets the approximate maximum current.
- unsigned int [getNumberOfCycles](#) ()
get the value set for the number of cycles
- void [setNumberOfCycles](#) (unsigned int cycles)
set the number of cycles to oscillate between the first voltage-limit and the second voltage-limit.

17.25.1 Constructor & Destructor Documentation

17.25.1.1 AisStaircasePotentialVoltammetryElement() [1/2]

```
AisStaircasePotentialVoltammetryElement::AisStaircasePotentialVoltammetryElement (
    double startVoltage,
    double firstVoltageLimit,
    double secondVoltageLimit,
    double endVoltage,
    double stepSize,
    double stepDuration,
    double samplingInterval)
```

Parameters

<i>startVoltage</i>	The starting voltage in volts.
<i>firstVoltageLimit</i>	The first voltage limit in volts.
<i>secondVoltageLimit</i>	The second voltage limit in volts.
<i>endVoltage</i>	The ending voltage in volts.
<i>stepSize</i>	The potential step size in volts.
<i>stepDuration</i>	The potential step duration in seconds.
<i>samplingInterval</i>	The potential sampling interval in seconds.

17.25.1.2 AisStaircasePotentialVoltammetryElement() [2/2]

```
AisStaircasePotentialVoltammetryElement::AisStaircasePotentialVoltammetryElement (
    const AisStaircasePotentialVoltammetryElement & other)
```

Parameters

<i>other</i>	The AisStaircasePotentialVoltammetryElement to copy.
--------------	--

17.25.2 Member Function Documentation**17.25.2.1 getApproxMaxCurrent()**

```
double AisStaircasePotentialVoltammetryElement::getApproxMaxCurrent () const
```

Returns

The approximate maximum current.

17.25.2.2 getCategory()

```
QStringList AisStaircasePotentialVoltammetryElement::getCategory () const [override]
```

Returns

The category of the element.

17.25.2.3 getEndVoltage()

```
double AisStaircasePotentialVoltammetryElement::getEndVoltage () const
```

Returns

The ending voltage in volts.

17.25.2.4 getFirstVoltageLimit()

```
double AisStaircasePotentialVoltammetryElement::getFirstVoltageLimit () const
```

Returns

The first voltage limit in volts.

17.25.2.5 getName()

```
QString AisStaircasePotentialVoltammetryElement::getName () const [override]
```

Returns

The name of the element.

17.25.2.6 getNumberOfCycles()

```
unsigned int AisStaircasePotentialVoltammetryElement::getNumberOfCycles ()
```

Returns

the number of cycles set.

17.25.2.7 getQuietTime()

```
double AisStaircasePotentialVoltammetryElement::getQuietTime () const
```

Returns

The quiet time duration in seconds.

17.25.2.8 getQuietTimeSamplingInterval()

```
double AisStaircasePotentialVoltammetryElement::getQuietTimeSamplingInterval () const
```

Returns

samplingInterval The quiet time sampling interval to set in seconds.

17.25.2.9 getSamplingInterval()

```
double AisStaircasePotentialVoltammetryElement::getSamplingInterval () const
```

Returns

The potential sampling interval in seconds.

17.25.2.10 getSecondVoltageLimit()

```
double AisStaircasePotentialVoltammetryElement::getSecondVoltageLimit () const
```

Returns

The second voltage limit in volts.

17.25.2.11 getStartVoltage()

```
double AisStaircasePotentialVoltammetryElement::getStartVoltage () const
```

Returns

The starting voltage in volts.

17.25.2.12 getStepDuration()

```
double AisStaircasePotentialVoltammetryElement::getStepDuration () const
```

Returns

The potential step duration in seconds.

17.25.2.13 getStepSize()

```
double AisStaircasePotentialVoltammetryElement::getStepSize () const
```

Returns

The potential step size in volts.

17.25.2.14 isAutorange()

```
bool AisStaircasePotentialVoltammetryElement::isAutorange () const
```

Returns

True if autorange is enabled, false otherwise.

17.25.2.15 isEndVoltageVsOCP()

```
bool AisStaircasePotentialVoltammetryElement::isEndVoltageVsOCP () const
```

Returns

True if the ending voltage is with respect to the open circuit mode, false otherwise.

17.25.2.16 isFirstVoltageLimitVsOCP()

```
bool AisStaircasePotentialVoltammetryElement::isFirstVoltageLimitVsOCP () const
```

Returns

True if the first voltage limit is with respect to the open circuit mode, false otherwise.

17.25.2.17 isSecondVoltageLimitVsOCP()

```
bool AisStaircasePotentialVoltammetryElement::isSecondVoltageLimitVsOCP () const
```

Returns

True if the second voltage limit is with respect to the open circuit mode, false otherwise.

17.25.2.18 isStartVoltageVsOCP()

```
bool AisStaircasePotentialVoltammetryElement::isStartVoltageVsOCP () const
```

Returns

True if the starting voltage is with respect to the open circuit mode, false otherwise.

17.25.2.19 operator=()

```
AisStaircasePotentialVoltammetryElement & AisStaircasePotentialVoltammetryElement::operator= (
    const AisStaircasePotentialVoltammetryElement & other)
```

Parameters

<i>other</i>	The AisStaircasePotentialVoltammetryElement to assign.
--------------	--

Returns

Reference to this [AisStaircasePotentialVoltammetryElement](#).

17.25.2.20 setApproxMaxCurrent()

```
void AisStaircasePotentialVoltammetryElement::setApproxMaxCurrent (
    double approxMaxCurrent)
```

Parameters

<i>approxMaxCurrent</i>	The approximate maximum current to set.
-------------------------	---

17.25.2.21 setEndVoltage()

```
void AisStaircasePotentialVoltammetryElement::setEndVoltage (
    double endVoltage)
```

Parameters

<i>endVoltage</i>	The ending voltage to set in volts.
-------------------	-------------------------------------

17.25.2.22 setEndVoltageVsOCP()

```
void AisStaircasePotentialVoltammetryElement::setEndVoltageVsOCP (  
    bool endVsOCP)
```

Parameters

<i>endVsOCP</i>	True to set the ending voltage with respect to the open circuit mode, false otherwise.
-----------------	--

17.25.2.23 setFirstVoltageLimit()

```
void AisStaircasePotentialVoltammetryElement::setFirstVoltageLimit (  
    double firstVoltageLimit)
```

Parameters

<i>firstVoltageLimit</i>	The first voltage limit to set in volts.
--------------------------	--

17.25.2.24 setFirstVoltageLimitVsOCP()

```
void AisStaircasePotentialVoltammetryElement::setFirstVoltageLimitVsOCP (  
    bool firstVoltageLimitVsOCP)
```

Parameters

<i>firstVoltageLimitVsOCP</i>	True to set the first voltage limit with respect to the open circuit mode, false otherwise.
-------------------------------	---

17.25.2.25 setNumberOfCycles()

```
void AisStaircasePotentialVoltammetryElement::setNumberOfCycles (  
    unsigned int cycles)
```

Parameters

<i>cycles</i>	the number of cycles to set
---------------	-----------------------------

17.25.2.26 setQuietTime()

```
void AisStaircasePotentialVoltammetryElement::setQuietTime (  
    double quietTime)
```


Parameters

<i>quietTime</i>	The quiet time duration to set in seconds.
------------------	--

17.25.2.27 setQuietTimeSamplingInterval()

```
void AisStaircasePotentialVoltammetryElement::setQuietTimeSamplingInterval (
    double quietTimeSamplingInterval)
```

Parameters

<i>quietTimeSamplingInterval</i>	The quiet time sampling interval to set in seconds.
----------------------------------	---

17.25.2.28 setSamplingInterval()

```
void AisStaircasePotentialVoltammetryElement::setSamplingInterval (
    double samplingInterval)
```

Parameters

<i>samplingInterval</i>	The potential sampling interval to set in seconds.
-------------------------	--

17.25.2.29 setSecondVoltageLimit()

```
void AisStaircasePotentialVoltammetryElement::setSecondVoltageLimit (
    double secondVoltageLimit)
```

Parameters

<i>secondVoltageLimit</i>	The second voltage limit to set in volts.
---------------------------	---

17.25.2.30 setSecondVoltageLimitVsOCP()

```
void AisStaircasePotentialVoltammetryElement::setSecondVoltageLimitVsOCP (
    bool secondVoltageLimitVsOCP)
```

Parameters

<i>secondVoltageLimitVsOCP</i>	True to set the second voltage limit with respect to the open circuit mode, false otherwise.
--------------------------------	--

17.25.2.31 setStartVoltage()

```
void AisStaircasePotentialVoltammetryElement::setStartVoltage (
    double startVoltage)
```

Parameters

<i>startVoltage</i>	The starting voltage to set in volts.
---------------------	---------------------------------------

17.25.2.32 setStartVoltageVsOCP()

```
void AisStaircasePotentialVoltammetryElement::setStartVoltageVsOCP (
    bool startVsOCP)
```

Parameters

<i>startVsOCP</i>	True to set the starting voltage with respect to the open circuit mode, false otherwise.
-------------------	--

17.25.2.33 setStepDuration()

```
void AisStaircasePotentialVoltammetryElement::setStepDuration (
    double stepDuration)
```

Parameters

<i>stepDuration</i>	The potential step duration to set in seconds.
---------------------	--

17.25.2.34 setStepSize()

```
void AisStaircasePotentialVoltammetryElement::setStepSize (
    double stepSize)
```

Parameters

<i>stepSize</i>	The potential step size to set in volts.
-----------------	--

The documentation for this class was generated from the following file:

- AisStaircasePotentialVoltammetryElement.h

17.26 AisSteppedCurrentElement Class Reference

A class representing an experiment to apply the stepped current.

```
#include <AisSteppedCurrentElement.h>
```

```
#include <AisSteppedCurrentElement.h>
```

Public Member Functions

- [AisSteppedCurrentElement](#) (double startCurrent, double endCurrent, double stepSize, double stepDuration, double samplingInterval)
Constructs a Stepped Current element.
- [AisSteppedCurrentElement](#) (const [AisSteppedCurrentElement](#) &)
copy constructor for the [AisSteppedCurrentElement](#) object.
- [AisSteppedCurrentElement](#) & **operator=** (const [AisSteppedCurrentElement](#) &)
overload equal to operator for the [AisSteppedCurrentElement](#) object.
- QString [getName](#) () const override
get the name of the element.
- QStringList [getCategory](#) () const override
get a list of applicable categories of the element.
- double [getSamplingInterval](#) () const
get how frequently we are sampling the data.
- void [setSamplingInterval](#) (double samplingInterval)
set how frequently we are sampling the data.
- double [getEndCurrent](#) () const
Gets the ending current value for the stepped experiment.
- void [setEndCurrent](#) (double iEnd)
Sets the ending current value for the stepped experiment.
- double [getStepSize](#) () const
Gets the size of each current step in the stepped experiment.
- void [setStepSize](#) (double iStep)
Sets the size of each current step in the stepped experiment.
- double [getStartCurrent](#) () const
Gets the starting current value for the stepped experiment.
- void [setStartCurrent](#) (double iStart)
Sets the starting current value for the stepped experiment.
- double [getStepDuration](#) () const
Gets the duration of each current step in the stepped experiment.
- void [setStepDuration](#) (double tStep)
Sets the duration of each current step in the stepped experiment.
- double [getApproxMaxVoltage](#) () const
get the value set for the expected maximum voltage.
- void [setApproxMaxVoltage](#) (double approxMaxVoltage)
set maximum voltage expected, for manual voltage range selection.
- double [getApproxMinVoltage](#) () const
get the value set for the expected minimum voltage.
- void [setApproxMinVoltage](#) (double approxMinVoltage)
set minimum voltage expected, for manual voltage range selection.

17.26.1 Constructor & Destructor Documentation

17.26.1.1 AisSteppedCurrentElement()

```
AisSteppedCurrentElement::AisSteppedCurrentElement (
    double startCurrent,
    double endCurrent,
    double stepSize,
    double stepDuration,
    double samplingInterval) [explicit]
```

This constructor initializes the Stepped Current element with the specified parameters.

Parameters

<i>startCurrent</i>	The initial current value in amperes.
<i>endCurrent</i>	The final current value in amperes.
<i>stepSize</i>	The size of each current step in amperes.
<i>stepDuration</i>	The duration of each current step in seconds.
<i>samplingInterval</i>	The data sampling interval value in seconds.

17.26.2 Member Function Documentation

17.26.2.1 getApproxMaxVoltage()

```
double AisSteppedCurrentElement::getApproxMaxVoltage () const
```

Returns

the value set for the expected maximum Voltage in volt.

Note

if nothing was manually set, the device will auto-select the voltage range and the return value will be positive infinity.

17.26.2.2 getApproxMinVoltage()

```
double AisSteppedCurrentElement::getApproxMinVoltage () const
```

Returns

the value set for the expected maximum Voltage in volt.

Note

if nothing was manually set, the device will auto-select the voltage range and the return value will be positive infinity.

17.26.2.3 getCategory()

```
QStringList AisSteppedCurrentElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Galvanostatic Control").

17.26.2.4 getEndCurrent()

```
double AisSteppedCurrentElement::getEndCurrent () const
```

Returns

The ending current value in amperes.

17.26.2.5 getName()

```
QString AisSteppedCurrentElement::getName () const [override]
```

Returns

The name of the element: "SteppedCurrent".

17.26.2.6 getSamplingInterval()

```
double AisSteppedCurrentElement::getSamplingInterval () const
```

Returns

the data sampling interval value in seconds.

17.26.2.7 getStartCurrent()

```
double AisSteppedCurrentElement::getStartCurrent () const
```

Returns

The starting current value in amperes.

17.26.2.8 getStepDuration()

```
double AisSteppedCurrentElement::getStepDuration () const
```

Returns

The duration of each current step in seconds.

17.26.2.9 getStepSize()

```
double AisSteppedCurrentElement::getStepSize () const
```

Returns

The size of each current step in amperes.

17.26.2.10 setApproxMaxVoltage()

```
void AisSteppedCurrentElement::setApproxMaxVoltage (  
    double approxMaxVoltage)
```

This is an **optional** parameter. If nothing is set, the device will auto-select the voltage range.

Parameters

<i>approxMaxVoltage</i>	the value for the maximum current expected in V.
-------------------------	--

17.26.2.11 setApproxMinVoltage()

```
void AisSteppedCurrentElement::setApproxMinVoltage (
    double approxMinVoltage)
```

This is an **optional** parameter. If nothing is set, the device will auto-select the voltage range.

Parameters

<i>approxMinVoltage</i>	the value for the minimum current expected in V.
-------------------------	--

17.26.2.12 setEndCurrent()

```
void AisSteppedCurrentElement::setEndCurrent (
    double iEnd)
```

Parameters

<i>iEnd</i>	The ending current value in amperes.
-------------	--------------------------------------

17.26.2.13 setSamplingInterval()

```
void AisSteppedCurrentElement::setSamplingInterval (
    double samplingInterval)
```

Parameters

<i>samplingInterval</i>	the data sampling interval value in seconds.
-------------------------	--

17.26.2.14 setStartCurrent()

```
void AisSteppedCurrentElement::setStartCurrent (
    double iStart)
```

Parameters

<i>iStart</i>	The starting current value in amperes.
---------------	--

17.26.2.15 setStepDuration()

```
void AisSteppedCurrentElement::setStepDuration (
    double tStep)
```

Parameters

<i>tStep</i>	The duration of each current step in seconds.
--------------	---

17.26.2.16 setStepSize()

```
void AisSteppedCurrentElement::setStepSize (
    double iStep)
```

Parameters

<i>iStep</i>	The size of each current step in amperes.
--------------	---

Note

Regardless of iStep's sign, the device will determine the step direction based on the start and end current.

The documentation for this class was generated from the following file:

- AisSteppedCurrentElement.h

17.27 AisSteppedVoltageElement Class Reference

A class representing an experiment to apply the stepped volatge.

```
#include <AisSteppedVoltageElement.h>
```

```
#include <AisSteppedVoltageElement.h>
```

Public Member Functions

- [AisSteppedVoltageElement](#) (double startVoltage, double endVoltage, double voltageStep, double voltage↵StepDuration, double samplingInterval)
Constructor for the [AisSteppedVoltageElement](#) element.
- [AisSteppedVoltageElement](#) (const [AisSteppedVoltageElement](#) &other)
Copy constructor for the [AisSteppedVoltageElement](#) object.
- [AisSteppedVoltageElement](#) & operator= (const [AisSteppedVoltageElement](#) &other)
Overloaded assignment operator for the [AisSteppedVoltageElement](#) object.
- ~[AisSteppedVoltageElement](#) () override
Destructor for the [AisSteppedVoltageElement](#) object.
- QString [getName](#) () const override
Get the name of the element.
- QStringList [getCategory](#) () const override
Get a list of applicable categories of the element.
- double [getStartVoltage](#) () const
Get the starting voltage for the experiment.
- void [setStartVoltage](#) (double vStart)

- Set the starting voltage for the experiment.*
 - double [getEndVoltage](#) () const
- Get the ending voltage for the experiment.*
 - void [setEndVoltage](#) (double vEnd)
- Set the ending voltage for the experiment.*
 - double [getStepSize](#) () const
- Get the voltage step for each iteration.*
 - void [setStepSize](#) (double vStep)
- Set the voltage step for each iteration.*
 - double [getStepDuration](#) () const
- Get the time step for each iteration.*
 - void [setStepDuration](#) (double duration)
- Set the duration of each step.*
 - double [getSamplingInterval](#) () const
- Get the data sampling interval.*
 - void [setSamplingInterval](#) (double samplingInterval)
- Set the data sampling interval.*
 - double [getApproxMaxCurrent](#) () const
- Get the approximate maximum current.*
 - void [setApproxMaxCurrent](#) (double approxMaxCurrent)
- Set the approximate maximum current.*
 - bool [isStartVoltageVsOCP](#) () const
- Check if the experiment starts with the open circuit potential.*
 - void [setStartVoltageVsOCP](#) (bool startVsOCP)
- Set whether the experiment starts with the open circuit potential.*
 - bool [isEndVoltageVsOCP](#) () const
- Check if the experiment ends with the open circuit potential.*
 - void [setEndVoltageVsOCP](#) (bool endVsOCP)
- Set whether the experiment ends with the open circuit potential.*
 - bool [isAutoRange](#) () const
- Check if current autoranging is enabled.*
 - void [setCurrentAutorange](#) ()
- Enable current autoranging for the experiment.*

17.27.1 Constructor & Destructor Documentation

17.27.1.1 AisSteppedVoltageElement() [1/2]

```
AisSteppedVoltageElement::AisSteppedVoltageElement (
    double startVoltage,
    double endVoltage,
    double voltageStep,
    double voltageStepDuration,
    double samplingInterval) [explicit]
```

This constructor initializes the [AisSteppedVoltageElement](#) element with the specified parameters.

Parameters

<i>startVoltage</i>	The initial voltage value in volts.
<i>endVoltage</i>	The final voltage value in volts.
<i>voltageStep</i>	The size of each voltage step in volts.
<i>voltageStepDuration</i>	The duration of each voltage step in seconds.
<i>samplingInterval</i>	The data sampling interval value in seconds.

17.27.1.2 AisSteppedVoltageElement() [2/2]

```
AisSteppedVoltageElement::AisSteppedVoltageElement (  
    const AisSteppedVoltageElement & other) [explicit]
```

Parameters

<i>other</i>	The AisSteppedVoltageElement object to be copied.
--------------	---

17.27.2 Member Function Documentation

17.27.2.1 getApproxMaxCurrent()

```
double AisSteppedVoltageElement::getApproxMaxCurrent () const
```

Returns

The approximate maximum current in Amps.

17.27.2.2 getCategory()

```
QStringList AisSteppedVoltageElement::getCategory () const [override]
```

Returns

A list of applicable categories: ("Potentiostatic Control").

17.27.2.3 getEndVoltage()

```
double AisSteppedVoltageElement::getEndVoltage () const
```

Returns

The ending voltage in volts.

17.27.2.4 getName()

```
QString AisSteppedVoltageElement::getName () const [override]
```

Returns

The name of the element: "Stepped Voltage".

17.27.2.5 getSamplingInterval()

```
double AisSteppedVoltageElement::getSamplingInterval () const
```

Returns

The data sampling interval in seconds.

17.27.2.6 getStartVoltage()

```
double AisSteppedVoltageElement::getStartVoltage () const
```

Returns

The starting voltage in volts.

17.27.2.7 getStepDuration()

```
double AisSteppedVoltageElement::getStepDuration () const
```

Returns

The time step in seconds.

17.27.2.8 getStepSize()

```
double AisSteppedVoltageElement::getStepSize () const
```

Returns

The voltage step in volts.

17.27.2.9 isAutoRange()

```
bool AisSteppedVoltageElement::isAutoRange () const
```

Returns

True if current autoranging is enabled, false otherwise.

17.27.2.10 isEndVoltageVsOCP()

```
bool AisSteppedVoltageElement::isEndVoltageVsOCP () const
```

Returns

True if the experiment ends with open circuit potential, false otherwise.

17.27.2.11 isStartVoltageVsOCP()

```
bool AisSteppedVoltageElement::isStartVoltageVsOCP () const
```

Returns

True if the experiment starts with open circuit potential, false otherwise.

17.27.2.12 operator=()

```
AisSteppedVoltageElement & AisSteppedVoltageElement::operator= (
    const AisSteppedVoltageElement & other)
```

Parameters

<i>other</i>	The AisSteppedVoltageElement object to be assigned.
--------------	---

Returns

A reference to the assigned [AisSteppedVoltageElement](#) object.

17.27.2.13 setApproxMaxCurrent()

```
void AisSteppedVoltageElement::setApproxMaxCurrent (
    double approxMaxCurrent)
```

Parameters

<i>approxMaxCurrent</i>	The approximate maximum current in Amps.
-------------------------	--

17.27.2.14 setEndVoltage()

```
void AisSteppedVoltageElement::setEndVoltage (
    double vEnd)
```

Parameters

<i>vEnd</i>	The ending voltage in volts.
-------------	------------------------------

17.27.2.15 setEndVoltageVsOCP()

```
void AisSteppedVoltageElement::setEndVoltageVsOCP (
    bool endVsOCP)
```

Parameters

<i>endVsOCP</i>	True to end with open circuit potential, false otherwise.
-----------------	---

17.27.2.16 setSamplingInterval()

```
void AisSteppedVoltageElement::setSamplingInterval (
    double samplingInterval)
```

Parameters

<i>samplingInterval</i>	The data sampling interval in seconds.
-------------------------	--

17.27.2.17 setStartVoltage()

```
void AisSteppedVoltageElement::setStartVoltage (
    double vStart)
```

Parameters

<i>vStart</i>	The starting voltage in volts.
---------------	--------------------------------

17.27.2.18 setStartVoltageVsOCP()

```
void AisSteppedVoltageElement::setStartVoltageVsOCP (
    bool startVsOCP)
```

Parameters

<i>startVsOCP</i>	True to start with open circuit potential, false otherwise.
-------------------	---

17.27.2.19 setStepDuration()

```
void AisSteppedVoltageElement::setStepDuration (
    double duration)
```

Parameters

<i>duration</i>	The step duration in seconds.
-----------------	-------------------------------

17.27.2.20 setStepSize()

```
void AisSteppedVoltageElement::setStepSize (
    double vStep)
```

Parameters

<i>vStep</i>	The voltage step in volts.
--------------	----------------------------

Note

Regardless of *vStep*'s sign, the device will determine the step direction based on the start and end voltage.

The documentation for this class was generated from the following file:

- AisSteppedVoltageElement.h

Chapter 18

Examples

18.1 advancedControlFlow.cpp

This example shows how we can setup an advanced control flow for a sequence of experiments. For simplicity we will be using an incremented integer as an external condition to control the workflow. This logic can be replaced with any sort of logic pertaining to external instruments, sensors, or any other conditions.

```
#include "AisExperiment.h"
#include "AisDeviceTracker.h"
#include "AisInstrumentHandler.h"
#include "experiments/builder_elements/AisConstantCurrentElement.h"
#include "experiments/builder_elements/AisConstantPotElement.h"
#include <QCoreApplication>
#include <QTimer>
#include <qdebug.h>

int main()
{
    // Environment Setup
    char** test = nullptr;
    int args;
    QCoreApplication a(args, test); // this creates a non-GUI Qt application

    // constructing a constant potential element with required arguments
    AisConstantPotElement cvElement(
        5, // voltage: 1v
        1, // sampling interval: 1s
        10 // duration: 10s
    );

    // constructing a constant current element with required arguments
    AisConstantCurrentElement ccElement(
        0.002, // current: 2mA
        1, // sampling interval: 1s
        10 // duration: 10s
    );

    auto experimentA = std::make_shared<AisExperiment>(); // create a custom experiment
    experimentA->appendElement(cvElement, 1); // append to experimentA, the created CV element and set it to
        run 1 time

    auto experimentB = std::make_shared<AisExperiment>(); // create a second experiment
    experimentB->appendElement(ccElement, 1); // append to experimentB, the created CC element and set it to
        run 1 time

    auto experimentC = std::make_shared<AisExperiment>(); // create a third experiment
    experimentC->appendElement(cvElement, 2); // append to experimentC, the created CV element and set it to
        run 2 times

    /*
    * Now we have the experiments set up. Next we will create the logic for the sequence of experiments.
    * We will be using timers as external conditions to control the workflow. You may substitute that with
    * your own conditions.
    *
    * The following lambda function creates a logic and assigns it to the given handler.
    * We will call this function after a newDeviceConnected signal has been emitted and a handler has been
    * created.
    * The logic controls the workflow as follows:
    */
}
```

```

* # Start the first timer
* # once the timer times out, start Experiment A
* # once Experiment A completes, start the second timer
* # once the second timer times out, start Experiment B
* # start a third timer to stop Experiment B early
* # once the third timer out, stop Experiment B
* # start a fourth timer
* # once the fourth timer times out, start Experiment C
* # once Experiment C completes, start Experiment B
*/

// Lambda function
auto createLogic = [&](const AisInstrumentHandler* handler) {
    QTimer* timer1 = new QTimer(); // the first timer is used in lieu of the first external condition
    timer1->setSingleShot(true);
    timer1->start(1000);

    QObject::connect(timer1, &QTimer::timeout, [=]() {
        qDebug() << "Initial condition met. Starting Experiment A ";
        handler->uploadExperimentToChannel(0, experimentA);
        handler->startUploadedExperiment(0);

        // once the first experiment is completed (Experiment A), start the next experiment (Experiment
        B).
        // this signal will be emitted for any experiment not just A so, we will track of the sequence
        with experimentStep
        // once an experiment has completed or has been stopped, continue to the next experimentStep
        QObject::connect(handler, &AisInstrumentHandler::experimentStopped, [&](uint8_t channel) {
            static int experimentStep = 0;
            qDebug() << "Experiment Step " << experimentStep << " Completed";

            experimentStep++; //increment the experiment step
            if (experimentStep == 1) {
                // Wait for external start condition
                QTimer* timer = new QTimer(); // the timer is used in lieu of an external condition
                timer->setSingleShot(true);
                timer->start(10000); // when this timer times out, the next experiment (Experiment B)
                will start

                // Create an external condition that will stop the upcoming experiment early
                QTimer* StopEarlyTimer = new QTimer();
                StopEarlyTimer->setSingleShot(true);
                QObject::connect(StopEarlyTimer, &QTimer::timeout, [&]() {
                    qDebug() << "External early stop condition met";
                    handler->stopExperiment(0); // Once the external condition is met, experiment B will
                    stop, and the experimentCompleted signal will be emitted
                });

                QObject::connect(timer, &QTimer::timeout, [&, StopEarlyTimer]() {
                    qDebug() << "External condition met, starting experiment B";
                    handler->uploadExperimentToChannel(0, experimentB); // start Experiment B
                    handler->startUploadedExperiment(0);
                    StopEarlyTimer->start(2000);
                });
            } else if (experimentStep == 2) {
                QTimer* timer = new QTimer(); // the timer is used in lieu of an external condition
                timer->setSingleShot(true);
                timer->start(10000); // when this timer times out, the next experiment (Experiment C)
                will start

                QObject::connect(timer, &QTimer::timeout, [&]() {
                    qDebug() << "External condition met, starting Experiment C ";
                    handler->uploadExperimentToChannel(0, experimentC); // start Experiment C
                    handler->startUploadedExperiment(0);
                });
            } else if (experimentStep == 3) {
                QTimer* timer = new QTimer(); // the timer is used in lieu of an external condition
                timer->setSingleShot(true);
                timer->start(10000); // when this timer times out, the next experiment (Experiment B)
                will start

                QObject::connect(timer, &QTimer::timeout, [&]() {
                    qDebug() << "External condition met, starting Experiment B ";
                    handler->uploadExperimentToChannel(0, experimentB); // start Experiment B
                    handler->startUploadedExperiment(0);
                });
            }
        });
    });
};

// this is a signal-slot connection where the slot assigns the logic to the device handler when
newDeviceConnected signal is emitted.
auto tracker = AisDeviceTracker::Instance(); // create a tracker that tracks connected devices
QObject::connect(tracker, &AisDeviceTracker::newDeviceConnected, &a, [&](const QString& deviceName) {
    auto& handler = tracker->getInstrumentHandler(deviceName);
    createLogic(&handler); // controlling experiments is to be done only after a device handler has been

```



```

    assigned. That is why it is placed inside this slot.
});

// here we have a signal and slot for when a device has been disconnected
QObject::connect(tracker, &AisDeviceTracker::deviceDisconnected, &a, [=](const QString& deviceName) {
    qDebug() << deviceName << "is disconnected ";
});

AisErrorCode error = tracker->connectToDeviceOnComPort("COM18"); // change the port number to your
device. For example, in windows, you can find it from the device manager
if (error != error.Success) {
    qDebug() << error.message();
}
a.exec();
}

```

18.2 advancedExperiment.cpp

```

#include "AisDeviceTracker.h"
#include "AisExperiment.h"
#include "AisInstrumentHandler.h"

#include "experiments/builder_elements/AisOpenCircuitElement.h"
#include "experiments/builder_elements/AisConstantPotElement.h"
#include "experiments/builder_elements/AisEISGalvanostaticElement.h"
#include "experiments/builder_elements/AisConstantCurrentElement.h"

#include <QCoreApplication>
#include <QDebug>

// Define relevant device information, for easy access
#define COMPORT "COM5"
#define CHANNEL 0
#define INSTRUMENT_NAME "PLus1366"

int main()
{
    char** test = nullptr;
    int args;

    QCoreApplication a(args, test);

    auto tracker = AisDeviceTracker::Instance();

    bool success = true;
    auto customExperiment = std::make_shared<AisExperiment>();

    AisOpenCircuitElement ocpElement(1, 10);
    success &= customExperiment->appendElement(ocpElement);

    int voltage = 0;
    for (int i = 0; i < 4; i++) {
        AisConstantPotElement cvElement(voltage, 0.1, 5);
        success &= customExperiment->appendElement(cvElement, 1);
        voltage += 0.1; // Adding 100 mV
    }

    AisExperiment eisSubExperiment;

    AisEISGalvanostaticElement galvEISElement(10, 10000, 10, 0.01, 0.1);
    AisOpenCircuitElement ocpElement2(1, 5);

    success &= eisSubExperiment.appendElement(galvEISElement, 1);
    success &= eisSubExperiment.appendElement(ocpElement2, 1);

    success &= customExperiment->appendSubExperiment(eisSubExperiment, 3);

    AisConstantCurrentElement ccElement(0.1, 1, 10);
    success &= customExperiment->appendElement(ccElement, 2);

    if (!success) {
        qDebug() << "Error building experiment";
        return 0;
    }

    auto connectSignals = [=](const AisInstrumentHandler& handler) {
        QObject::connect(&handler, &AisInstrumentHandler::activeDCDataReady, [=](uint8_t channel, const
        AisDCData& data) {
            qDebug() << "Timestamp: " << data.timestamp << " Current: " << data.current << " Voltage: " <<
            data.workingElectrodeVoltage << " CE Voltage : " << data.counterElectrodeVoltage;
        });
        QObject::connect(&handler, &AisInstrumentHandler::experimentNewElementStarting, [=](uint8_t channel,
        const AisExperimentNode& info) {

```

```

        qDebug() << "New element starting: " << info.stepName;
    });
    QObject::connect(&handler, &AisInstrumentHandler::experimentStopped, [=](uint8_t channel, const
QString& reason) {
        qDebug() << "Experiment Stopped Signal " << channel << "Reason : " << reason;
    });
    QObject::connect(&handler, &AisInstrumentHandler::deviceError, [=](uint8_t channel, const QString&
error) {
        qDebug() << "Device Error: " << error;
    });
};

// When device is connected, setup connections, and upload/start the experiment
QObject::connect(tracker, &AisDeviceTracker::newDeviceConnected, [=](const QString& deviceName) {
    qDebug() << "New Device Connected: " << deviceName;
    auto& handler = tracker->getInstrumentHandler(INSTRUMENT_NAME);

    connectSignals(handler);

    AisErrorCode error = handler.uploadExperimentToChannel(CHANNEL, customExperiment);
    if (error) {
        qDebug() << error.message();
        return 0;
    }

    error = handler.startUploadedExperiment(CHANNEL);
    if (error) {
        qDebug() << error.message();
        return 0;
    }
});

AisErrorCode error = tracker->connectToDeviceOnComPort(COMPORT);
if (error != error.Success) {
    qDebug() << error.message();
    return 0;
}

return a.exec();
}

```

18.3 basicExperiment.cpp

```

#include "AisDeviceTracker.h"
#include "AisExperiment.h"
#include "AisInstrumentHandler.h"

#include "experiments/builder_elements/AisConstantPotElement.h"

#include <QCoreApplication>
#include <QDebug>

// Define relevant device information, for easy access
#define COMPORT "COM1"
#define CHANNEL 0

int main()
{
    char** test = nullptr;
    int args;

    QCoreApplication a(args, test);

    auto tracker = AisDeviceTracker::Instance();

    //      Voltage = 1V, Sampling Interval = 1s, Duration = 30s
    AisConstantPotElement cvElement(1, 1, 30);

    auto customExperiment = std::make_shared<AisExperiment>();
    // Append the constant potential element, and tell it to execute that element 1 time
    customExperiment->appendElement(cvElement, 1);
    auto connectSignals = [=](const AisInstrumentHandler& handler) {
        QObject::connect(&handler, &AisInstrumentHandler::activeDCDataReady, [=](uint8_t channel, const
AisDCData& data) {
            qDebug() << "Timestamp: " << data.timestamp << " Current: " << data.current << " Voltage: " <<
data.workingElectrodeVoltage << " CE Voltage : " << data.counterElectrodeVoltage;
        });
        QObject::connect(&handler, &AisInstrumentHandler::activeACDataReady, [=](uint8_t channel, const
AisACData& data) {
            qDebug() << "Timestamp: " << data.timestamp << " Frequency: " << data.frequency << " " <<
data.absoluteImpedance;
        });
    };
}

```

```

// Whenever a new node in the element starts, note: some Ais Elements contain multiple logical nodes
// i.e AisCyclicVoltammetryElement contains 4 nodes for each linear segment of each cycle plus a
quiet time node if enabled
// So this lambda would be executed atleast 4 times for each cycle
QObject::connect(&handler, &AisInstrumentHandler::experimentNewElementStarting, [=](uint8_t channel,
const AisExperimentNode& info) {
    qDebug() << "New element starting: " << info.stepName;
});
// Whenever an experiment completes or is manually stopped, this will execute
QObject::connect(&handler, &AisInstrumentHandler::experimentStopped, [=](uint8_t channel, const
QString& reason) {
    qDebug() << "Experiment Stopped Signal " << channel << "Reason : " << reason;
});
QObject::connect(&handler, &AisInstrumentHandler::deviceError, [=](uint8_t channel, const QString&
error) {
    qDebug() << "Device Error: " << error;
});
};
QObject::connect(tracker, &AisDeviceTracker::deviceDisconnected, [=](const QString& deviceName) {
    qDebug() << "New Device Connected: " << deviceName;
});
QObject::connect(tracker, &AisDeviceTracker::newDeviceConnected, [=](const QString& deviceName) {
    qDebug() << "New Device Connected: " << deviceName;
    auto& handler = tracker->getInstrumentHandler(deviceName);

    connectSignals(handler);

    AisErrorCode error = handler.uploadExperimentToChannel(CHANNEL, customExperiment);
    if (error) {
        qDebug() << error.message();
        return;
    }

    // Start the previously uploaded experiment on the same channel
    error = handler.startUploadedExperiment(CHANNEL);

    // Exit the application if there is any error starting the experiment
    if (error) {
        qDebug() << error.message();
        return;
    }
});
AisErrorCode error = tracker->connectToDeviceOnComPort(COMPORT);
if (error != error.Success) {
    qDebug() << error.message();
    return 0;
}

// Returning a.exec() executes the event loop, which continues running until the application is exited
return a.exec();
}

```

18.4 dataOutput.cpp

This example shows how to capture data from the device, and write it to a CSV file using Qt's QFile and QTextStream classes.

```

#include "AisExperiment.h"
#include "AisDeviceTracker.h"
#include "AisInstrumentHandler.h"
#include "experiments/builder_elements/AisConstantCurrentElement.h"
#include "experiments/builder_elements/AisConstantPotElement.h"

#include <QCoreApplication>
#include <QTimer>
#include <QDebug>
#include <QFile>
#include <QStandardPaths>

// Define relevant device information, for easy access
#define COMPORT "COM1"
#define CHANNEL 0

int main()
{
    char** test = nullptr;
    int args;
    QCoreApplication a(args, test);

    auto tracker = AisDeviceTracker::Instance();

    // Build the experiment

```

```

AisConstantPotElement cvElement(
    1,      // voltage: 1v
    1,      // sampling interval: 1s
    30      // duration: 30s
);
AisConstantCurrentElement ccElement(
    0.001, // current: 1mA
    1,     // sampling interval: 1s
    60     // duration: 60s
);
auto customExperiment = std::make_shared<AisExperiment>();
customExperiment->appendElement(ccElement, 1);
customExperiment->appendElement(cvElement, 1);

// Static QString variable to store the current elements file path
static QString filePath;

// Create a lambda function to connect signals to the handler
auto connectSignals = [=](const AisInstrumentHandler& handler) {
    QObject::connect(&handler, &AisInstrumentHandler::experimentNewElementStarting, [=](uint8_t channel,
        const AisExperimentNode& node) {
        // Create a unique file name for each element, with the format
        "stepNumber_stepName_expStartTime.csv"
        static int fileNum = 1;
        auto name = "/" + QString::number(fileNum) + "_" + QString::number(node.stepNumber) + "_" +
        node.stepName + ".csv";
        filePath = QString(QStandardPaths::writableLocation(QStandardPaths::DesktopLocation)) + name;

        QFile file(filePath);
        if (!file.open(QIODevice::WriteOnly | QIODevice::Text))
            return;

        // Writing headers to file
        QTextStream out(&file);
        out << "Time Stamp,"
            << "Counter Electrode Voltage,"
            << "Working Electrode Voltage,"
            << "Current"
            << "\n";
        file.close();

        qDebug() << "New element beginning: " << node.stepName << "step: " << node.stepNumber;
    });

    QObject::connect(&handler, &AisInstrumentHandler::activeDCDataReady, [=](uint8_t channel, const
    AisDCData& data) {
        qDebug() << "current : " << data.current << " voltage: " << data.workingElectrodeVoltage << "
        counter electrode : " << data.counterElectrodeVoltage << " timestamp : " << data.timestamp;

        // Save the DC data to the file created at element beginning
        QFile file(filePath);
        if (!file.open(QIODevice::Append | QIODevice::WriteOnly | QIODevice::Text))
            return;
        QTextStream out(&file);
        out << data.timestamp << ", "
            << data.counterElectrodeVoltage << ", "
            << data.workingElectrodeVoltage << ", "
            << data.current
            << "\n";
        file.close();
    });

    // The experiment created only uses DC elements, so this will not be executed.
    // However, ac data can be handled and written to a csv like seen above with dc data
    QObject::connect(&handler, &AisInstrumentHandler::activeACDataReady, [=](uint8_t channel, const
    AisACData& data) {
        qDebug() << data.frequency << " " << data.absoluteImpedance << " " << data.phaseAngle;
    });

    QObject::connect(&handler, &AisInstrumentHandler::experimentStopped, [=](uint8_t channel, const
    QString& reason) {
        qDebug() << "Experiment Completed Signal " << channel << "Reason : " << reason;
    });
};

// When device is connected, setup connections, and upload/start the experiment
QObject::connect(tracker, &AisDeviceTracker::newDeviceConnected, [=](const QString& deviceName) {
    auto& handler = tracker->getInstrumentHandler(deviceName);

    connectSignals(handler);

    AisErrorCode error = handler.uploadExperimentToChannel(0, customExperiment);
    if (error) {
        qDebug() << error.message();
        return 0;
    }
};

```

```

    }

    error = handler.startUploadedExperiment (CHANNEL);
    if (error) {
        qDebug() << error.message();
        return 0;
    }
});

AisErrorCode error = tracker->connectToDeviceOnComPort (COMPORT);
if (error != error.Success) {
    qDebug() << error.message();
    return 0;
}

return a.exec();
}

```

18.5 firmwareUpdate.cpp

```

#include "AisInstrumentHandler.h"
#include "AisDeviceTracker.h"

#include <QCoreApplication>
#include <QThread>
#include <QDebug>

#define COMPORT "COM1"

int main()
{
    int args;
    QCoreApplication a(args, nullptr);

    auto tracker = AisDeviceTracker::Instance();

    QObject::connect(tracker, &AisDeviceTracker::firmwareUpdateNotification, [=](const QString& message) {
        qDebug() << message;
        if (message.contains("firmware is updated.")) {
            // Now instrument is ready to go
        }
    });

    // Attempt to connect to the device
    auto error = tracker->connectToDeviceOnComPort (COMPORT);
    if (error == AisErrorCode::FirmwareNotSupported) {
        error = tracker->updateFirmwareOnComPort (COMPORT);

        // Some other error occurred
        if (error != AisErrorCode::Success) {
            qDebug() << "Error: " << error.message();
        }
    } else if (error != AisErrorCode::Success) {
        qDebug() << "Error: " << error.message();
    } else {
        qDebug() << "Device firmware is up to date";
    }

    return a.exec();
}

```

18.6 linkedChannels.cpp

This example will show you how linked channels can be used to combine the multiple channels on a device, in order to amplify the output for a single experiment. [AisInstrumentHandler::setLinkedChannels](#) MUST be called before each experiment that uses paralleled channels. Once linked, these channels must be controlled by ONLY the master channel, which is returned by the [AisInstrumentHandler::setLinkedChannels](#) function.

Note

This feature is only available on Cyclar models.

```
#include "AisDeviceTracker.h"
#include "AisExperiment.h"
#include "AisInstrumentHandler.h"

#include "experiments/builder_elements/AisConstantCurrentElement.h"

#include <QCoreApplication>
#include <QDebug>

// Define relevant device information, for easy access
#define COMPORT "COM1"

int main()
{
    char** test = nullptr;
    int args;

    QCoreApplication a(args, test);

    auto tracker = AisDeviceTracker::Instance();

    // Create an experiment
    AisConstantCurrentElement ccElement(10, 1, 30);
    auto customExperiment = std::make_shared<AisExperiment>();
    customExperiment->appendElement(ccElement, 1);

    auto connectSignals = [=](const AisInstrumentHandler& handler) {
        QObject::connect(&handler, &AisInstrumentHandler::activeDCDataReady, [=](uint8_t channel, const
        AisDCData& data) {
            qDebug() << "Timestamp: " << data.timestamp << " Current: " << data.current << " Voltage: " <<
            data.workingElectrodeVoltage << " CE Voltage : " << data.counterElectrodeVoltage;
        });
        QObject::connect(&handler, &AisInstrumentHandler::deviceError, [=](uint8_t channel, const QString&
        error) {
            qDebug() << "Device Error: " << error;
        });
    };

    QObject::connect(tracker, &AisDeviceTracker::newDeviceConnected, [=](const QString& deviceName) {
        qDebug() << "New Device Connected: " << deviceName;

        auto& handler = tracker->getInstrumentHandler(deviceName);

        // Here we want to link channels 0 and 1 together, so we pass in a vector of the channels to link
        // It will return which of the channels is the master channel, this should be used to control the
        experiment
        int8_t masterChannel = handler.setLinkedChannels({ 0, 1 });

        connectSignals(handler);

        AisErrorCode error = handler.uploadExperimentToChannel(masterChannel, customExperiment);
        if (error) {
            qDebug() << error.message();
        }

        // Start the previously uploaded experiment on the master channel
        error = handler.startUploadedExperiment(masterChannel);
        if (error) {
            qDebug() << error.message();
            return 0;
        }
    });

    AisErrorCode error = tracker->connectToDeviceOnComPort (COMPORT);
    if (error != error.Success) {
        qDebug() << error.message();
        return 0;
    }
    return a.exec();
}
```

18.7 manualExperiment.cpp

```
#include "AisDeviceTracker.h"
#include "AisInstrumentHandler.h"

#include <QCoreApplication>
```

```

#include <QDebug>

#include <QTimer> // For QTimer::singleShot

// Define relevant device information, for easy access
#define COMPORT "COM1"
#define CHANNEL 0

int main()
{
    char** test = nullptr;
    int args;

    QCoreApplication a(args, test);

    auto tracker = AisDeviceTracker::Instance();

    // Create a lambda function to connect signals to the handler
    auto connectSignals = [=](const AisInstrumentHandler& handler) {
        QObject::connect(&handler, &AisInstrumentHandler::activeDCDataReady, [=](uint8_t channel, const
        AisDCData& data) {
            qDebug() << "Timestamp: " << data.timestamp << " Current: " << data.current << " Voltage: " <<
            data.workingElectrodeVoltage << " CE Voltage : " << data.counterElectrodeVoltage;
        });
        QObject::connect(&handler, &AisInstrumentHandler::deviceError, [=](uint8_t channel, const QString&
        error) {
            qDebug() << "Device Error: " << error;
        });
        QObject::connect(&handler, &AisInstrumentHandler::experimentStopped, [=](uint8_t channel, const
        QString& reason) {
            qDebug() << reason;
            qDebug() << "Experiment has ended. Closing application.";
            QCoreApplication::quit();
        });
    };

    // Create a lambda function to run the experiment
    auto runExperiment = [=](const AisInstrumentHandler& handler) {
        //The default starting mode is always Open Circuit Potential
        qDebug() << "Starting manual mode at open circuit potential";
        AisErrorCode error = handler.startManualExperiment(CHANNEL);
        if (error != AisErrorCode::Success) {
            qDebug() << error.message();
            QCoreApplication::quit();
        }

        // In this section we connect singleshot QTimer to lambda functions that call our mode changing
        functions.
        // These lambdas are called asynchronously when the QTimer expires.

        // 5 seconds after starting experiment, change to Constant Current at .1A
        QTimer::singleShot(5000, [=, &handler]() {
            qDebug() << "Switching to constant current at .1A";
            AisErrorCode error = handler.setManualModeConstantCurrent(CHANNEL, .1);
            if (error != AisErrorCode::Success) {
                qDebug() << error.message();
            }
        });

        // 15 seconds after starting experiment, change to Constant Voltage at 1V
        QTimer::singleShot(15000, [=, &handler]() {
            qDebug() << "Switching to constant voltage at 1V";
            AisErrorCode error = handler.setManualModeConstantVoltage(CHANNEL, 1);
            if (error != AisErrorCode::Success) {
                qDebug() << error.message();
            }
        });

        // 25 seconds after starting experiment, change back into Open Circuit Potential mode.
        QTimer::singleShot(25000, [=, &handler]() {
            qDebug() << "Switching to open circuit potential";
            AisErrorCode error = handler.setManualModeOCP(CHANNEL);
            if (error != AisErrorCode::Success) {
                qDebug() << error.message();
            }
        });

        // Stop the experiment after 30 seconds
        QTimer::singleShot(30000, [=, &handler]() {
            qDebug() << "Stopping experiment";
            if (handler.stopExperiment(CHANNEL) != AisErrorCode::Success) {
                qDebug() << error.message();
            }
        });
    };

    QObject::connect(tracker, &AisDeviceTracker::newDeviceConnected, [=](const QString& deviceName) {

```

```

qDebug() << "New Device Connected: " << deviceName;

// When instrument is connected, grab the handler and setup/start the experiment
auto& handler = tracker->getInstrumentHandler(deviceName);

connectSignals(handler);
runExperiment(handler);
});

AisErrorCode error = tracker->connectToDeviceOnComPort(COMPORT);
if (error != error.Success) {
    qDebug() << error.message();
    return 0;
}

return a.exec();
}

```

18.8 nonblockingExperiment.cpp

This non blocking example show how we can start and run an experiment, while still being able to process other events. Say there is other logic that needs to be processed while an experiment is running, this example shows how we can continue to process events while our experiment and relevant logic is running. The Admiral Instruments API gives more control of the device and gives you the tools to integrate running our experiments in your pipeline and automating your workflow. Our API lets you programmatically start an experiment, pause an experiment and stop an experiment. For example, you may want to start an experiment, and stop after certain time, and exit the program.

Below is an example for start a manual experiment, stop after 25 s, and exit the program after.

```

#include "AisExperiment.h"
#include "AisDeviceTracker.h"
#include "AisInstrumentHandler.h"
#include "experiments/builder_elements/AisConstantCurrentElement.h"
#include "experiments/builder_elements/AisConstantPotElement.h"
#include <QCoreApplication>
#include <QDebug>
#include <QTimer>

// Define relevant device information, for easy access
#define COMPORT "COM1"
#define CHANNEL 0
#define INSTRUMENT_NAME "Plus2001"

int main()
{
    char** test = nullptr;
    int args;
    QCoreApplication a(args, test);

    auto tracker = AisDeviceTracker::Instance();

    auto createLogic = [=] (const AisInstrumentHandler* handler) {
        QObject::connect(handler, &AisInstrumentHandler::activeDCDataReady, [=](uint8_t channel, const
        AisDCData& data) {
            qDebug() << "Timestamp : " << data.timestamp << " Current : " << data.current << " Voltage: " <<
            data.workingElectrodeVoltage << " CE Voltage: " << data.counterElectrodeVoltage ;
        });

        QObject::connect(handler, &AisInstrumentHandler::activeACDataReady, [=](uint8_t channel, const
        AisACData& data) {
            qDebug() << "Frequency: " << data.frequency << " Absolute Impedance:" << data.absoluteImpedance << "
            Phase Angle" << data.phaseAngle;
        });

        QObject::connect(handler, &AisInstrumentHandler::experimentNewElementStarting, [=](uint8_t channel,
        const AisExperimentNode& nodeInfo) {
            qDebug() << "New Node begining ";
        });

        QObject::connect(handler, &AisInstrumentHandler::experimentStopped, [=](uint8_t channel, const
        QString& reason) {
            qDebug() << "Experiment Completd Signal " << channel << "Reason: " << reason;
        });
    };

    QObject::connect(tracker, &AisDeviceTracker::newDeviceConnected, [=, &a](const QString& deviceName) {

```



```

qDebug() << "New Device Connected: " << deviceName;

auto& handler = tracker->getInstrumentHandler(INSTRUMENT_NAME);

createLogic(&handler);

auto error = handler.startManualExperiment(CHANNEL);
if (error) {
    qDebug() << error.message();
    return;
}
error = handler.setManualModeSamplingInterval(CHANNEL, 2);
if (error) {
    qDebug() << error.message();
    return;
}

error = handler.setManualModeConstantVoltage(CHANNEL, 2);
if (error) {
    qDebug() << error.message();
    return;
}
QTimer::singleShot(25000, [=, &handler]() {
    auto error = handler.stopExperiment(CHANNEL);
    if (error) {
        qDebug() << error.message();
        return;
    }
});

// Allow the event loop to process other events while the current experiment is running
// isChannelBusy will return true while the channel is running an experiment
while (handler.isChannelBusy(CHANNEL)) {
    a.processEvents();
}

// Process any remaining events
a.processEvents();
});

auto connectdevices = tracker->connectAllPluggedInDevices();
if (connectdevices == 0)
{
    qDebug() << "No devices connected";
    return 0;
}

return 0;
}

```

18.9 pulseData.cpp

This example demonstrates how to use the [AisDataManipulator](#) class to calculate advanced parameters from raw DC data recieved from pulse experiments.

```

#include "AisDeviceTracker.h"
#include "AisInstrumentHandler.h"
#include "AisExperiment.h"
#include "AisDataManipulator.h"

#include "experiments/builder_elements/AisDiffPulseVoltammetryElement.h"

#include <QCoreApplication>
#include <QTimer>
#include <QDebug>

// Define relevant device information, for easy access
#define COMPORT "COM1"
#define CHANNEL 0

int main()
{
    char** test = nullptr;
    int args;
    QCoreApplication a(args, test);
    auto tracker = AisDeviceTracker::Instance();

    // Create the AisDiffPulseVoltammetryElement element and add it to the experiment
    std::shared_ptr<AisExperiment> experiment = nullptr;
    AisDiffPulseVoltammetryElement dpv_element(-0.4, 0.5, 0.01, 0.1, 0.1, 0.45, 0.1);
    dpv_element.setStartVoltageVsOCP(false);

```

```

dpv_element.setEndVoltageVsOCP(false);
experiment = std::make_shared<AisExperiment>();
experiment->appendElement(dpv_element, 1);

// Create an `AisDataManipulator` class for calculating advance parameters.
std::shared_ptr<AisDataManipulator> dataManipulator = std::make_shared<AisDataManipulator>(dpv_element);
//Pass the pulse element to the manipulator

auto connectSignals = [=](const AisInstrumentHandler* handler) {
    QObject::connect(handler, &AisInstrumentHandler::activeDCDataReady, [=](uint8_t channel, const
    AisDCData& data) {
        auto utcTime = handler->getExperimentUTCStartTime(0);

        dataManipulator->loadPrimaryData(data);

        if (dataManipulator->isPulseCompleted()) {
            static bool writeheader = true;

            if (writeheader) {
                qDebug() << "time"
                    << ", Pulse_current"
                    << ", base_current"
                    << ", diff_current"
                    << ", base_voltage"
                    << ", pulse_volatge";
                writeheader = false;
            }
            qDebug() << data.timestamp << ", " << dataManipulator->getPulseCurrent() << ", " <<
            dataManipulator->getBaseCurrent() << ", " << dataManipulator->getPulseCurrent() -
            dataManipulator->getBaseCurrent() << ", " << dataManipulator->getBaseVoltage() << ", " <<
            dataManipulator->getPulseVoltage();
        }
    });

    QObject::connect(handler, &AisInstrumentHandler::activeACDataReady, [=](uint8_t channel, const
    AisACData& data) {
        qDebug() << "channel: " << (int)channel << "frequency: " << data.frequency << " absoluteImpedance:
        " << data.absoluteImpedance << " phaseAngle: " << data.phaseAngle << " timestamp: " << data.timestamp;
    });

    QObject::connect(handler, &AisInstrumentHandler::experimentNewElementStarting, [=](uint8_t channel,
    const AisExperimentNode& data) {
        qDebug() << "New Node beginning " << data.stepName << " step number " << data.stepNumber << " step
        sub: " << data.substepNumber;
    });

    QObject::connect(handler, &AisInstrumentHandler::experimentStopped, [=](uint8_t channel, const
    QString& reason) {
        qDebug() << "Experiment Completed Signal " << channel << "Reason: " << reason;
    });
    QObject::connect(handler, &AisInstrumentHandler::experimentPaused, [=](uint8_t channel) {
        qDebug() << "Experiment Paused " << channel;
    });
    QObject::connect(handler, &AisInstrumentHandler::experimentResumed, [=](uint8_t channel) {
        qDebug() << "Experiment Resume " << channel;
    });
});

QObject::connect(tracker, &AisDeviceTracker::newDeviceConnected, &a, [=](const QString& deviceName) {
    auto& handler = tracker->getInstrumentHandler(deviceName);
    connectSignals(&handler);
    {
        std::shared_ptr<AisExperiment> deleteExp = std::make_shared<AisExperiment>(*experiment);
        auto error = handler.uploadExperimentToChannel(CHANNEL, deleteExp);
        if (error) {
            qDebug() << "Error: " << error.message();
        }
    }
    auto error = handler.startUploadedExperiment(0);
    if (error) {
        qDebug() << "Error: " << error.message();
    }
});

QObject::connect(tracker, &AisDeviceTracker::deviceDisconnected, &a, [=](const QString& deviceName) {
    qDebug() << deviceName << "is disconnected ";
});

auto error = tracker->connectToDeviceOnComPort(COMPORT);
if (error) {
    qDebug() << "Error: " << error.message();
}

a.exec();
}

```

18.10 advancedExperiment.py

```

00001 """! @example advancedExperiment.py """
00002 #! [Setup]
00003 import sys
00004 from PySide6.QtCore import QApplication
00005 from SquidstatPyLibrary import AisDeviceTracker, AisExperiment, AisInstrumentHandler, AisErrorCode,
00006     AisOpenCircuitElement, AisConstantPotElement, AisConstantCurrentElement, AisEISGalvanostaticElement
00007 #Define relavant device information, for easy access
00008 COMPORT = "COM1"
00009 CHANNEL = 0
00010 INSTRUMENT_NAME = "Plus2000"
00011
00012 app = QApplication()
00013
00014 tracker = AisDeviceTracker.Instance()
00015
00016 success = True
00017
00018 customExperiment = AisExperiment()
00019 # Step 1
00020
00021 ocpElement = AisOpenCircuitElement(1, 10)
00022 success &= customExperiment.appendElement(ocpElement)
00023
00026 voltage = 0
00027 for i in range(0, 4):
00028     cvElement = AisConstantPotElement(voltage, 0.1, 5)
00029     success &= customExperiment.appendElement(cvElement, 1)
00030     voltage = voltage + 0.1
00031
00034 eisSubExperiment = AisExperiment()
00035
00036 galvEISElement = AisEISGalvanostaticElement(10, 10000, 10, 0.01, 0.1)
00037 ocpElement2 = AisOpenCircuitElement(1, 5)
00038
00039 success &= eisSubExperiment.appendElement(galvEISElement, 1)
00040 success &= eisSubExperiment.appendElement(ocpElement2, 1)
00041
00042 success &= customExperiment.appendSubExperiment(eisSubExperiment, 3)
00043
00046 ccElement = AisConstantCurrentElement(0.1, 1, 10)
00047 success &= customExperiment.appendElement(ccElement, 2)
00048
00049 if not success:
00050     print("Error building experiment")
00051     sys.exit()
00052
00053
00054 # When device is connected, setup connections, and upload/start the experiment
00055 def connectSignals(handler):
00056     handler.activeDCDataReady.connect(lambda channel, data: print(f"Timestamp: {data.timestamp}
00057         Current: {data.current} Voltage: {data.workingElectrodeVoltage} CE Voltage :
00058         {data.counterElectrodeVoltage}"))
00059     handler.activeACDataReady.connect(lambda channel, data: print(f"Timestamp: {data.timestamp}
00060         Frequency: {data.frequency} Absolute Impedance: {data.absoluteImpedance}"))
00061     handler.experimentNewElementStarting.connect(lambda channel, info: print(f"New element starting:
00062         {info.stepName}"))
00063     handler.experimentStopped.connect(lambda channel, reason: (print(f"Experiment Stopped Signal
00064         {channel}, {reason}"), app.quit()))
00065     handler.deviceError.connect(lambda channel, error: print(f"Device Error: {error}"))
00066
00067
00068 def startExperiment():
00069     handler = tracker.getInstrumentHandler(INSTRUMENT_NAME)
00070
00071     connectSignals(handler)
00072
00073     error = handler.uploadExperimentToChannel(CHANNEL, customExperiment)
00074     if error.value() != AisErrorCode.Success:
00075         print(error.message())
00076         app.quit()
00077
00078     error = handler.startUploadedExperiment(CHANNEL)
00079     if error.value() != AisErrorCode.Success:
00080         print(error.message())
00081         app.quit()
00082
00083 tracker.newDeviceConnected.connect(startExperiment)
00084
00085 error = tracker.connectToDeviceOnComPort(COMPORT)
00086 if error.value() != AisErrorCode.Success:
00087     print(error.message())
00088     sys.exit()
00089
00090

```

```
00086 sys.exit(app.exec())
```

18.11 async.py

```
00001 """! @example async.py """
00002 import sys
00003 import struct
00004 import asyncio
00005 from PySide6.QtCore import QApplication
00006 from SquidstatPyLibrary import AisDeviceTracker
00007 from SquidstatPyLibrary import AisCompRange
00008 from SquidstatPyLibrary import AisDCData
00009 from SquidstatPyLibrary import AisACData
00010 from SquidstatPyLibrary import AisExperimentNode
00011 from SquidstatPyLibrary import AisErrorCode
00012 from SquidstatPyLibrary import AisExperiment
00013 from SquidstatPyLibrary import AisInstrumentHandler
00014 from SquidstatPyLibrary import AisConstantPotElement
00015 from SquidstatPyLibrary import AisEISPotentiostaticElement
00016 from SquidstatPyLibrary import AisConstantCurrentElement
00017
00018 # initialize the application
00019 app = QApplication([])
00020
00021 # Add delay before quitting
00022 async def delayed_quit():
00023     await asyncio.sleep(5)
00024     app.quit()
00025
00026 # function to print when experiment has completed
00027 def experiment_complete_handler(channel, reason):
00028     print(f"Experiment Completed on channel {channel} , {reason}")
00029     asyncio.run(delayed_quit())
00030
00031
00032 # main function setup as async
00033 async def main():
00034     # initialize a device tracker
00035     tracker = AisDeviceTracker.Instance()
00036     # connect to device associated with the tracker
00037     # print device serial number
00038     tracker.newDeviceConnected.connect(lambda deviceName: print(f"Connected Device: {deviceName}"))
00039     # connect to device on com port 4
00040     error = tracker.connectToDeviceOnComPort("COM4")
00041     if error.value() != AisErrorCode.Success:
00042         print(error.message())
00043         app.quit()
00044
00045     # Add initial delay before asking for handler (5 s)
00046     await asyncio.sleep(5)
00047
00048     # use serial number to get handler for instrument
00049     handler = tracker.getInstrumentHandler("Cycler1518")
00050     # manages DC data input and output
00051     # add more variables if you want to print more data to the console
00052     handler.activeDCDataReady.connect(lambda channel, data: print("timestamp:",
"{:.9f}".format(data.timestamp), "workingElectrodeVoltage: ",
"{:.9f}".format(data.workingElectrodeVoltage)))
00053     # manages AC data input and output
00054     # add more variables if you want to print more data to the console
00055     handler.activeACDataReady.connect(lambda channel, data: print("frequency:",
"{:.9f}".format(data.frequency), "absoluteImpedance: ", "{:.9f}".format(data.absoluteImpedance),
"phaseAngle: ", "{:.9f}".format(data.phaseAngle)))
00056     # print when a new node starts to the console
00057     handler.experimentNewElementStarting.connect(lambda channel, data: print("New Node beginning:",
data.stepName, "step number: ", data.stepNumber, " step sub : ", data.substepNumber))
00058     # called when an experiment has completed
00059     handler.experimentStopped.connect(experiment_complete_handler)
00060
00061     # initialize an experiment
00062     experiment = AisExperiment()
00063     # define a constant potential experiment at 0.2 V, with 1 s sampling time, and a duration of 10 s
00064     cvElement = AisConstantPotElement(0.2, 1, 10)
00065     # define a constant current experiment at 0.1 A, with 0.1 s sampling time, and a duration of 5 s
00066     ccElement = AisConstantCurrentElement(0.1, 0.1, 5)
00067
00068     success = True
00069
00070     # initialize a sub experiment
00071     subExperiment = AisExperiment()
00072     # add constant current experiment to position 1 of the sub experiment
00073     # this experiment will run 1 time
00074     success &= subExperiment.appendElement(ccElement, 1)
```

```

00075     # add constant potential experiment to positions 2 and 3 of the sub experiment
00076     # this experiment will run 2 times
00077     success &= subExperiment.appendElement(cvElement, 2)
00078
00079     # add constant current experiment to position 1 and 2 of the main experiment
00080     # this experiment will run 2 times
00081     success &= experiment.appendElement(ccElement, 2)
00082     # add constant potential experiment to position 3 of the main experiment
00083     # this experiment will run 1 time
00084     success &= experiment.appendElement(cvElement, 1)
00085
00086     # add the sub experiment to the main experiment
00087     # the sub experiment will run 2 times
00088     success &= experiment.appendSubExperiment(subExperiment, 2)
00089
00090     if not success:
00091         print("Error building experiment")
00092         app.quit()
00093
00094     # upload experiment list to the given channel
00095     error = handler.uploadExperimentToChannel(0, experiment)
00096     if error.value() != AISErrorCode.Success:
00097         print(error.message())
00098         app.quit()
00099
00100     # start the expiment on channel
00101     error = handler.startUploadedExperiment(0)
00102     if error.value() != AISErrorCode.Success:
00103         print(error.message())
00104         app.quit()
00105
00106     # Add initial delay (5 s)
00107     await asyncio.sleep(5)
00108
00109     # setup main to be ran as async
00110     startFunc = main()
00111     asyncio.run(startFunc)
00112
00113     # exit application
00114     sys.exit(app.exec()) # Start the event loop
00115

```

18.12 basicExperiment.py

```

00001 """! @example basicExperiment.py """
00002
00003 import sys
00004 from PySide6.QtCore import QApplication
00005 from SquidstatPyLibrary import AISDeviceTracker
00006 from SquidstatPyLibrary import AISExperiment, AISErrorCode
00007 from SquidstatPyLibrary import AISInstrumentHandler
00008 from SquidstatPyLibrary import AISConstantPotElement
00009
00010 # Define relavant device information, for easy access
00011 COMPORT = "COM16"
00012 CHANNEL = 0
00013
00014 app = QApplication()
00015
00016 tracker = AISDeviceTracker.Instance()
00017
00018
00019
00021 cvElement = AISConstantPotElement(1, 1, 30)
00022
00023 # After this point, the experiment is empty, so we need to add some elements to it
00024 experiment = AISExperiment()
00025 # Append the constant potential element, and tell the experiment to execute that element 1 time
00026 success = experiment.appendElement(cvElement, 1)
00027
00028 # Check if the element was added successfully
00029 if not success:
00030     print("Error adding element to experiment")
00031     app.quit()
00032
00033
00034
00035 def connectSignals(handler):
00036
00037     handler.activeDCDataReady.connect(lambda channel, data: print(f"Timestamp: {data.timestamp}
Current: {data.current} Voltage: {data.workingElectrodeVoltage} CE Voltage :
{data.counterElectrodeVoltage}"))
00038     handler.activeACDataReady.connect(lambda channel, data: print(f"Timestamp: {data.timestamp}
Frequency: {data.frequency} Absolute Impedance: {data.absoluteImpedance}"))

```

```

00039
00041
00042     # Whenever a new node in the element starts, note: some Ais Elements contain multiple logical
    nodes
00043     # i.e AisCyclicVoltammetryElement contains 4 nodes for each linear segment of each cycle plus a
    quiet time node if enabled
00044     # So this lambda would be executed atleast 4 times for each cycle
00045     handler.experimentNewElementStarting.connect(lambda channel, info: print(f"New element starting:
    {info.stepName}"))
00046     # Whenever an experiment completes or is manually stopped, this will execute
00047     handler.experimentStopped.connect(lambda channel, reason: (print(f"Experiment Stopped Signal
    {channel}, {reason}"), app.quit()))
00048     handler.deviceError.connect(lambda channel, error: print(f"Device Error: {error}"))
00049
00051
00052 def startExperiment(deviceName):
00053     print(f"New Device Connected: {deviceName}")
00054
00055     handler = tracker.getInstrumentHandler(deviceName)
00056
00057     connectSignals(handler)
00058
00059     error = handler.uploadExperimentToChannel(CHANNEL, experiment)
00060     # Exit the application if there is any error uploading experiment
00061     if error.value() != AIS_ERROR_CODE.Success:
00062         print(error.message())
00063         app.quit()
00064
00065     error = handler.startUploadedExperiment(CHANNEL)
00066     # Exit the application if there is any error starting experiment
00067     if error.value() != AIS_ERROR_CODE.Success:
00068         print(error.message())
00069         app.quit()
00070
00071
00072
00073 tracker.newDeviceConnected.connect(startExperiment)
00074 tracker.deviceDisconnected.connect(lambda deviceName: print(f"Device Disconnected: {deviceName}"))
00075
00076
00077
00078 error = tracker.connectToDeviceOnComPort(COMPORT)
00079 # Check if connection was successful
00080 if error.value() != AIS_ERROR_CODE.Success:
00081     print(error.message())
00082     sys.exit()
00083
00084
00085 # Calling sys.exit(app.exec()) will keep the program running until the application is exited
00086 sys.exit(app.exec())

```

18.13 dataOutput.py

```

00001 """! @example dataOutput.py """
00002 import sys
00003 from PySide6.QtCore import QApplication
00004 from PySide6.QtCore import QTextStream, QFile, QStandardPaths, QIODevice
00005 from SquidstatPyLibrary import AISDeviceTracker, AIS_ERROR_CODE
00006 from SquidstatPyLibrary import AISExperiment
00007 from SquidstatPyLibrary import AISInstrumentHandler
00008 from SquidstatPyLibrary import AISConstantPotElement
00009 from SquidstatPyLibrary import AISConstantCurrentElement
00010
00011 # Define relevant device information, for easy access
00012 COMPORT = "COM1"
00013 CHANNEL = 0
00014
00015 app = QApplication()
00016
00017 tracker = AISDeviceTracker.Instance()
00018
00019 cvElement = AISConstantPotElement(1, 1, 30)
00020 ccElement = AISConstantCurrentElement(0.001, 1, 60)
00021
00022 experiment = AISExperiment()
00023
00024 success = True
00025
00026 success &= experiment.appendElement(cvElement, 1)
00027 success &= experiment.appendElement(ccElement, 1)
00028
00029 if not success:
00030     print("Error building experiment")

```

```

00031     sys.exit()
00032
00033     filePath = ""
00034     fileNum = 1
00035
00036     def onNewElementStarting(channel, info):
00037         name = f"/{fileNum}_{info.stepNumber}_{info.stepName}.csv"
00038         filePath = QStandardPaths.writableLocation(QStandardPaths.DesktopLocation) + name
00039
00040         file = QFile(filePath)
00041         if not file.open(QIODevice.WriteOnly | QIODevice.Text):
00042             return
00043
00044         # Writing headers to file
00045         out = QTextStream(file)
00046         out << "Time Stamp, Counter Electrode Voltage, Working Electrode Voltage, Current \n"
00047         file.close()
00048
00049         print(f"New element beginning: {info.stepName} step: {info.stepNumber}")
00050
00051     def onActiveDCDataReady(channel, data):
00052         print(f"current: {data.current} voltage: {data.workingElectrodeVoltage} counter electrode:
00053         {data.counterElectrodeVoltage} timestamp: {data.timestamp}")
00054
00055         # Save the DC data to the file created at element beginning
00056         file = QFile(filePath)
00057         if not file.open(QIODevice.Append | QIODevice.WriteOnly | QIODevice.Text):
00058             return
00059
00060         out = QTextStream(file)
00061         out << data.timestamp << ", "<< data.counterElectrodeVoltage << ", " << data.workingElectrodeVoltage <<
00062         ", " << data.current << "\n"
00063         file.close()
00064
00065     def connectSignals(handler):
00066         handler.activeDCDataReady.connect(onActiveDCDataReady)
00067         handler.activeACDataReady.connect(lambda channel, data: print(f"Timestamp: {data.timestamp}
00068         Frequency: {data.frequency} Absolute Impedance: {data.absoluteImpedance}"))
00069         handler.experimentNewElementStarting.connect(onNewElementStarting)
00070         handler.experimentStopped.connect(lambda channel, reason: (print(f"Experiment Stopped Signal
00071         {channel}, {reason}"), app.quit()))
00072         handler.deviceError.connect(lambda channel, error: print(f"Device Error: {error}"))
00073
00074     def runExperiment(deviceName):
00075         handler = tracker.getInstrumentHandler(deviceName)
00076
00077         connectSignals(handler)
00078
00079         error = handler.uploadExperimentToChannel(CHANNEL, experiment)
00080         # Exit the application if there is any error uploading experiment
00081         if error.value() != AisErrorCode.Success:
00082             print(error.message())
00083             app.quit()
00084
00085         error = handler.startUploadedExperiment(CHANNEL)
00086         # Exit the application if there is any error starting experiment
00087         if error.value() != AisErrorCode.Success:
00088             print(error.message())
00089             app.quit()
00090
00091         tracker.newDeviceConnected.connect(runExperiment)
00092         tracker.deviceDisconnected.connect(lambda deviceName: print(f"Device Disconnected: {deviceName}"))
00093
00094         error = tracker.connectToDeviceOnComPort(COMPORT)
00095         # Check if connection was successful
00096         if error.value() != AisErrorCode.Success:
00097             print(error.message())
00098             sys.exit()
00099
00100         sys.exit()
00101
00102     # Calling sys.exit(app.exec()) will keep the program running until the application is exited
00103     sys.exit(app.exec())

```

18.14 firmwareUpdate.py

```

00001 """! @example firmwareUpdate.py """
00002
00003 import sys
00004 from PySide6.QtCore import QCoreApplication
00005
00006 from SquidstatPyLibrary import AisDeviceTracker
00007 from SquidstatPyLibrary import AisInstrumentHandler
00008 from SquidstatPyLibrary import AisErrorCode
00009

```

```

00010 # Define relavant device information, for easy access
00011 COMPORT = "COM16"
00012
00013 app = QCoreApplication()
00014
00015 tracker = AisDeviceTracker.Instance()
00016
00017 def onProgressMessage(message):
00018     print(message)
00019     if message.__contains__("firmware is updated"):
00020         app.quit()
00021
00022
00023 tracker.firmwareUpdateNotification.connect(onProgressMessage)
00024
00025
00026 # Attempt to connect to the device
00027 error = tracker.connectToDeviceOnComPort(COMPORT)
00028 if error.value() == AisErrorCode.FirmwareNotSupported:
00029     error = tracker.updateFirmwareOnComPort(COMPORT)
00030
00031 # Some other error occurred
00032 if error.value() != AisErrorCode.Success:
00033     print(f"Error: {error.message()}")
00034     sys.exit()
00035 elif error.value() != AisErrorCode.Success:
00036     print(f"Error: {error.message()}")
00037     sys.exit()
00038 else:
00039     print("Device is already up to date.")
00040     sys.exit()
00041
00042 sys.exit(app.exec())
00043

```

18.15 linkedChannels.py

This example will show you how linked channels can be used to combine the multiple channels on a device, in order to amplify the output for a single experiment.

This example will show you how linked channels can be used to combine the multiple channels on a device, in order to amplify the output for a single experiment. [AisInstrumentHandler.setLinkedChannels](#) MUST be called before each experiment that uses paralleled channels. Once linked, these channels must be controlled by ONLY the master channel, which is returned by the `AisInstrumentHandler::setLinkedChannels` function.

Note

This feature is only available on Cyclor models.

```

00001 """! @example linkedChannels.py
00002     This example will show you how linked channels can be used to combine the multiple channels on a
00003     device, in order to amplify the output for a single experiment.
00004     AisInstrumentHandler::setLinkedChannels MUST be called before each experiment that uses paralleled
00005     channels.
00006     Once linked, these channels must be controlled by ONLY the master channel, which is returned by
00007     the AisInstrumentHandler::setLinkedChannels function.
00008
00009     @note This feature is only available on Cyclor models.
00010 """
00011 import sys
00012 from PySide6.QtCore import QCoreApplication
00013 from SquidstatPyLibrary import AisDeviceTracker, AisInstrumentHandler, AisErrorCode,
00014     AisConstantCurrentElement, AisExperiment
00015
00016 # Define relavant device information, for easy access
00017 COMPORT = "COM1"
00018
00019 app = QCoreApplication()
00020
00021 tracker = AisDeviceTracker.Instance()
00022
00023 # Build an experiment
00024 ccElement = AisConstantCurrentElement(10, 1, 30)
00025 experiment = AisExperiment()
00026 success = experiment.appendElement(ccElement)
00027

```



```

00024 if not success:
00025     print("Error adding element to experiment")
00026     app.quit()
00027
00028 def connectSignals(handler):
00029     handler.activeDCDataReady.connect(lambda channel, data: print(f"Timestamp: {data.timestamp}
Current: {data.current} Voltage: {data.workingElectrodeVoltage} CE Voltage :
{data.counterElectrodeVoltage}"))
00030     handler.deviceError.connect(lambda channel, error: print(f"Device Error: {error}"))
00031     handler.experimentStopped.connect(lambda channel, reason: (print(f"Experiment Stopped Signal
{channel}, {reason}"), app.quit()))
00032
00033 def startExperiment(deviceName):
00034     handler = tracker.getInstrumentHandler(deviceName)
00035
00036     # Here we want to link channels 0 and 1 together, so we pass in a vector of the channels to link
00037     # It will return which of the channels is the master channel, this should be used to control the
experiment
00038     masterChannel = handler.setLinkedChannels([ 0, 1 ])
00039
00040     connectSignals(handler)
00041
00042     error = handler.uploadExperimentToChannel(masterChannel, experiment)
00043     if error.value() != AISERRORCODE.Success:
00044         print(error.message())
00045         app.quit()
00046
00047     # Start the previously uploaded experiment on the master channel
00048     error = handler.startUploadedExperiment(masterChannel)
00049     if error.value() != AISERRORCODE.Success:
00050         print(error.message())
00051         app.quit()
00052
00053
00054 tracker.newDeviceConnected.connect(startExperiment)
00055 tracker.deviceDisconnected.connect(lambda deviceName: print(f"Device Disconnected: {deviceName}"))
00056
00057 error = tracker.connectToDeviceOnComPort(COMPORT)
00058 if error.value() != AISERRORCODE.Success:
00059     print(error.message())
00060     sys.exit()
00061 # Calling sys.exit(app.exec()) will keep the program running until the application is exited
00062 sys.exit(app.exec())

```

18.16 manualExperiment.py

```

00001 """! @example manualExperiment.py """
00002 import sys
00003 from PySide6.QtCore import QCoreApplication
00004 from PySide6.QtCore import QTimer
00005 from SquidstatPyLibrary import AISDeviceTracker
00006 from SquidstatPyLibrary import AISInstrumentHandler
00007 from SquidstatPyLibrary import AISERRORCODE
00008
00009
00010 # Define relevant device information, for easy access
00011 COMPORT = "COM1"
00012 CHANNEL = 0
00013
00014 app = QCoreApplication()
00015
00016 tracker = AISDeviceTracker.Instance()
00017
00018 def handleStopExperiment(handler, reason):
00019     print(reason)
00020     print("Experiment has ended. Closing application.")
00021     app.quit()
00022
00023
00024 def connectSignal(handler):
00025     handler.activeDCDataReady.connect(lambda channel, data: print(f"Timestamp: {data.timestamp}
Current: {data.current} Voltage: {data.workingElectrodeVoltage} CE Voltage :
{data.counterElectrodeVoltage}"))
00026     handler.deviceError.connect(lambda channel, error: print(f"Device Error: {error}"))
00027     handler.experimentStopped.connect(handleStopExperiment)
00028
00029 def startExperiment(deviceName):
00030     handler = tracker.getInstrumentHandler(deviceName)
00031
00032     connectSignal(handler)
00033
00034
00035

```

```

00036     # The default starting mode is always Open Circuit Potential.
00037
00038     print("Starting manual mode at open circuit potential")
00039     error = handler.startManualExperiment(CHANNEL)
00040     if error.value() != AIS.ERRORCODE.SUCCESS:
00041         print(error.message())
00042         app.quit()
00043
00044
00045
00046
00047     # In this section we create wrapper functions for the manual mode changing functions.
00048     # These wrappers are called asynchronously when singleshoot QTimer.s expire.
00049
00050     # This function changes the instrument to Constant Current at .1A
00051     def setConstantCurrent(channel):
00052         print("Switching to constant current at .1A")
00053         error = handler.setManualModeConstantCurrent(channel, .1)
00054         if error.value() != AIS.ERRORCODE.SUCCESS:
00055             print(error.message())
00056     # It is called 5 seconds after the experiment starts
00057     QTimer.singleShot(5000, lambda:setConstantCurrent(CHANNEL))
00058
00059     # This function changes the instrument to Constant Voltage at 1V
00060     def setConstantVoltage(channel):
00061         print("Switching to constant voltage at 1V")
00062         error = handler.setManualModeConstantVoltage(channel, 1)
00063         if error.value() != AIS.ERRORCODE.SUCCESS:
00064             print(error.message())
00065     # It is called 15 seconds after the experiment starts
00066     QTimer.singleShot(15000, lambda:setConstantVoltage(CHANNEL))
00067
00068     # This function changes the instrument to Open Circuit Potential
00069     def setOpenCircuit(channel):
00070         print("Switching to open circuit potential")
00071         error = handler.setManualModeOCP(channel)
00072         if error.value() != AIS.ERRORCODE.SUCCESS:
00073             print(error.message())
00074     # It is called 25 seconds after the experiment starts
00075     QTimer.singleShot(25000, lambda:setOpenCircuit(CHANNEL))
00076
00077
00078
00079
00080
00081     # Stop experiment after 30 seconds
00082     def stopExperiment(channel):
00083         print("Stopping experiment.")
00084         error = handler.stopExperiment(channel)
00085         if error.value() != AIS.ERRORCODE.SUCCESS:
00086             print(error.message())
00087     QTimer.singleShot(30000, lambda:stopExperiment(CHANNEL))
00088
00089
00090
00091 tracker.newDeviceConnected.connect(startExperiment)
00092 tracker.deviceDisconnected.connect(lambda deviceName: print(f"Device Disconnected: {deviceName}"))
00093
00094 error = tracker.connectToDeviceOnComPort(COMPORT)
00095 if error.value() != AIS.ERRORCODE.SUCCESS:
00096     print(error.message())
00097     sys.exit()
00098 # Calling sys.exit(app.exec()) will keep the program running until the application is exited
00099 sys.exit(app.exec())

```

18.17 nonblockingExperiment.py

```

00001 """! @example nonblockingExperiment.py """
00002 import sys
00003 from PySide6.QtCore import QApplication
00004 from PySide6.QtCore import QTimer
00005 from SquidstatPyLibrary import AISDeviceTracker
00006 from SquidstatPyLibrary import AISInstrumentHandler
00007 from SquidstatPyLibrary import AISERRORCODE
00008
00009
00010 # Define relevant device information, for easy access
00011
00012 CHANNEL = 0
00013
00014 app = QApplication()
00015
00016 tracker = AISDeviceTracker.Instance()

```

```

00017
00018 def connectSignals(handler):
00019     handler.activeDCDataReady.connect(lambda channel, data: print(f"Timestamp: {data.timestamp}
Current: {data.current} Voltage: {data.workingElectrodeVoltage} CE Voltage :
{data.counterElectrodeVoltage}"))
00020     handler.activeACDataReady.connect(lambda channel, data: print(f"Frequency: {data.frequency}
Absolute Impedance: {data.absoluteImpedance} Phase Angle: {data.phaseAngle}"))
00021     handler.experimentNewElementStarting.connect(lambda channel, nodeInfo: print(f"New Node
Beginning"))
00022     handler.experimentStopped.connect(lambda channel, reason: print(f"Experiment Stopped Signal
{channel}, {reason}"))
00023
00024 def startExperiment(deviceName):
00025     handler = tracker.getInstrumentHandler(deviceName)
00026
00027     connectSignals(handler)
00028
00029     error = handler.startManualExperiment(CHANNEL)
00030     if error.value() != AIS.ERROR_CODE.SUCCESS:
00031         print(error.message())
00032         app.quit()
00033
00034     error = handler.setManualModeSamplingInterval(CHANNEL, 2)
00035     if error.value() != AIS.ERROR_CODE.SUCCESS:
00036         print(error.message())
00037         app.quit()
00038
00039     error = handler.setManualModeConstantVoltage(CHANNEL, 2)
00040     if error.value() != AIS.ERROR_CODE.SUCCESS:
00041         print(error.message())
00042         app.quit()
00043
00044     error = handler.setManualModeOCP(CHANNEL)
00045     if error.value() != AIS.ERROR_CODE.SUCCESS:
00046         print(error.message())
00047         app.quit()
00048
00049     def stopExperiment():
00050         error = handler.stopExperiment(CHANNEL)
00051         if error.value() != AIS.ERROR_CODE.SUCCESS:
00052             print(error.message())
00053             app.quit()
00054
00055     QTimer.singleShot(25000, stopExperiment)
00056
00057     # Allow the recent loop to process other events while the current experiment is running
00058     while handler.isChannelBusy(CHANNEL):
00059         app.processEvents()
00060
00061     # Process any remaining events
00062     app.processEvents()
00063
00064 tracker.newDeviceConnected.connect(startExperiment)
00065
00066 numDevices = tracker.connectAllPluggedInDevices()
00067 if numDevices == 0:
00068     print("No devices connected")
00069     sys.exit()
00070
00071 # Calling sys.exit(app.exec()) will keep the program running until the application is exited
00072 sys.exit(app.exec())

```

18.18 pulseData.py

```

00001 """! @example pulseData.py """
00002
00003 import sys
00004 from PySide6.QtCore import QApplication
00005 from SquidstatPyLibrary import AISDeviceTracker
00006 from SquidstatPyLibrary import AISCompRange
00007 from SquidstatPyLibrary import AISDCData
00008 from SquidstatPyLibrary import AISACData
00009 from SquidstatPyLibrary import AISExperimentNode
00010 from SquidstatPyLibrary import AISERROR_CODE
00011 from SquidstatPyLibrary import AISExperiment
00012 from SquidstatPyLibrary import AISInstrumentHandler
00013 from SquidstatPyLibrary import AIScyclicVoltammetryElement
00014 from SquidstatPyLibrary import AISDiffPulseVoltammetryElement
00015 from SquidstatPyLibrary import AISDataManipulator
00016 from SquidstatPyLibrary import AISPulseType
00017
00018 # do you want headers in your file?
00019 write_header = True

```

```

00020 # instantiate the data manipulator
00021 data_manipulator = AisDataManipulator()
00022 # setup COM port
00023 COMPORT = "COM5"
00024 CHANNEL = 0
00025
00026 def create_logic(handler):
00027     def on_active_dc_data_ready(channel, data):
00028         # define global parameters
00029         global write_header, data_manipulator
00030
00031         # convert time to UTC
00032         # utc_time = handler.getExperimentUTCStartTime(0)
00033         # read data via data manipulator
00034         data_manipulator.loadPrimaryData(data)
00035
00036         # upon completed pulse, write data with header
00037         if data_manipulator.isPulseCompleted():
00038
00039             # if write_header = true
00040             # prints header for every pulse data point
00041             if write_header:
00042                 print("time, Pulse_current, base_current, diff_current, base_voltage, pulse_voltage")
00043                 write_header = False
00044
00045             # will print data as defined by the header print statement above
00046             print(f"{data.timestamp}, {data_manipulator.getPulseCurrent()},
{data_manipulator.getBaseCurrent()}, "
f"{data_manipulator.getPulseCurrent() - data_manipulator.getBaseCurrent()}, "
f"{data_manipulator.getBaseVoltage()}, {data_manipulator.getPulseVoltage()}")
00047
00048             # function for printing ac data when it is received
00049             def on_active_ac_data_ready(channel, data):
00050                 print(f"channel: {channel}, frequency: {data.frequency}, absoluteImpedance:
{data.absoluteImpedance}, "
f"phaseAngle: {data.phaseAngle}, timestamp: {data.timestamp}")
00051
00052             # function for printing a new node when it begins
00053             def on_experiment_new_element_starting(channel, data):
00054                 print(f"New Node beginning {data.stepName}, step number {data.stepNumber}, step sub:
{data.substepNumber}")
00055
00056             # function for printing an experiment has stopped
00057             def on_experiment_stopped(channel, reason):
00058                 print(f"Experiment has completed on channel {channel}, {reason}")
00059                 QCoreApplication.quit()
00060
00061             # function to print an experiment has paused
00062             def on_experiment_paused(channel):
00063                 print(f"Experiment on channel {channel} has been paused")
00064
00065             # function to print an experiment has resumed
00066             def on_experiment_resumed(channel):
00067                 print(f"Experiment resumed on channel {channel}")
00068
00069             # pass dc data to data manipulator function
00070             handler.activeDCDataReady.connect(on_active_dc_data_ready)
00071
00072             # pass ac data to print function
00073             handler.activeACDataReady.connect(on_active_ac_data_ready)
00074
00075             # pass new node starting to print function
00076             handler.experimentNewElementStarting.connect(on_experiment_new_element_starting)
00077
00078             # pass experiment stopped to print function
00079             handler.experimentStopped.connect(on_experiment_stopped)
00080
00081             # pass experiment paused to print function
00082             handler.experimentPaused.connect(on_experiment_paused)
00083             handler.experimentResumed.connect(on_experiment_resumed)
00084
00085 # setup experiment
00086 def main():
00087     # initialize the application
00088     app = QCoreApplication()
00089
00090     # get a device tracker
00091     tracker = AisDeviceTracker.Instance()
00092
00093     # Create the AisDiffPulseVoltammetryElement pulse experiment.
00094     # startPotential (V), endPotential (V), potentialStep (V), pulseHeight (V), pulseWidth (s),
00095     # pulsePeriod (s)
00096     dpv_element = AisDiffPulseVoltammetryElement(-0.115, 0.115, 0.005, 0.01, 0.02, 0.2)
00097
00098     # set start voltage VS reference
00099     dpv_element.setStartVoltageVsOCP(False)
00100
00101
00102

```

```

00103     # set end voltage VS reference
00104     dpv_element.setEndVoltageVsOCP(False)
00105
00106     # set current range
00107     # will range up, but will not range down
00108     dpv_element.setApproxMaxCurrent(0.2)
00109
00110     # initialize an experiment
00111     experiment = AisExperiment()
00112
00113     # append differential pulse element to the experiment list
00114     # will run 1 time
00115     success = experiment.appendElement(dpv_element, 1)
00116     if not success:
00117         print("Error building experiment")
00118         sys.exit()
00119
00120     # Create an `AisDataManipulator` class for calculating advance parameters.
00121     data_manipulator.setPulseType(AisPulseType.DifferentialPulse, dpv_element.getPulseWidth(),
dpv_element.getPulsePeriod())
00122
00123     # if device is connected, get name and use logic function to handle events
00124     def on_new_device_connected(device_name):
00125
00126         # get instrument handler using device name
00127         handler = tracker.getInstrumentHandler(device_name)
00128
00129         # create the required connections for the handler.
00130         create_logic(handler)
00131
00132         # upload experiment to device.
00133         error = handler.uploadExperimentToChannel(CHANNEL, experiment)
00134         if error.value() != AisErrorCode.ErrorCode.Success:
00135             print(f"Error: {error.message()}")
00136             app.quit()
00137
00138         # start experiment on device on defined channel
00139         error = handler.startUploadedExperiment(CHANNEL)
00140         if error.value() != AisErrorCode.ErrorCode.Success:
00141             print(f"Error: {error.message()}")
00142             app.quit()
00143
00144         # connect call back handler which is called on connection of device.
00145         tracker.newDeviceConnected.connect(on_new_device_connected)
00146
00147         # print which device has disconnected to terminal
00148         tracker.deviceDisconnected.connect(lambda device_name: print(f"{device_name} has been
disconnected"))
00149
00150         # if error is encountered, print to terminal
00151         error = tracker.connectToDeviceOnComPort(COMPORT)
00152         if error.value() != AisErrorCode.ErrorCode.Success:
00153             print(f"Error: {error.message()}")
00154             sys.exit()
00155
00156         # exit application
00157         sys.exit(app.exec())
00158
00159 # run main
00160 if __name__ == "__main__":
00161     main()

```

18.19 tcpClient.py

This example file shows how to create a TCP client which can send commands to a server that is connected to a Squidstat.

This example file shows how to create a TCP client which can send commands to a server that is connected to a Squidstat. You can build experiments and receive data from the server without having to interact with the instrument directly.

```

00001 """! @example tcpClient.py This example file shows how to create a TCP client which can send commands
to a server that is connected to a Squidstat.
00002
00003 You can build experiments and receive data from the server without having to interact with the
instrument directly.
00004 """
00005
00006 import os
00007 import socket

```

```

00008 import threading
00009 import time
00010
00011 # Create a TCP/IP socket
00012 client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
00013 activeSockets = [client_socket]
00014
00015 # Define the server address and port
00016 SERVER_HOST = "localhost"
00017 SERVER_PORT = 12345
00018
00019 # Function to send a command to the server
00020 def send_command(command):
00021     # Send the command to the server
00022     try:
00023         client_socket.send(command.encode())
00024     except:
00025         print("Connection was closed by host")
00026         os._exit(1)
00027
00028     # Receive and print the response from the server
00029     response = client_socket.recv(1024).decode()
00030     print("Server response:", response)
00031
00032 # listens for <CTRL>+c to stop the client script
00033 def interrupt_listener():
00034     print("Press <CTRL>+c to stop the program at any time.")
00035     try:
00036         while True:
00037             input()
00038     except (EOFError, KeyboardInterrupt):
00039         pass
00040     for socket in activeSockets:
00041         socket.close()
00042     os._exit(1)
00043
00044 # Try and open a socket to the server
00045 try:
00046     client_socket.connect((SERVER_HOST, SERVER_PORT))
00047 except Exception as ex:
00048     print("Unable to establish connection to server:\n%s" % ex)
00049     exit()
00050
00051 print("Connected to the server.")
00052
00053 # Get a duration from the user
00054 duration = 0
00055 while duration == 0:
00056     try:
00057         duration = int(input("Enter a duration for Open Circuit Potential: "))
00058     except ValueError:
00059         duration = 0
00060     if duration < 1:
00061         print("Invalid entry.")
00062         duration = 0
00063
00064 # Send the start command to the server with the duration
00065 send_command(f'startExperiment {duration}')
00066
00067 interrupt_thread = threading.Thread(target=interrupt_listener)
00068 interrupt_thread.start()
00069
00070 # Listen for information from the server, which at this point will be data and the experiment stop
00071 message
00072 while True:
00073     try:
00074         data = client_socket.recv(1024).decode()
00075     except (ConnectionAbortedError, BrokenPipeError):
00076         # This exception will be raised when the user presses <ENTER>
00077         print("Finishing connection")
00078         break
00079     except ConnectionResetError:
00080         print("The server closed the connection suddenly.")
00081         break
00082
00083     if not data:
00084         break
00085
00086     # Handle the data that was received.
00087     print(data)
00088
00089     if("Experiment Completed: " in data):
00090         break
00091
00092 os._exit(1)

```

18.20 tcpServer.py

This example file shows how to create a server that connects to an instrument.

This example file shows how to create a server that connects to an instrument. It can receive TCP commands from a client to trigger experiments and send data back to the client.

```

00001 """! @example tcpServer.py
00002 @brief This example file shows how to create a server that connects to an instrument.
00003
00004 It can receive TCP commands from a client to trigger experiments and send data back to the client.
00005 """
00006 import os
00007 import socket
00008 import threading
00009 from PySide6.QtCore import QApplication
00010 from SquidstatPyLibrary import AisDeviceTracker
00011 from SquidstatPyLibrary import AisExperiment
00012 from SquidstatPyLibrary import AisOpenCircuitElement
00013 from SquidstatPyLibrary import AisErrorCode
00014
00015 # Define the server address and port
00016 HOST = 'localhost'
00017 PORT = 12345
00018
00019 # The COM port the Squidstat is connected to
00020 SQUIDCOMPORT = "COM1"
00021 SQUIDNAME = "Plus2000"
00022
00023 # Create the QT application
00024 app = QApplication([])
00025 activeSockets = []
00026
00027 # This will build and start the Open Circuit Potential experiment
00028 def start_ocp_experiment(handler, durationSec=60):
00029     # Create an experiment with elements
00030     experiment = AisExperiment()
00031     ocpElement = AisOpenCircuitElement(durationSec, 1)
00032
00033     success = experiment.appendElement(ocpElement, 1)
00034     if not success:
00035         print("Error adding element to experiment")
00036         return error.ExperimentNotUploaded
00037
00038     # Upload the experiment to channel 0
00039     error = handler.uploadExperimentToChannel(0, experiment)
00040     if error.value() != AisErrorCode.ErrorCode.Success:
00041         return error
00042
00043     # Start the experiment
00044     return(handler.startUploadedExperiment(0))
00045
00046 # Send a specified command to our Squidstat
00047 def command_to_device(command, handler):
00048     # Check if we had an argument associated with the command
00049     splitCommand = command.split(" ")
00050     action = splitCommand[0]
00051     actionArg = 0
00052     if(len(splitCommand) > 1):
00053         try:
00054             actionArg = int(splitCommand[1])
00055         except:
00056             actionArg = 0
00057
00058     # Here you can add various commands which can be send from the tcpClient to directly interact with
    the Squidstat
00059     response = None
00060     if action == 'startExperiment':
00061         response = start_ocp_experiment(handler, actionArg)
00062     elif action == 'stopExperiment':
00063         response = handler.stopExperiment(0)
00064     else:
00065         #print("Invalid command:", command)
00066         pass
00067     return response
00068
00069 # Handle commands from the client
00070 def handle_command(command, handler, client_socket):
00071     # Send a response back to the client
00072     responseMsg = "Unknown Command"
00073     response = command_to_device(command, handler)
00074     if(response != None):
00075         responseMsg = response.message()
00076     response = "{}".format(responseMsg)
00077     client_socket.send(response.encode())

```

```

00078
00079 # Listen for the client's messages, and disconnect signals and terminate program when finished
00080 def handle_client(handler, client_socket):
00081     print("Client connected")
00082
00083     while True:
00084         # Receive data from the client
00085         try:
00086             data = client_socket.recv(1024).decode()
00087         except ConnectionResetError:
00088             break
00089
00090         # Check if the client has closed the connection
00091         if not data:
00092             break
00093
00094         # Handle the command
00095         handle_command(data, handler, client_socket)
00096
00097     handler.activeDCDataReady.disconnect()
00098     handler.activeACDataReady.disconnect()
00099     handler.experimentNewElementStarting.disconnect()
00100     handler.experimentStopped.disconnect()
00101     command_to_device("stopExperiment", handler)
00102     # Close the client socket
00103     client_socket.close()
00104     print("Client disconnected")
00105     os._exit(1)
00106
00107
00108 # Send data the the client based on the type of event (Hooked up to signals)
00109 def send_data_to_client(client_socket, event_type, data):
00110     if event_type == "DCData":
00111         message = "timestamp: {:.9f}, workingElectrodeVoltage: {:.9f}".format(data.timestamp,
data.workingElectrodeVoltage)
00112     elif event_type == "ACData":
00113         message = "frequency: {:.9f}, absoluteImpedance: {:.9f}, phaseAngle:
{:.9f}".format(data.frequency, data.absoluteImpedance, data.phaseAngle)
00114     elif event_type == "NewElement":
00115         message = "New Node beginning: {}, step number: {}, step sub: {}".format(data.stepName,
data.stepNumber, data.substepNumber)
00116     elif event_type == "ExperimentCompleted":
00117         message = "Experiment Completed: {}".format(data)
00118     else:
00119         return
00120
00121     client_socket.send(message.encode())
00122
00123 def terminate_program():
00124     print("Press <CTRL>+c to close the server")
00125     try:
00126         while True:
00127             input()
00128     except (EOFError, KeyboardInterrupt):
00129         pass
00130     for socket in activeSockets:
00131         socket.close()
00132     app.quit()
00133     os._exit(1)
00134
00135
00136 # Create the device tracker and connect to the Squidstat we will be using
00137 print(f"Attempting to connect to the Squidstat {SQUIDNAME} on {SQUIDCOMPORT}...")
00138 tracker = AisDeviceTracker.Instance()
00139 tracker.newDeviceConnected.connect(lambda deviceName: print("Device is Connected: %s" % deviceName))
00140 error = tracker.connectToDeviceOnComPort(SQUIDCOMPORT)
00141
00142 if error.value() != AisErrorCode.ErrorCode.Success:
00143     print(error.message())
00144     exit()
00145
00146 # Create the instrument handler
00147 handler = tracker.getInstrumentHandler(SQUIDNAME)
00148 print("Connection successful\n")
00149
00150 # Create the TCP/IP socket and bind it to our host
00151 print("Starting server...")
00152 server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
00153 activeSockets.append(server_socket)
00154 server_socket.bind((HOST, PORT))
00155
00156 # Listen for incoming connections
00157 server_socket.listen(1)
00158
00159 print("Server started successfully. Waiting for client connection...")
00160
00161 terminal_thread = threading.Thread(target=terminate_program)

```



```

00162 terminal_thread.start()
00163
00164 # Accept a client connection
00165 client_socket, client_address = server_socket.accept()
00166 activeSockets.append(client_socket)
00167
00168 # Connect the signals to send data to the client
00169 handler.activeDCDataReady.connect(lambda channel, data: send_data_to_client(client_socket, "DCData",
data))
00170 handler.activeACDataReady.connect(lambda channel, data: send_data_to_client(client_socket, "ACData",
data))
00171 handler.experimentNewElementStarting.connect(lambda channel, data: send_data_to_client(client_socket,
"NewElement", data))
00172 handler.experimentStopped.connect(lambda channel: send_data_to_client(client_socket,
"ExperimentCompleted", channel))
00173
00174 # Start the listening process in a separate thread
00175 listening_thread = threading.Thread(target=handle_client, args=(handler, client_socket))
00176 listening_thread.start()
00177
00178 # Start the QT event loop
00179 app.exec()

```

18.21 writeinCSV.py

```

00001 """! \example writeinCSV.py """
00002
00003 """
00004 This is an example of the writeinCSV.py file, which helps control Squidstat in parallel with other
devices.
00005 In this example, we inform other devices using SerialPortReader::writeData when a new element starts
executing
00006 inside Squidstat, and also print the data received from the other device using the
SerialPortReader::dataReceived signal.
00007 All operations occur in parallel with the Squidstat operation.
00008
00009 The Squidstat data is also written to a CSV file.
00010
00011 In detail:
00012 1. The `SerialPortReader` class handles reading from and writing to a serial port of other device.
00013 2. The `WriteCSV` class is responsible for writing data to a CSV file which is received from
Squidstat.
00014 3. The `writingThread` class is used to manage the data writing in csv file process in a separate
thread.
00015 """
00016
00017 import sys
00018 from PySide6.QtCore import QIODevice, QThread, QObject, Signal
00019 from PySide6.QtSerialPort import QSerialPort
00020 from PySide6.QtCore import QApplication
00021 from SquidstatPyLibrary import AisDeviceTracker
00022 from SquidstatPyLibrary import AISErrorCode
00023 from SquidstatPyLibrary import AISExperiment
00024 from SquidstatPyLibrary import AISConstantCurrentElement
00025 from SquidstatPyLibrary import AISOpenCircuitElement
00026
00027 # initialize the application
00028 app = QApplication([])
00029
00030 # convert incoming data to string with single line
00031 def convert_to_csv_line(data_list):
00032     return ','.join(str(item) for item in data_list)
00033
00034 # \cond EXCLUDE_FROM_DOX
00035 # class for reading and writing data from serial port of other device.
00036 class SerialPortReader(QObject):
00037     # define the Qt signal.
00038     dataReceived = Signal(str)
00039
00040     # initialize self and port
00041     def __init__(self, port):
00042         super().__init__()
00043         self.port = port
00044
00045     # open port if closed
00046     # get data and decode
00047     # emit data
00048     def run(self):
00049         if not self.port.isOpen():
00050             self.port.open(QIODevice.ReadOnly)
00051         while self.port.isOpen():
00052             if self.port.waitForReadyRead():
00053                 data = self.port.readAll().data().decode()

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00054         self.dataReceived.emit(data)
00055
00056     # open port if closed
00057     # write data and encode
00058     def writeData(self, data):
00059         if not self.port.isOpen():
00060             successfullyOpen = self.port.open(QIODevice.WriteOnly)
00061             if not successfullyOpen:
00062                 print("USB port is not open.")
00063             self.port.write(data.encode())
00064
00065     # check if port is open, close port if open
00066     def closePort(self):
00067         if self.port.isOpen():
00068             self.port.close()
00069
00070
00071 # class for writing data to a csv file. data received from Squidstat.
00072 class WriteCSV:
00073     # init filename and file
00074     def __init__(self, filename):
00075         self.filename = filename
00076         self.file = None
00077
00078     # open file and write headers
00079     def write_header(self, header):
00080         if self.file is None:
00081             self.file = open(self.filename, 'w')
00082             self.file.write(convert_to_csv_line(header) + '\n')
00083
00084     # write data to file
00085     def write_data(self, data):
00086         if self.file is not None:
00087             self.file.write(convert_to_csv_line(data) + '\n')
00088
00089     # close file when we are done
00090     def close(self):
00091         if self.file is not None:
00092             self.file.close()
00093
00094
00095 # class to handle the write functionality on seprate thread in pareller operation of Squidstat and
# other device.
00096 class writingThread(QThread):
00097     # define the signal.
00098     writeData = Signal(float, float)
00099     stopTowrite = Signal()
00100
00101     # init self with values to be written
00102     def __init__(self, csv_writer):
00103         super().__init__()
00104         self.timestamps = []
00105         self.voltages = []
00106         self.csv_writer = csv_writer
00107
00108     # setup data file with headers and connect the call back function on emitting of Qt signal.
00109     def run(self):
00110         self.csv_writer.write_header(['Timestamp', 'Working Electrode Voltage'])
00111         self.writeData.connect(self.add_data)
00112         self.stopTowrite.connect(self.close)
00113
00114     # add data into list as well as call back handler you can use to write the data in csv file.
00115     def add_data(self, timestamp, voltage):
00116         self.timestamps.append(timestamp)
00117         self.voltages.append(voltage)
00118         self.csv_writer.write_data([timestamp, voltage])
00119
00120     # close writer for a channel
00121     def close(self):
00122         self.csv_writer.close()
00123 # \endcond
00124
00125 # setup serial port of other device.
00126 serialPort = QSerialPort("COM3")
00127 # setup baud rate of other device.
00128 serialPort.setBaudRate(QSerialPort.Baud9600)
00129 # set up communication data type of other device.
00130 serialPort.setDataBits(QSerialPort.Data8)
00131 # define a serial port reader thread.
00132 serialPortReader = SerialPortReader(serialPort)
00133
00134
00135 # function to write data from the serial port of other device.
00136 def writeDataToPort(data):
00137     serialPortReader.writeData(data)
00138
00139 # instantiate a Squidstat device tracker

```

```

00140 tracker = AisDeviceTracker.Instance()
00141
00142 # interact with data and send experiments to Squidstat device
00143 def onNewDeviceConnected(deviceName):
00144     # print which device has been connected
00145     print(f"Connected to: {deviceName}")
00146     # get handler using device name.
00147     handler = tracker.getInstrumentHandler(deviceName)
00148     # if handler is present for the particular device then we can interact with the data and
upload/start/stop/puase/resume experiments
00149     if handler:
00150         # setup file name
00151         csv_writer = WriteCSV('dataFile.csv')
00152         # add csv file to writing thread.
00153         writingThread = writingThread(csv_writer)
00154         # start sub thread for write functionality.
00155         writingThread.start()
00156
00157         # manages DC data input and output
00158         # add more variables if you want to print more data to the console
00159         # send the signal to writing thread to write the information in csv file.
00160         handler.activeDCDataReady.connect(lambda channel, data: (
00161             print("timestamp:", "{:.9f}".format(data.timestamp), "workingElectrodeVoltage: ",
00162                 "{:.9f}".format(data.workingElectrodeVoltage)),
00163             writingThread.writeData.emit(data.timestamp, data.workingElectrodeVoltage)
00164         ))
00165
00166         # manages AC data input and output
00167         # add more variables if you want to print more data to the console
00168         handler.activeACDataReady.connect(lambda channel, data: print("frequency:",
"{:.9f}".format(data.frequency),
00169                                     "absoluteImpedance: ",
00170                                     "{:.9f}".format(
00171                                     data.absoluteImpedance),
00172                                     "phaseAngle: ",
00173                                     "{:.9f}".format(data.phaseAngle)))
00174         # write when new node is starting
00175         handler.experimentNewElementStarting.connect(lambda channel, data:
writeDataToPort(data.stepName))
00176         # print when experiment has stopped, stop writeting thread
00177         # send the signal to writing thread experiment is completed, which will close the csv file.
00178         handler.experimentStopped.connect(lambda channel: (print(f"Experiment completed on channel
{channel}"), writingThread.stopTowrite.emit(), app.quit()))
00179
00180         # initialize an experiment
00181         experiment = AisExperiment()
00182
00183         # define a constant current experiment at 0.1 A, with 1 s sampling time, and a duration of 10
s
00184         ccElement = AisConstantCurrentElement(0.1, 1, 10)
00185         # define an open circuit experiment with a duration of 10 s and a sampling time of 2 s
00186         opencircuitElement = AisOpenCircuitElement(10, 2)
00187
00188         # add constant current as the first element in the list
00189         # element runs 1 time
00190         successfullyadd = experiment.appendElement(ccElement, 1)
00191         # add open circuit as the second and thirds elements in the list
00192         # element runs 2 times
00193         successfullyadd |= experiment.appendElement(opencircuitElement, 2)
00194
00195         if not successfullyadd:
00196             print("Error adding element to experiment")
00197             app.quit()
00198
00199         # upload experiment to channel 1
00200         error = handler.uploadExperimentToChannel(0, experiment)
00201         if error.value() != AisErrorCode.Success:
00202             print(error.message())
00203             app.quit()
00204
00205         # start experiment on channel 1
00206         error = handler.startUploadedExperiment(0)
00207         if error.value() != AisErrorCode.Success:
00208             print(error.message())
00209             app.quit()
00210
00211         # connect to device associated with the tracker
00212         tracker.newDeviceConnected.connect(onNewDeviceConnected)
00213
00214         # Request the device to connect using com port 4
00215         error = tracker.connectToDeviceOnComPort("COM4")
00216         if error:
00217             print(error.message())
00218             sys.exit()
00219
00220         # print the data which is received from another device (other then Squidstat)

```

```
00219 serialPortReader.dataReceived.connect(lambda data: print("Received data from COM port 3:", data))
00220 # setup a sub thread for read and write the information from the other device.
00221 serialPortThread = QThread()
00222 # pushes object to another thread
00223 serialPortReader.moveToThread(serialPortThread)
00224 # start the sub thread of other device.
00225 serialPortThread.start()
00226 # exit program
00227 sys.exit(app.exec())
```

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